

CLASSIFICATION MADE SIMPLE

An Introduction to
Knowledge Organisation
and Information Retrieval

THIRD EDITION



ERIC J. HUNTER

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and Information Retrieval

Third Edition

ERIC J. HUNTER

Liverpool John Moores University, UK



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Introduction

This work was written in an attempt to simplify the initial study of classification as used for knowledge organisation and information retrieval.

The text adopts a gradual progression from very basic principles, a progression that, hopefully, will enable the reader to gain a firm grasp of one idea before proceeding to the next. It describes the different types of classification scheme, the way in which they are compiled, and the features that may be required. Practical examples are provided as illustration and reinforcement at each stage.

This revision takes account of developments that have taken place since the previous edition. Where published classification schemes are concerned, for instance, a 22nd edition of the *Dewey Decimal Classification* appeared in 2003 and a new edition of the *London Classification of Business Studies* in 2005. There are also other important, recently developed schemes designed for the electronic transfer of subject and business information. Examples are *BIC Subject Categories* and *BISAC Subject Headings*, intended for use in the book trade, and *UNSPSC (United Nations Standard Products and Services Code)*.

It must, however, be stressed that the book constitutes an introductory study only; as such it should provide a gateway to more advanced treatises and to a more detailed study of specific schemes or information handling processes.

In information work, classification may be used in various ways: for coding in computerised data processing; for the organisation of files and catalogues; and for shelf arrangement in libraries and information services. It can constitute a basis for the production of alphabetical authority lists of subject terms, or thesauri, and can be applied to other subject indexing and search techniques, including those used in online systems and the World Wide Web. Where the latter is concerned, as noted above, classification can facilitate e-commerce; the accurate coding of goods and services for electronic interchange is becoming essential and some form of categorisation can also be very helpful to the Internet shopper. The approach adopted here is a wide one and is not limited to classification for one specific purpose.



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1 What is Classification?

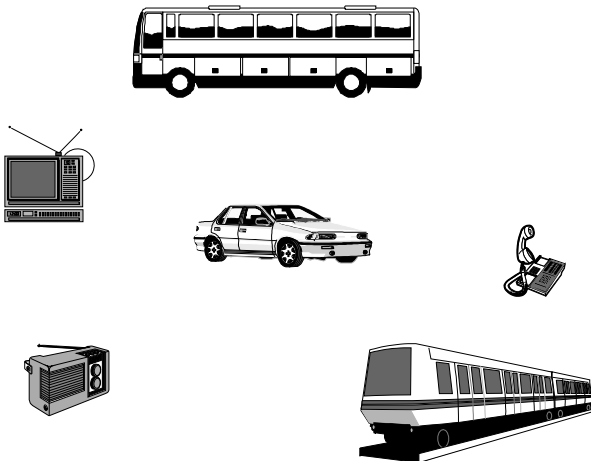
We all have to process information. Consider the following images:



Could you tell the difference? Of course you could; you would be in a certain amount of trouble if you couldn't. This means that you are capable of processing information and, in order to do that, you must make use of *classification*.

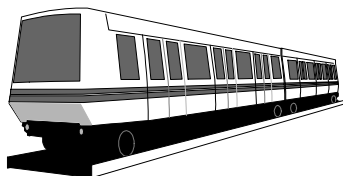
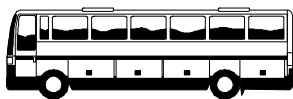
In essence the process of classification simply means the grouping together of like things according to some common quality or characteristic. This automatically implies the separation of the unlike as in the above example.

As human beings, we are able to recognise a member of a particular class because it displays certain characteristics common to that class but not to others. Examine the following collection:

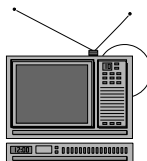


From observation, one is able to immediately identify like entities:

Forms of transport:



Means of communication:



In making the above distinction and grouping the things as shown, a process of classification has been carried out in that like things have been grouped together and unlike things separated. It is essential for a person to be able to think logically in this way and to make such distinctions – classification ‘lies at the base of every well-managed life and occupation’ (Maltby, 1975). Otherwise one would be sitting on the television set waiting to be transported to work! Beam me up, Scotty!

Wherever one looks, examples of the use of classification can be found. Take, for instance, the case of Harrison Ford, alias Indiana Jones. Ford hasn’t always been a film actor. In the early seventies he became seriously dissatisfied with the sameness and blandness of the work that he was being offered so he gave up acting and became a carpenter instead (Glock, 2008). Apparently he became a very good carpenter and examples of his work are to be found to this day in many a home in Beverly Hills (Norman, 1997). When Ford was whittling wood, he kept all of his carpentry tools laid out in order of type and size right though his entire workshop (Courtenay, 1994). He realised that this would make his work that much easier. By putting all of his *cutting* tools together, Ford would separate them immediately from his *drilling* tools and from his *smoothing* tools. Clearly, he recognises the enormous value of classification.

What would Ford have to say do you think, if he visited his local supermarket and found the goods arranged as shown below.

	Liqueurs	Lamb	Bread		
Beans	Margarine		Wine	Potatoes	
	Cakes	Beer	Beef		
Tea	Butter	Cabbage	Pies	Spirits	
	Carrots		Coffee		
Cheese	Cocoa	Pork	Chicken		

It would be **chaos!** Shopping would be very difficult and time consuming.

No doubt Ford would much prefer, as we all would, to find similar products shelved together and differing products separated:

Lamb	Bread	Liqueurs	Beans	Tea	Margarine
Beef	Cakes	Wine	Potatoes	Coffee	Butter
Pork	Pies	Beer	Cabbage	Cocoa	Cheese
Chicken		Spirits	Carrots		

It is now quite simple to process the information that we are being given. **Classification** makes shopping, and indeed all aspects of our lives, very much easier.

References

- Courtenay, Chris (1994), 'Indiana Jones and His Homely Crusade', *Daily Mail Weekend* (16 July), pp. 4-5.
- Glock, Allison (2008), 'Blade Runner', *Telegraph Magazine* (10 May), pp.46-50. ,
- Maltby, Arthur (1975), *Sayers' Manual of Classification for Librarians*, 5th ed., Deutsch, London, p. 15.
- Norman, Barry (1997), 'Why Ford Sticks To What He Does Best', *RadioTimes* (12-18 April), p. 46.

2 Classification in an Information System

Specifically – from the point of view of this book – classificatory techniques may be applied in an information system in order to facilitate access, organisation, use, and retrieval. The information system could be a manual filing system or a computerised data processing system, in which classification might be employed for coding entities. Where the recent phenomenon of the Internet is concerned, search engines and other facilities also make use of classification. In fact, it can be demonstrated, as we shall see, that whatever method of indexing or retrieval is used, just as in life itself, classification cannot be ignored.

Library and information services

Probably the most common example of a system that makes use of classification is the library or information service, where classification by subject is extensively used to arrange books and other materials on the shelves or to make entries for them in catalogues and indexes. If, for instance, we visited the local public library we would find the sciences:

Mathematics

Astronomy

Physics

Chemistry

Geology

shelved adjacent to each other, brought together by the classification scheme in use, possibly the *Dewey Decimal Classification* (see page 49) or *Library of Congress Classification* (see page 53).

In Chapter 1, mention was made of Harrison Ford giving up acting at one time. Having found acting dull, ‘he took up carpentry, teaching himself from library books’ (Glock, 2008) ‘before proceeding to totally strip and gut his house from top to bottom ... and rebuild the entire house to his own specification’ (Sellers, 1993). When he was first shown where the carpentry books were located on the library shelves, he found them nestling amongst other works which might also be of use to him – books on bricklaying, plastering, plumbing, etc. Again, these related subjects had been brought together by means of *classification*.

Data processing

Where data processing is concerned, classification is one of the most important ways used to deal with complexity of information. To take an example, in a business such as insurance, decisions have to be made about insurance cover for millions of individual people and ‘some way has to be found of estimating an individual’s lifespan, their likelihood of having a car accident or their probability of being burgled’ (Harry, 1994). ‘What insurance companies do is the same as we do as individuals, they classify their experience so that they can use commonly shared characteristics as a basis for simplifying the processing of a vast amount of information. Once you know that a particular person falls into the classes of ‘male’ and ‘aged 54’ you can use the results of processing the data about previous 54 year old males to estimate how long you expect the individual to live’ (*ibid.*). That information is then used to calculate the premiums that would be required from the man in order to guarantee a certain payout when he dies. Note that ‘projected lifespan’ of the man is not a characteristic of *every* individual man but the information that results from the processing of component data. Such data is more easily manipulated if it is coded. Clifton and Sutcliffe in their *Business Information Systems* (1994) maintain that ‘For practical purposes, it is impossible to identify, uniquely and unerringly, even as few as a thousand different entities if only their descriptions are available. In a computer-based business system, it is inevitable that code numbers (which can include alphabetic characters) are needed’.

One organisation which makes extensive use of coding is NATO (North Atlantic Treaty Organisation), which uses a classification designed to facilitate the function of supply management (see also pages 76-9).

The Internet

When retrieving information from the Internet, ‘there is a bland assumption that the machine does all that. Have these people never evaluated the quality of the material that Google drags up for them’ (Trickey, 2004). Google is, of course, currently one of the major search engines and later we will examine the part that classification can play in engines such as this or in other Internet services. For example, when making a purchase on the Internet, just as on the high street, the supplier will classify the items that are for sale to make it easier for the customer to identify his or her requirements. A clothes retailer might, for example, divide first by sex: ‘Men’s clothes’ or ‘Women’s clothes’. One of the divisions of Women’s clothes could be ‘Sports and leisure wear’ and this

could be further divided into ‘Trainers’, ‘Socks’, ‘Trousers’, ‘Tops’, and so on.

Subjects

The reader may have noted that the ‘things’ listed on page 2 in Chapter 1 are all concrete entities, whilst some of the ‘things’ noted in this chapter, sciences such as ‘mathematics’ or activities such as ‘life insurance’ involve abstract ideas. For example, a television is concrete but the activity of insuring the television against theft is abstract. In this text, for simplicity, both concrete entities and abstract ideas, or combinations of both (for example open-heart surgery), will usually be referred to as subjects.

Complexity of subjects

In an information system a classification scheme must not only cater for simple subjects which consist of a single concept, for example:

Sex
Children
Psychology
Teaching
Steel
Wires
Plating

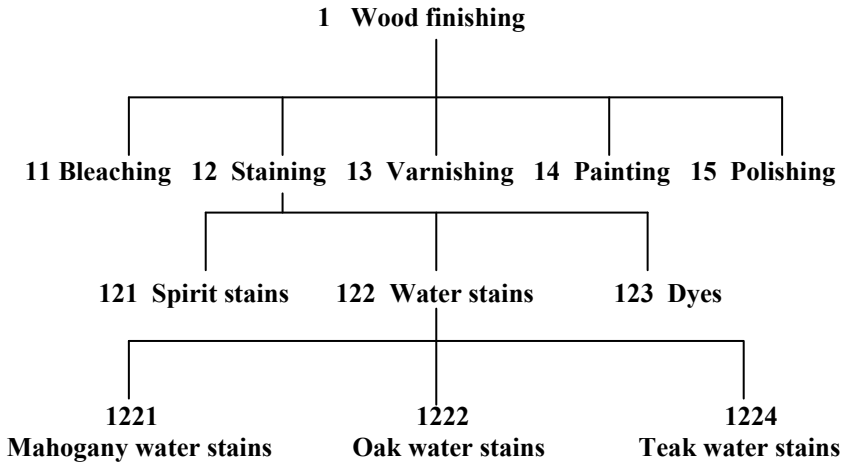
but also for more complex subjects which are formed by combining these concepts, for example:

Psychology of sex
Child psychology
Teaching children about sex
Teaching psychology
Teaching child psychology
Steel wires
Plating steel wires

In order to achieve this objective the system of classification, as used in an information system, may adopt a ‘hierarchical’ or ‘faceted’ approach, or a mixture of both. The former, as the name implies, utilises a ‘top down’ approach, a process of division producing a series of subject classes in successive subordination. Faceted classification, on the other hand, makes use of a ‘bottom up’ technique, constituent parts of subjects

being used as nuts and bolts to produce whatever subject classes are required.

The subject 'Wood finishing', for example, might be divided hierarchically by process. Each process would be further subdivided. It might be thought suitable to divide 'staining' by the 'type' of stain, e.g. 'dyes' and the 'type' of stain by 'shade': 'mahogany', 'oak', 'teak', and so on, as shown below for 'Water stains'.



Thus the subject 'Mahogany water stains for wood' is listed within the hierarchy and would be classified as 1221.

A faceted classification, on the other hand, would simply list within facets, or 'groups', the terms that together make up the subject, e.g.

1 Wood finishing

Process facet

- 1 Bleaching
- 2 Staining
- 3 Varnishing
- 4 Painting
- 5 Polishing

Type facet

- 1 Spirit
- 2 Water
- 3 Dye

Shade facet

- 1 Mahogany
- 2 Oak
- 3 Teak

The subject ‘Mahogany water stains for wood’ is not listed; it would have to be ‘reconstructed’ by combining the constituent components: ‘Staining’, ‘Water’ and ‘Mahogany’, to give the classification number 1221.

Although only partially developed, the above examples attempt to illustrate the basic difference between the two methods. The nature of faceted classification enables it ‘to be more easily interpretable by both human beings and computers’ (Clifton, 1994) and therefore this methodology will be the first to be more fully described.

References

- Clifton, H.O. and Sutcliffe, A.G. (1994), *Business Information Systems*, 5th ed., Prentice Hall, London, p. 320.
- Glock, Allison (2008), ‘Blade Runner’, *Telegraph Magazine* (10 May), pp.46-50.
- Harry, Mike (1994), *Information Systems in Business*, Pitman, London, p. 131.
- Sellers, Robert (1993), *Harrison Ford: a Biography*, Hale, London, p. 33.

3 Faceted Classification

A facet is defined in the Oxford dictionary as ‘one side of a many sided body’. Subjects can be broken down into constituent sides or parts and the word faceted, as we saw in Chapter 2, may therefore be applied to a classification which works on an analytico-synthetic principle. Fundamental concepts are analysed and grouped together into facets. These concepts can then be combined or synthesised as necessary to form more complex subjects. The following further example should help to make this principle clearer.

A brewing company sells canned and bottled beers (mild, bitter, lager, and stout) in various sizes from 330 ml to 3½ litres. The company wishes to devise a faceted classification scheme for coding items in its computerised sales system. In order to do this the relevant concepts, ‘330 ml’, ‘cans’, ‘bitter’ and so on, are grouped into facets using the characteristics of ‘capacity’, ‘container’ and ‘beer type’, that is:

Capacity facet	Container facet	Beer type facet
330 ml	Cans	Mild
440 ml	Bottles	Bitter
550 ml		Lager
1 litre		Stout
3½ litre		

Notation

Each concept within the classification scheme must be allocated a notational symbol which will enable the representation of that concept by a ‘shorthand’ code or classification number. Numeric digits could perform this function. Here is such a notation added to the above scheme:

Capacity facet	Container facet	Beer type facet
1 330 ml	1 Cans	1 Mild
2 440 ml	2 Bottle	2 Bitter
3 550 ml		3 Lager
4 1 litre		4 Stout
5 3½ litre		

The process of classification

The process of classification is achieved by first ascertaining the notation for each constituent concept and then combining (synthesising) these notations to form a coding or classification number for the complete subject. For example, to classify ‘440 ml cans of lager’ the subject is first broken down into its constituent concepts:

440 ml cans lager

and the relevant notations ascertained from the classification scheme:

2 1 3

These separate notations are then combined or synthesised to give the classification number:

213

Similarly:

1 litre bottles of mild = **421**
3½ litre cans of bitter = **512**

In order to avoid confusion, when using a notation such as that shown here, the notational length (that is the number of digits) must be constant and concepts from facets must be combined or cited in a consistent order. In the above examples the order is that in which the facets have been written down, namely:

Capacity → Container → Type of beer

If a coding in a computerized system were all that was required, then the above scheme would be satisfactory but for other purposes, for example a manual filing system, it would be inadequate. The reasons for this are explained at a later point in this text (page 17).

The initial steps involved in compiling a faceted scheme

Let us now review the steps that have been taken in order to compile the faceted scheme.

Step 1 Analysis into relevant concepts, that is:

440 ml
bitter

cans
lager
1 litre
etc

Step 2 Grouping into facets according to appropriate characteristics, for example:

Capacity facet	Container facet	Beer type facet
440 ml	Cans	Bitter
1 litre	Bottles	Lager
etc		

Step 3 Adding a notation, for example:

2 440 ml
4 1 litre
etc

Step 4 Choosing a citation order, for example:

Capacity → Container → Type of beer

A further decision, which has not been specifically referred to, also had to be taken. This relates to the way in which the scheme is to be written down or printed, that is the schedule order. In the above example the schedule order is:

Capacity → Container → Type of beer

(which is the same as the citation order but it need not necessarily be so; this point is explained more fully later – pages 81 and 85). Schedule order would therefore become Step 5:

Step 5 Deciding upon the schedule order

For very brief schedules a tabular form of presentation can be used, as shown on page 9, but for lengthier schemes a continuous form is more appropriate, for example:

Capacity facet

- 1 300 ml
- 2 440 ml
- 3 550 ml
- 4 1 litre
- 5 3½ litre

Container facet

- 1 Cans
- 2 Bottles

Type of beer facet

- 1 Mild
- 2 Bitter
- 3 Lager
- 4 Stout

Illustrative faceted scheme for a real estate agent's records

Here is a further example of a simple faceted classification scheme, which the reader can attempt to compile, as a practical exercise, if desired, before turning to page 14 to discover the answer.

A real estate agent, or broker, is computerising his records and wishes to devise a coding for the various dwellings that he has for sale. There are four pieces of information that are relevant: (1) number of bedrooms; (2) type of accommodation; (3) area; (4) price. The following are typical examples of his advertisements. Analyse these advertisements in order to obtain a list of relevant concepts and from these compile a faceted scheme that will satisfy the requirement.

4 bedroomed semi-detached bungalow. Atwell. £235,000.

3 bedroomed detached house. Denby. £259,000.

1 bedroomed apartment. Crosswood. £95,000.

5 bedroomed detached house. Blanford. £349,000.

3 bedroomed semi-detached house. Denby. £169,000.

2 bedroomed detached bungalow. Atwell. £189,000.

3 bedroomed terraced house. Crosswood. £119,000.

Where the price facet is concerned it is considered sufficient to indicate broad bands, that is, £50-100,000, £100-150,000, £150-200,000 and so on. In the type of accommodation facet, concepts such as ‘detached’ and ‘house’ could be considered separately but for simplicity, at this stage ‘detached house’ is to be regarded as a single concept.



Terraced houses



Semi-detached houses



Detached house



Bungalow

The terms in the advertisements on the previous page are those used in the United Kingdom and may be defined as follows:

Apartment	A room or set of rooms fitted with housekeeping facilities, usually on one floor, comprising a single residence in a larger building. Also referred to as a flat.
Terraced house	One of a row of houses in a single block. Also referred to as a town house.
Semi-detached house	One of two houses joined to each other by a party wall on one side only.
Detached house	A separate building standing by itself, not sharing a wall with another building.
Bungalow	A one-storied house with a low roof.

The first step is to analyse the advertisements in order to obtain a list of relevant concepts, that is:

4 bedroomed
semi-detached bungalow
Atwell
£200-250,000
3 bedroomed
detached house
Denby
£250-300,000
1 bedroomed
apartment
Crosswood
£50-100,000
5 bedroomed
Blanford
£300-350,000
semi-detached house
£150-200,000
2 bedroomed
detached bungalow
terraced house
£100-150,000

Note that once a concept is listed, it is not repeated when found in another advertisement.

The second step is to group these concepts into facets according to relevant characteristics, which, in this case, are: number of bedrooms; type of accommodation; area; price range. For example, the first concept is ‘4 bedroomed’ and looking down the list other concepts with these same characteristics can be found, for example ‘3 bedroomed’, ‘1 bedroomed’ and so on. When this process is complete, the result would be:

No. of bedrooms	Type of accommodation	Area	Price range (£)
4 bed.	Semi-det. bungalow	Atwell	200-250,000
3 bed.	Detached house	Denby	250-300,000
1 bed.	Apartment	Crosswood	50-100,000
5 bed.	Semi-det. house	Blanford	300-350,000
2 bed.	Detached bungalow		150-200,000
	Terraced house		100-150,000

Readers may prefer to change the currency to that used in their own country, for example, in the US the price could be indicated in dollars (\$).

The third step is to add a notation, for example:

No. of bedrooms	Type of accommodation	Area	Price range (£)
1 4 bed.	1 Semi-det. bungalow	1 Atwell	1 200-250,000
2 3 bed.	2 Detached house	2 Denby	2 250-300,000
3 1 bed.	3 Apartment	3 Crosswood	3 50-100,000
4 5 bed.	4 Semi-det. house	4 Blanford	4 300-350,000
5 2 bed.	5 Detached bungalow		5 150-200,000
	6 Terraced house		6 100-150,000

The fourth and fifth steps involve choosing a citation order and then a schedule order. Supposing that the order shown above is used for both, then the scheme is complete and the original advertisements can be classified, for example:

2 bedroomed detached bungalow. Atwell. £189,000 = **5515**

1 bedroomed apartment. Crosswood. £95,000 = **3333**

5 bedroomed detached house. Blanford. £349,000 = **4244**

3 bedroomed semi-detached house. Denby. £169,000 = **2425**

This may well be the sort of result that you have obtained. The scheme would certainly work, in this form, in a computerised system. However, when written down, it is somewhat confusing because, as the reader will have observed, the order of concepts within facets is not helpful. This order (often referred to as 'order in array') should be one that is of maximum assistance to the user of the classification.

For example the concepts of capacity in the classification for beer sales are best listed in increasing order of size, as shown on page 9.

There are many examples of possible helpful orders. In 'Mathematics', one might choose an order of increasing complexity, for example:

Arithmetic
Algebra
Calculus

For the subject 'Hand-guns' an evolutionary order might be chosen, for example:

Matchlock
Wheel-lock
Flintlock

Looking at the facets in the scheme for the estate agent, certain helpful orders do suggest themselves. The number of bedrooms facet and the price range facet are best arranged in numerical order. In the type of accommodation facet, one possible order is to begin with the apartment and ‘one-floor’ accommodation and then progress to ‘more-than-one-floor’ accommodation, that is, the house. No particular order is immediately apparent for the area facet; an alphabetical order could be used. Revising the scheme in this way would result in the following:

No. of bedrooms	Type of accommodation	Area	Price range (£)
1 1 bed.	1 Apartment	1 Atwell	1 50-100,000
2 2 bed.	2 Semi-det. bungalow	2 Blanford	2 100-150,000
3 3 bed.	3 Detached bungalow	3 Crosswood	3 150-200,000
4 4 bed.	4 Terraced house	4 Denby	4 200-250,000
5 5 bed.	5 Semi-det. house		5 250-300,000
	6 Detached house		6 300-350,000

Obviously this arrangement is much more helpful to the user and sample classification numbers would now be:

4 bedroomed semi-detached bungalow.	Atwell.	£235,000	=	4214
3 bedroomed detached house.	Denby.	£259,000	=	3645
1 bedroomed apartment.	Crosswood.	£95,000	=	1131
5 bedroomed detached house.	Blanford.	£349,000	=	5626
3 bedroomed semi-detached house.	Denby.	£169,000	=	3543
2 bedroomed detached bungalow.	Atwell.	£189,000	=	2313
3 bedroomed terraced house.	Crosswood.	£119,000	=	3432

The impact of citation order

If the estate agent wished to apply this classification not only in his computerised information system but also to his manual filing system, the filing order would be:

1131
2313
3432
3543
3645
4214
5626

A careful examination of this arrangement will indicate that the chosen citation order brings together accommodation with the same number of bedrooms but *separates* similar dwellings (for example detached houses), *separates* areas and *separates* similar prices. Is this what the estate agent would require in his manual system? The estate agent is approached and he states that the main priority is to group first by area, then by dwelling, then by number of bedrooms and lastly by price. In order to achieve this, the citation order must be changed to:

Area → Type of accommodation → No. of bedrooms → Price

The schedule order could be left as shown above and the instruction relating to the citation order changed to:

3rd facet → 2nd facet → 1st facet → 4th facet

giving, for example, classification numbers such as:

4 bedroomed semi-detached bungalow. Atwell. £235,000	=	1244
3 bedroomed detached house. Denby. £259,000	=	4635
1 bedroomed apartment. Crosswood. £95,000	=	3111
5 bedroomed detached house. Blanford. £349,000	=	2656
3 bedroomed semi-detached house. Denby. £169,000	=	4533
2 bedroomed detached bungalow. Atwell. £189,000	=	1323
3 bedroomed terraced house. Crosswood. £119,000	=	3432

However, in practice it is more convenient to write down the scheme in the same order as the citation order or, as is the case with some schemes, in the reverse of the citation order. Here is the scheme in the same order:

Area	Type of accommodation	No. of bedrooms	Price range (£)
1 Atwell	1 Apartment	1 1 bed.	1 50-100,000
2 Blanford	2 Semi-det. bungalow	2 2 bed.	2 100-150,000
3 Crosswood	3 Detached bungalow	3 3 bed.	3 150-200,000
4 Denby	4 Terraced house	4 4 bed.	4 200-250,000
	5 Semi-det. house	5 5 bed.	5 250-300,000
	6 Detached house		6 300-350,000

Classification numbers would, of course, remain as above, for example:

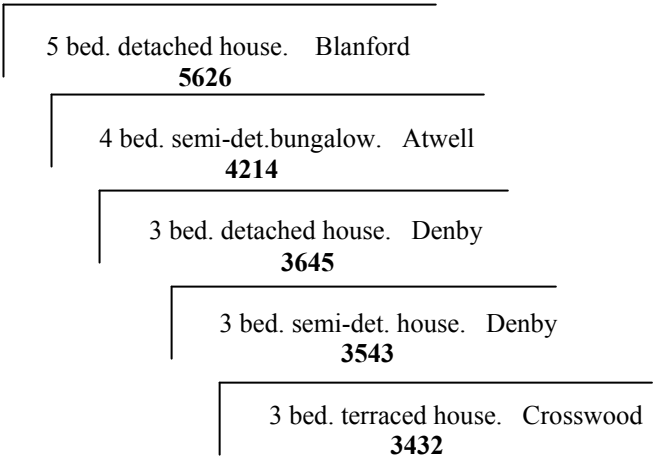
4 bedroomed semi-detached bungalow. Atwell. £235,000 = **1244**

and the filing order would be:

1244
1323
2656
3111
3432
4533
4635

As can be seen, this fulfils the estate agent's requirement. The Atwell, Crosswood and Denby residences have now been brought together and, within each area, similar accommodation will also be collocated.

Filing order before adjustment of the citation order



2 bed. det. bungalow. Atwell 2313

1 bed. apartment. Crosswood 1131
--

The decision regarding combination or citation order is clearly very important, for upon this decision will depend which aspects of a subject will be brought together and which will be separated. User needs, therefore, must obviously be taken into account.

Filing order after adjustment of the citation order

Denby. Detached house . . . 4635
--

Denby. Semi-det. House . . . 4533

Crosswood. Terraced house . . . 3432
--

Crosswood. Apartment . . . 3111

Blanford. Detached house . . . 2656

Atwell. Detached bungalow . . . 1323
--

Atwell. Semi-det.bungalow . . . 1244
--

In a computerized information system it would be comparatively easy to search for subjects which did not contain concepts from all facets. The machine could be programmed to find only the concepts required. For example, a 'string' search could be conducted for classification numbers in which the first digit was 2 and the second 1, or for classification numbers in which the first digit was 2, ignoring any further digits. In a manual system, however, it would be necessary to 'build in' the ability to classify such subjects, for example:

Apartments in Blanford
or Blanford

 This could be achieved by using a zero when a concept from a particular facet was not required, thus:

Blanford	would be classified as	2000
Apartments in Blanford	would be classified as	2100

 This method would work in a very simple scheme but not in a more complex one. The use of zeros would also have to be carefully monitored in a machine-based system. The computer has a habit of deleting leading zeros from numeric strings. An alternative would be to devise another notational system. For example, a different symbol could be used for each facet in order to distinguish between them. Here is the scheme with upper case letters used for the area facet, lower case letters for the accommodation facet, numeric digits for the number of bedrooms facet and numbers preceded by the letter P for the price facet. (The use of the letter P would then have to be avoided in the area facet.)

Area	Type of Accommodation	No. of bedrooms	Price range (£)
A Atwell	a Apartment	1 1 bed.	P1 50-100,000
B Blanford	b Semi-det. bungalow	2 2 bed.	P2 100-150,000
C Crosswood	c Detached bungalow	3 3 bed.	P3 150-200,000
D Denby	d Terraced house	4 4 bed.	P4 200-250,000
	e Semi-det. house	5 5 bed.	P5 250-300,000
	f Detached house		P6 300-350,000

 Whatever the notation chosen, it must reflect and fix the order previously selected for the concepts in each facet (order in array). This can be achieved by using numeric digits or the letters of the alphabet.

 Note that a mnemonic, or ‘memory aid’, element has been introduced into certain facets, for example A = Atwell and B = Blanford. (When the notational symbol is the same as the first letter of the concept, this is known as a ‘literal’ mnemonic.) The reader will already have noticed a mnemonic feature in relation to the ‘no. of bedrooms’ facet: 1 = 1 bedroomed, 2 = 2 bedroomed, and so on. Sample classification numbers would be:

1 bedroomed apartment. Crosswood. £95,000	=	CalP1
Apartments in Blanford	=	Ba
Blanford	=	B

It is also possible to use one type of symbol, for example upper case letters, for all facets and adopt a facet linking device such as the oblique stroke (/) to avoid confusion. Here is the scheme with a notation of this nature:

Area	Type of accommodation	No. of bedrooms	Price range (£)
AA Atwell	BA Apartment	CA 1 bed.	DA 50-100,000
AB Blanford	BB Semi-det. bungalow	CB 2 bed.	DB 100-150,000
AC Crosswood	BC Detached bungalow	CC 3 bed.	DC 150-200,000
AD Denby	BD Terraced house	CD 4 bed.	DD 200-250,000
	BE Semi-det. house	CE 5 bed.	DE 250-300,000
	BF Detached house		DF 300-350,000

Sample classification numbers would be:

1 bedroomed apartment. Crosswood. £95,000 = **AC/BA/CA/DA**

Apartments in Blanford = **AB/BA**

Blanford = **AB**

A one-letter notation could be used but only if different letters were allocated to each facet, for example A-D for first facet, E to begin the second facet, and so on. If each facet were to begin at A, it would be unclear whether A alone was 'Atwell' or 'Apartment', or '1 bedroomed' or '£50-100,000'. Similarly, A/A could be 'Atwell apartments' or '1 bedroomed apartments' and so on.

One other requirement needs to be mentioned and that is when two concepts from the same facet are both part of the subject, for example 'Detached houses in Blanford over £100,000 but under £200,000'. A further special connector must be used in order to accomplish classification of such subjects; a plus sign (+) is one possibility, for example:

Detached houses in Blanford over £100,000 but under £200,000

= **AB/BF/DB+DC**

or = **AB/BF/DB+C**

This device would also enable 'houses' to be classified, that is, 'terraced houses plus semi-detached houses plus detached houses', for example:

Blanford houses over £300,000 = **AB/BD+E+F/DF**

Now that the reader realises that concepts from all facets need not necessarily be present in a subject and hence in its related classification number, it will be appreciated that the concepts ‘terraced’, ‘semi-detached’ and ‘detached’ could have been placed in a separate facet. Here is the scheme compiled in this way and presented in the more usual continuous rather than tabular form:

Area

AA Atwell
AB Blanford
AC Crosswood
AD Denby

Type of accommodation

BA Apartment
BB Bungalow
BC House

Physical situation

CA Terraced
CB Semi-detached
CC Detached

Number of bedrooms

DA 1 bedroomed
DB 2 bedroomed
DC 3 bedroomed
DD 4 bedroomed
DE 5 bedroomed

Price range (£)

EA 50-100,000
EB 100-150,000
EC 150-200,000
ED 200-250,000
EE 250-300,000
EF 300-350,000

Sample classification numbers would be:

4 bedroomed semi-detached bungalow. Atwell. £235,000 =
AA/BB/CB/DD/ED

1 bedroomed apartment. Crosswood. £95,000 =
AC/BA/DA/EA

Apartments in Blanford = **AB/BA**

Blanford = **AB**

Notations using primarily capital letters, similar to that shown above, are used, in practice, in schemes such as the *London Classification of Business Studies* (see pages 31-3). In other classification systems, a mix of upper case letters for main classes and numeric digits for subdivisions is preferred, as in *Uniclass* (see pages 29-31).

As classification numbers become more complex, this can be a problem in a manual system but it matters little in a computer-based environment. The machine can handle very complex notations quite easily. However it should be pointed out that for the coding of manufacturing components and other items in industry a purely numerical and uniform length notation is sometimes preferred as in the classification of machine bolts described on pages 25-6.

The designer of a scheme must decide, at the concept listing and grouping stage, on the degree to which pre-combination (for example, 'detached house' rather than 'detached' and 'house') is desirable, bearing in mind the aims and objectives of the scheme and the type or types of information system in which it may be used.

The index and introduction

In the above examples it was possible to classify subjects by scanning the classification schedules. However, as the classification scheme develops into a more exhaustive and more complex system, such a scanning process becomes difficult, if not impossible. The scheme will therefore require an alphabetical index to make it more intelligible; that is so that the place in the schedules of a particular concept, and hence its notation, can be ascertained. Extracts from the indexes to faceted schemes are given later (pages 31, 33, 35 and 38). In addition, an introduction to the scheme would have to be written. This introduction will explain how the scheme may be used and will provide examples to illustrate this explanation. In a faceted scheme, among the more important things to be dealt with are the citation order and the method of notational combination. The introduction, however, should be as brief as possible subject to clarity and explicitness.

The complete scheme – Compilation procedure

Before outlining the compilation procedure for a complete scheme, one final, important point needs to be made. As the reader will probably have

noted, after examining the illustrative, alphabetic notations on pages 20-22, the citation and schedule orders could have a strong influence upon, and must take precedence over, the allocation of notational symbols. Therefore this allocation could, in many instances, more usefully be undertaken *after* decisions relating to those orders have been made. Taking this into account, the full procedure, therefore, would entail the following eight steps:

- 1 Analysis**
- 2 Grouping**
- 3 Order in array**
- 4 Citation order**
- 5 Schedule order**
- 6 Notation**
- 7 Index**
- 8 Introduction**

As we have seen, all of these would not be essential in every case, for example in a simple scheme for coding entities in a computerized information system.

Where analysis is concerned, appropriate concepts or terms are selected by an examination of the literature of the subject field or fields. It may be sufficient to use sources such as glossaries, dictionaries, encyclopaedias, indexing and abstracting journals and other classification schemes but it might become necessary to look at actual texts when such sources are inadequate or for current terms. The scheme for the estate agent, for example, used terms taken from actual advertisements.

4 Practical Examples of Faceted Classification Schemes

Seven schemes are examined briefly here, including the ‘parent’ of faceted classification, Ranganathan’s *Colon Classification*. Since that scheme was devised in the 1920s, it has led to many more general and special schemes and other information management applications, e.g. ‘chain indexing’ (see pages 115-6).

Example 4.1 Classification of machine bolts (Clifton, 1994)

This is a faceted classification of machine bolts for use in a computer based business system. It enables each type of bolt to be uniquely identified by a coding. Such bolts are available in different materials, in different thread sizes, in different head shapes and with different finishes.

Material

- 1 stainless steel
- 2 brass
- 3 steel

Thread size

- 1 0 BA
- 2 1 BA
- 3 2 BA
- etc

Head shape

- 1 round
- 2 flat
- 3 hexagonal
- 4 square

Finish

- 1 unfinished
- 2 chromium plated
- 3 zinc plated
- 4 painted

The citation order is the same as the schedule order shown above so that sample classification numbers would be:

Chromium plated brass square head bolts of 2 BA thread = **2342**

Unfinished steel flat headed bolts of 1 BA thread = 3221

A brief introduction would be required in order to explain that the citation order is the same as the schedule order. As this is such a simple scheme, however, no index is required; classification can be done by scanning the schedules. It will also be apparent that little attention has been paid to order in array, except in the thread size facet where an ascending order has been used.

Example 4.2 *Brisch Building Classification* (E. G. Brisch, n.d.)

The building industry has been a major user of faceted classification for many years and various faceted schemes have been developed to cater for information handling in that industry. Probably the first such scheme was the *Brisch Building Classification*, which was devised by E.G. Brisch & Partners, in consultation with other appropriate bodies such as the British Standards Institution. According to Barbara Kyle (1956), Edward Brisch was able to apply facet-analysis with ‘unaccustomed simplicity and economy’ and his classification systems were ‘put to a practical use in a number of technical fields’. Kyle provides an example of how the *Brisch Building Classification* works. Each concept in the scheme is represented by a notation consisting of a pair of digits and a subject can be classified by from one to nine pairs. For example, ‘Modular design’ = 01, ‘Steel’ = 35 and ‘Framework’ = 63. Therefore:

Modular design of steel framework =

01	35	63

Each notation is entered in its appropriate place in one of the nine boxes at the head of a 'data sheet' (a pair beginning with 0 in the first box, with 1 in the second box, 2 in the third box, and so on), leaving empty any boxes that do not apply. The user could organise his collection of sheets under whichever concept interested him most, with cross-references from whichever of the other concepts were considered important enough. An index is not required as there are only ninety concepts in total. Although faceted in the sense that it lists individual concepts, this is a very simple scheme and, because of its nature, citation order is irrelevant (see also pages 91 and 113-4).

A further development was taken to three digits but the two digit system was considered sufficient for many purposes.

Example 4.3 *CI/SfB Construction Indexing Manual* (Ray-Jones, 1976 and 1991) and *Uniclass* (Crawford, 1997)

The later *CI/SfB* classification was used extensively for many years throughout the construction industry for organizing office libraries and for the coordination and communication of product information.

The terms found in the literature of the building industry are grouped into a number of separate facets. There are five broad categories of terms:

Table 0	Physical environment e.g: 71 Schools 94 Sanitary, hygiene facilities	0 to 9
Table 1	Elements e.g: (27) Roofs (71) Circulation fittings	(--) or (1-) to (9-)
Table 2	Constructions, forms e.g: S Rigid tile work	A to Z
Table 3	Materials e.g: n Rubbers, plastics, etc.	a to z
Table 4	Activities, requirements e.g: (Aq) Testing, evaluating (J3) Strength, resistance to deformation	(A) to (Z)

Classification is achieved, as previously explained, by subject analysis, followed by allocation of the relevant notations to constituent concepts and then by synthesis according to the scheme's instructions for citation order and notational combination. For example, to classify the subject:

Strength of plastic tiles in school roofs

Analysis into concepts:

Strength Plastic Tiles Schools Roofs

Allocation of notation:

(J3) n S 71 (27)

Using the schedule order as the citation order, the classification number is therefore:

71 (27) S n (J3)

This is the citation order to be used when there is no special requirement but *CI/SfB* recognizes that different users may have different priorities.

The notation is very ‘mixed’; it includes numeric digits, upper case letters, lower case letters and other symbols. In order to avoid confusion differentiation between facets is achieved either by a change of symbol or by enclosure within brackets. For example, 71 ‘Schools’ is distinguished from 71 ‘Circulation fittings’ by enclosing the latter in parentheses, that is (71).

A subject may contain not only concepts from different facets but also from the same facet, for example ‘Sanitary facilities in schools’ where ‘sanitary facilities’ and ‘schools’ are both from Table 0. A faceted scheme must cater for this eventuality. *CI/SfB* uses a colon to link two concepts from the same facet, eg:

71 : 94

but a colon is not necessary where brackets are already used, so that the subject ‘Testing the strength of tiles’ would have the classification number:

S (Aq) (J3) or, more correctly, S (J3) (Aq)

as *CI/SfB* specifies that the normal citation order within this facet should be: requirement (i.e. strength) – activity (i.e. testing).

CI/SfB enables the classification of very specific subjects but, although such specificity may be required in certain circumstances (for classifying product information for instance), it is recognized that many office libraries will have only a few documents on most subjects and such specificity may not be necessary. In this instance, *CI/SfB* recommends a ‘basic reference’ based mainly on Tables 1 and/or 2 and/or 3 (Ray-Jones, 1991, p.160). As a library expands, *CI/SfB* can provide the extra detail required.

To facilitate use of the classification an alphabetical subject index is provided. As is usual with a faceted scheme, this index ‘contains simple subjects only, and gives the class numbers for single terms such as blocks; bricks; maintenance, retaining walls, but not for compounds of these (e.g. maintenance of retaining walls)’ (Ray-Jones, 1976, p. 177).

In relation to the fact that one must never classify from the index alone (see page 43) each page of the *CI/SfB* index includes the following note: ‘Consult the schedules to check the context of items in this index’.

Uniclass (Unified classification for the construction industry) was first published in 1997 as the successor to *CI/SfB*. It was developed by consultants from NBS Services (NBS = National Building Specification) and has gradually gained acceptance in the industry. *Uniclass* codes appeared on trade literature along with the familiar *CI/SfB* codes and a number of institutions continued to do this until such time as natural wastage and careful weeding reduced the original *CI/SfB* stock to manageable proportions for conversion.

Uniclass, as the name implies, is not based solely on *CI/SfB* but also in part on other schemes. It follows the international framework set out in the 1994 ISO Technical Report 14177 *Classification of Information in the Construction Industry*. *Uniclass* is the UK implementation of British Standard BS ISO 12006-2:2001. A new code of practice, BS 1192:2007 *Collaborative production of architectural, engineering and construction information*, published in January 2008, recommends the use of *Uniclass*.

The scheme comprises 15 ‘tables’, each of which represents a different broad facet of construction information, as compared with five in *CI/SfB*, and the very mixed notation of *CI/SfB* has been greatly simplified. Each facet is allocated a single capital letter (except for classes J and K which have two initial capitals). Concepts within facets are identified by numeric digits. Examples are:

- A Form of information
 - eg A26 Patents
- B Subject disciplines
- C Management
 - eg C432 Marketing, Selling
- D Facilities
 - eg D32 Office facilities
 - D41 Medical facilities (hospitals)
 - D71 Education facilities (Schools)
 - D711 Nursery schools
 - D712 Primary schools
 - D713 Secondary schools
 - D714 Sixth form colleges
 - D731 Computing facilities, computer rooms
 - D94 Sanitary facilities
- E Construction entities
 - eg E2 Tunnels
 - E3 Embankments, retaining walls, etc
 - E5 Bridges
- F Spaces
- G Elements for buildings
 - eg G24 Roofs

	G331 Floor finishes
H	Elements for civil engineering works
J	Work sections for buildings
K	Work sections for civil engineering works
L	Construction products
	eg L521 Roof tiles
M	Construction aids
N	Properties and characteristics
	eg N311 Strength/stability of structures
	N312 Strength of materials, components of structures
P	Materials
	eg P71 Plastics

Each table can be used as a ‘stand-alone’ table or terms from different tables can be combined in order to classify complex subjects. Number building is facilitated by the use of three main signs of combination. These are the colon ‘:’, the slash ‘/’, and the plus sign ‘+’. Those readers who have some knowledge of the *Universal Decimal Classification* (see also pages 66-9) will recognise these signs and they are used in a similar way. The most used sign is the colon since this is the basic mechanism for number building and elements may be combined from the same or from different classes. For example:

Computing facilities in sixth form colleges = **D714:D731**

Plastic roof tiles = **L521:P71**

The default citation order is the same as the schedule order. However, as with *CI/SfB*, it is ‘possible for users to set their own citation order for particular uses, provided that they are not concerned about compatibility with class numbers that another organisation may assign’ (Crawford, 1997, p. 15). For example, the subject ‘floor finishes in hospitals’ would normally be assigned the code D41:G331 but, if a particular organisation wishes to give priority to the ‘element’ rather than to the ‘building’, then the code G331:D41 could be used.

The ‘/’ is used to indicate a range of *consecutive* elements and the ‘+’ sign is used to indicate a range of *non-consecutive* elements, for example:

Secondary schools and sixth form colleges = **D713/D714**

Computing facilities in secondary schools and sixth form colleges
= **D713/D714:D731**

Tunnels and bridges = **E2+E5**

There are also two specialist signs ‘not intended for general use’ (ibid. p. 13), namely ‘<’ for less than and ‘>’ greater than. These are used to indicate that one concept is a part of another. For example, D32<D71 means ‘an office facility which is part of a school’. Here the citation order assumes a critical importance because D71<D32 would mean ‘a school which is part of an office facility’.

As in *CI/SfB*, the alphabetical index to *Uniclass* lists concepts only, for example:

- Meters L7451
 - electricity L74521
 - gas L7135
 - parking L21531
 - water L7118
- Ventilation
 - natural L7535
 - tunnelling M616

A careful examination of these extracts will reveal that the index complements the schedules in that it brings together aspects of a subject that are separated by the scheme. The concept ‘Ventilation’, for example, can be found in different facets. The index helps to bring together, or ‘collocate’, the various aspects of the subject. Such an index is known as a ‘relative index’ and, to use professional jargon, it is said to ‘collocate distributed relatives’.

Example 4.4 *London Classification of Business Studies* (2005)

The first edition of this scheme was compiled as a result of the rapid expansion in the field of management education following the establishment of two graduate business schools in the UK (London and Manchester) in 1965. It has established an international reputation and has been used in a number of business libraries and information services throughout the world.

In the scheme, the field of business studies is broken down essentially into three main categories:

Management responsibility within the enterprise

Environmental studies (i.e. disciplines such as economics, sociology, psychology, education, law, political science and science and technology, all of which have some bearing on business and management)

Basic analytical techniques

Within these broad categories there are a number of main classes such as 'Marketing' (in the 'Management responsibility' category), 'Industries' (in the 'Environment' category) and 'Statistics' (in the 'Techniques' category). Individual concepts are listed within each main class and given a notation consisting of capital letters. The oblique stroke (/) is used as a linking device in order to avoid confusion between facets.

For example, in Class F, Human resource management, FF = 'Pay' and in Class K, Industries, KTC = 'Oil industry'. The recommended citation order is normally the same as the schedule order, that is, alphabetical, therefore:

Pay in the oil industry = **FF/KTC**

Here are some more sample numbers:

Financial management = **EE**

Financial modelling = **EE/TK**
(where TK = Mathematic models)

Internet banking = **ECB/WR**
(where ECB = Banking and WR = Internet)

Fire insurance fraud = **ECNK/LWJR**
(where ECNK = Fire insurance and LWJR = Fraud)

There are two exceptions to the alphabetical citation order noted above. Class A 'Management' is always cited last and class P 'Law' is always cited first. Therefore:

Management of the oil industry = **KTC/AA**
(where AA = Management)

and

Legislation relating to pay = **PA/FF**
(where PA = Law or Legislation)

The reasons for these departures from the normal citation order are to keep all law books together and to avoid too much material being classed in management, since much of the literature in business libraries is concerned with this subject (at the time of writing, a search in the London Business School catalogue under 'Management' produced over 4000 'hits'). However it must be emphasised that the recommended citation order is a *recommendation*, 'there is nothing to stop classifiers using whatever citation order they wish' (Vernon, 1979, p. 10).

A special Auxiliary schedule with a single number (the digit 2) may be added to the classification number for an industry to give the product of that industry. Therefore:

Oil = **KTC 2**

Similarly, if KSC is the 'road building industry', then:

Roads = **KSC 2**

Other Auxiliary schedules cater for various standard subdivisions such as 'People and occupational roles', 'Place', 'Time' and 'Form'. For example:

Pension schemes for self-employed people = **FGB 161**
(where FGB = Pension schemes and 161 = Self-employed people)

Housing in Singapore = **FHC 55362**
(where FHC =Housing and 55362 = Singapore)

Case studies in strategic management = **AD 79**
(where AD = Corporate strategy and 79 = Case studies)

An alphabetical list of subject terms acts not only as an index to the scheme, eg:

Closed shop	GEB
...	
Closed-circuit television	MPMF
...	
Clothing and footwear industries	KM
...	
Clubs and public houses	KWDC
...	
Cluster analysis	UKC

but also shows the hierarchical relationships that link terms together. This will be described and discussed more fully in Chapter 12.

Example 4.5 Broad System of Ordering (BSO) (1991)

In more recent years only two attempts have been made to produce a new *general* classification scheme based upon modern indexing and classification theory. One is the revision of the *Bliss Bibliographic Classification* (see pages 61-3 and 70-1) and the other is the *Broad System of Ordering (BSO)*. Prepared by the International Federation for

Documentation with the support of Unesco and under the auspices of UNISIST, it was devised essentially by a FID/BSO panel of three classification experts: E.J. Coates, G.A Lloyd and D. Simandl. Designed for information interchange, subject terms were included on the basis of 'organisational warrant'; a topic was listed only if there was an information centre, source or service devoted to it. It also had a low level of specificity (as the name suggests). Nonetheless, *BSO*, which has been revised and expanded following major testing, could prove very useful as a classification aid in any information system that covers the whole span of knowledge (see also page 135).

Here are some sample class numbers:

100	Knowledge generally
112	Philosophy
200	Science & Technology
250	Space & Earth sciences
252	Astronomy & Astrophysics
,21	Optical astronomy
,28	Satellite astronomy = Spacecraft astronomy
,40	Astrophysics
,70	Solar system astronomy
,72	Sun = Solar phenomena
450	Psychology
460	Education
535	Sociology
560	Law
580	Economics
740	Transport technology & services

Usually, elements of composite subjects within a particular subject field or 'combination area' are cited in reverse schedule order following instructions contained in the scheme, eg:

Satellite studies of solar phenomena = **252,72,28**

Codes for composite subjects with elements from different areas are formed by combining notations using a citation order determined by the relationship between elements and by the insertion of a dash or hyphen between them, eg:

Teaching of psychology	=	450-460
Philosophy of education	=	460-112
Educational psychology	=	460-450
Sociology of education	=	460-535

Educational legislation	=	460-560
Economics of transport	=	740-580

Generally applicable facets for time and place are provided, the former being introduced by -01 and the latter by -02. Sample numbers are:

Education since 1920	=	460-016
Education in the New World	=	460-024

As is the case with other general classification systems with synthetic features, the index does not list all of the possible subject combinations that *BSO* offers. However, some illustrative examples of combinations given in the schedule captions are included. In general structure it is a chain index (see pages 115-6). Here is a sample extract:

Administrative law 567,40
 Adolescents, Biomedical sciences 439,34
 Adolescents, Mental health & disorders 439,34,38
 Adolescents, Psychology 450,53
 Adolescents, Social groups 528,34
 Adolescents, Sociology 535,74,34
 Adoption, Child welfare 575,32,40
 Adornment, Fashions & modes 472,25
 Adornment & clothing 472
 Adornment & clothing, Customs 533,72
 Adornment & jewellery 472,80
 Adornment articles & clothing, Manufacture 890,472
 Adrenals, Biomedical sciences 435,67
 Adult and special education 465
 Adult education 465,40
 Adult language development 911,21,40
 Adults, Social groups 528,35
 Adventure literature 915,67
 Adverse reactions, Pharmacology, Biomedical sciences 425,50,60
 Advertising 158,25
 Advertising, Commercial 588,82
 Advertising & publicity, Commercial 588,80

Example 4.6 *Colon Classification* (Ranganathan, 1963 and 1987)

The last scheme that we are going to examine is a difficult scheme for the beginner to come to terms with. However, the *Colon Classification* is described briefly here because it was the first scheme to be based entirely on faceted principles and is therefore worthy of inclusion because of its pioneering nature. It is a general classification of the whole of knowledge

for use in libraries devised by S.R. Ranganathan. Ranganathan was one of the world's foremost classification theorists. He introduced a number of new ideas and terms, one of which was the term 'facet' and he can truthfully be described as the 'inventor' of faceted classification.

Ranganathan taught mathematics at Madras University before becoming the University Librarian of that Institution in 1924. He visited the United Kingdom for observational studies at the School of Librarianship, University College, London, and it was whilst he was there that his analytical mind began to question whether the enumeration of complex subjects was the best basis for a classification scheme. He noted, for example, that a subject such as 'Techniques for teaching mathematics in secondary schools' had to be classified by *Dewey* (at that time) either as 'Techniques for teaching mathematics' or as 'Techniques for teaching in secondary schools'. Inspiration came, apparently, when he visited a London department store and saw a meccano set for the first time. This led him to believe that a classification could be built around a similar methodology; numbers for any subject being constructed by combining the elements listed in the schedules. The first draft of a new classification based upon this idea was completed by 1925. It was called the *Colon Classification* because, initially, he decided to use the colon as a linking device.

In the *Colon Classification* a 'facet formula' for combining concepts is provided for each basic subject or main class and this is followed by lists of the concepts (called by Ranganathan 'isolates' or 'foci') in each facet within that basic subject. Here, for example, is the beginning of Class T Education:

CHAPTER T

EDUCATION

T[P]: [E], [2P], [2P2]

Foci in [P]

- | | |
|----|------------------|
| 1 | Pre-secondary |
| 13 | Pre-school child |
| 15 | Elementary |
| 2 | Secondary |
| 25 | Intermediate |
| 3 | Adult |
| 31 | Literate |
| 35 | Foreigner |

- 38 Illiterate
- 4 University

An examination of the concepts (or foci) in [P] and the first part of the 'facet formula' for the class, that is T[P], will indicate that the subject 'University education' will have the class number T4. The square brackets in the formula are ignored.

Foci in [E] cum [2P]

- 1 Nomenclature
- 2 Curriculum
- 3 Teaching technique

Again an examination of the concepts in [E] and a further look at the formula will indicate that the classification number for 'Teaching methods' is T:3 and for 'Teaching methods in universities' T4:3. The 'cum [2P]' tells us that the concepts in [E] should be linked whenever necessary to those in [2P] and the scheme states:

Foci in [2P]
to be got by (SD)

(SD) means subject device and this allows for the use of the classification number for any subject, from the whole of the *Colon* scheme, to be included here in parentheses. For example, 'Mathematics' is Class B and therefore the classification number for the subject 'The teaching of mathematics' would be T:3(B) and for 'The teaching of mathematics in universities' T4:3(B). The comma preceding [2P] is superfluous because of the parentheses.

Foci in [2P2]

- 1 Audio-visual
- 13 Audio
- 133 Gramophone
- 136 Radio
- 15 Visual

The concepts in the above facet would allow the classification of such subjects as 'The use of visual aids in the teaching of mathematics in universities', which would have the classification number T4:3 (B), 15.

Citation order in the *Colon Classification* always conforms to the one general formula, the PMEST formula, which stands for: Personality (relating to the most important facet which establishes the character of a

subject) – Matter (relating to materials) – Energy (relating to processes, activities or operations) – Space (relating to geographical areas) – Time (relating to time periods). This does not mean that the maximum number of facets is five, which would be insufficient for many subjects and multiple ‘personality’ or other types of facet may be required. In the above Education class formula, for instance, [2P] stands for a ‘second round’ of the personality facet and [2P2] stands for a second, subsidiary level, of this second round of personality facet. ‘Matter’ is not in fact present in the Education class.

‘Space’ and ‘Time’ may be obtained from special tables that are generally applicable. In these tables 56 is ‘Great Britain’ and N9 is the ‘1990s’. The connecting symbol for space is a ‘dot’, or period and for time is an inverted comma. Therefore:

The use of visual aids in the teaching of
mathematics in British universities in the 1990s = **T4:3(B),15.56’N9**

At first glance, an entry in the index to the *Colon Classification* seems almost unintelligible, for example:

University 2 [P], 34. T [P], 4

However this simply means that the concept ‘University’ appears in two main classes: first in Class 2 (Library science), where it is included in the ‘Personality’ facet with the notation 34; second in Class T (Education), where it is again included in the personality facet but with the notation 4. ‘University’ is therefore a ‘distributed relative’ which is collocated in this relative index.

An attempt has been made here to present relatively simple examples for illustration purposes. These are taken from the sixth edition (1963). The later, seventh edition (1987), ‘does not appear to have achieved wide acceptance in India, the one country where the scheme has been used extensively. The *Colon Classification* is becoming complex and difficult to use’ (Foskett, 2000, p.78). Unfortunately, ‘Ranganathan’s own concept of “user-friendliness”... does not appear to apply to his own scheme’s notation as it has now developed’ (*ibid.* p. 73). However, it is the principles upon which *Colon* is based that are important and which have led to many applications not only in librarianship but in other fields such as business systems and computing.

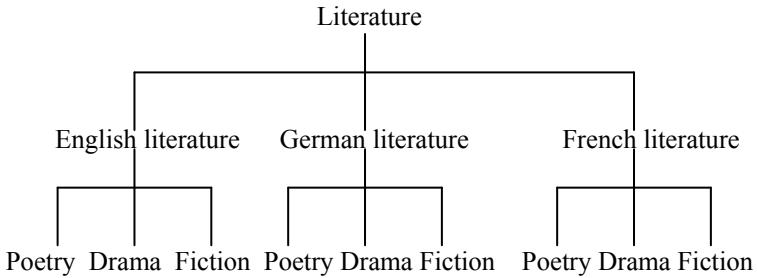
References

BSO: Broad System of Ordering (1991), FID/BSO Panel (Eric Coates, et al.), rev. ed.
Available, including 1994 update, at www.ucl.ac.uk.uk/fatks/bsa.

- The copyright of *BSO* is now held by University College London and the responsibility for its future management now lies with this institution.
- Clifton, H.O. and Sutcliffe, A.G. (1994), *Business Information Systems*, 5th ed., Prentice Hall, London, New York.
- Crawford, Marshall, Cann, John and O'Leary, Ruth (ed.) (1997), *Uniclass: Unified Classification for the Construction Industry*, RIBA, London.
- E.G. Brisch & Partners [n.d], *The Brisch Building Classification : prepared for the British Building Documentation Committee*, E.G. Brisch & Partners, [S.I.].
- Foskett, A.C. (2000), 'The Future of Faceted Classification', in Rita Marcella and Arthur Maltby (eds.), *The Future of Classification*, Gower, Aldershot, Hants, Brookfield, Vt., pp. 69-80.
- Kyle, B. (1956), 'E.G. Brisch: Something New in Classification', *Special Libraries* 47 (3), 100-105.
- London Business School Library (2005), *London Classification of Business Studies: Thesaurus*, The Library, 2 v. [Vol.1] Classified Sequence, [Vol. 2] Alphabetical Sequence. This is a new version of the scheme first devised by K.D.C. Vernon and Valerie Lang (see below).
- Ranganathan, S.R., (1987), *Colon Classification*, 7th ed., edited by M.A. Gopinath, Sarada Ranganathan Endowment for Library Science, Bangalore.
- Ranganathan, S.R. (1963), *Colon Classification*, 6th ed., with annexure, Asia Publishing House, Bombay; London.
- Ray-Jones, Alan and Clegg, David (1991), *CI/SfB Construction Indexing Manual*, RIBA, London. Abridged reprint of 1976 revision.
- Vernon, K.D.C. and Lang, Valerie (1979), *The London Classification of Business Studies : a Classification and Thesaurus for Business Libraries*, 2nd ed. rev., K.G.B. Bakewell and David A. Cotton, Aslib, London.

5 Hierarchical Classification

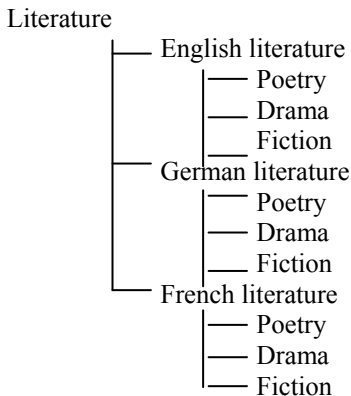
A ‘hierarchy’, as used in classification, is a series of classes or groups in successive subordination, for example:



Thus each entity (for example English poetry) falls into a subgroup of a larger group (for example English literature), which in turn forms part of an even larger group (for example Literature).

Such a classification is built up by a process of division, according to certain characteristics. Literature can be divided into sub-classes by the characteristic of language, that is English literature, German literature, French literature. Each of these classes can then be further subdivided by the characteristic of form of literature, that is, English poetry, English drama, English fiction; German poetry, German drama, German fiction, and so on. This is sometimes referred to as a ‘top down’ approach (as compared with that of a faceted system which utilises a ‘base-up’ technique).

The tree structure shown above could be presented thus:



This is near to the way in which the classification schedules would be written down or printed.

As the process of division continues, the hierarchical classification lists or ‘enumerates’ (hence ‘enumerative’ classification) complex subjects, for example ‘English poetry’, which comprises a language (English) and a literary form (poetry). This may be contrasted with the faceted approach, which would list ‘English’ and ‘poetry’ as separate concepts but not as a complete subject.

In a properly designed hierarchical classification each subject should have only one place where it fits into the system.

Order in array

In a hierarchical scheme the order of subjects at a particular level of division is important. This order corresponds to the order of concepts within facets and should be as helpful to the user as possible. For example, in literature, at the language level it might be helpful to place the ‘home’ language, for example English, first, then related European languages followed by other languages.

Citation order

The citation order in a hierarchical classification is derived from the order in which the characteristics of division are applied. In the literature example, above, the first characteristic of division is language and the second literary form. The citation order is therefore:

Language → Form

but this citation order is ‘in-built’ to classification numbers for complex subjects listed in the scheme.

Notation

Here is the above scheme with a simple notation consisting of numeric digits added:

- 1 Literature
- 11 English literature
- 111 Poetry
- 112 Drama
- 113 Fiction

12	German literature
121	Poetry
122	Drama
123	Fiction
13	French literature
131	Poetry
132	Drama
133	Fiction

This notation may be described as:

Pure	Only one type of symbol is used (the numeric digit).
Expressive	It reflects the structure of the classification; 111 'English poetry' is a division of 11 'English literature', which in turn is a division of 1 'Literature'.
Hospitable	A decimal notation such as this can be added to indefinitely and therefore, in theory, is infinitely expandable.
Mnemonic	'Poetry' is always represented by the digit 1; 'Drama' by the digit 2; 'Fiction' by the digit 3. This type of mnemonic, memory aid is referred to as a 'systematic' mnemonic.

Index

The printed index to an enumerative hierarchical scheme lists complex subjects, for example 'Drama: English literature' (unlike the index to a faceted scheme which lists concepts only). Here is an index to the above schedule:

Drama: English literature	112
Drama: French literature	132
Drama: German literature	122
English literature	11
Fiction: English literature	113
Fiction: French literature	133
Fiction: German literature	123
French literature	13
German literature	12
Literature	1
Poetry: English literature	111
Poetry: French literature	131
Poetry: German literature	121

The index brings together aspects of a subject that are separated in the scheme, for example poetry in various languages. It is therefore a relative index; it collocates distributed relatives.

Process of classification

The process of classification, in an enumerative scheme, is achieved first by consulting the index and then by proceeding to the location pinpointed in the schedules. However, one should *never classify from the index alone* as this can lead to error. The relevant schedules *must* be examined in order to find the correct number. To take a simple example, if a book which deals with the work of famous authors such as Shakespeare, Milton and Dickens were to be classified, the index entry for 'Literature' might be referred to and the class number 1 obtained. However after proceeding to 1 in the schedules it will be apparent immediately that 'Literature' is divided by language and the question of whether all the authors dealt with are English language authors must then be considered. Thus the structure of the classification scheme itself acts as an aid in classifying. As a classifier becomes more experienced and knowledgeable, he or she may not need to consult the index but will go straight to the schedules. The index will be used in emergency only. There are those who advocate that even learners should classify first by the schedules but this is difficult, of course, if the scheme is complex and the learner is not yet conversant with its structure.

Complexity of hierarchical schemes

If the faceted classification of machine bolts shown on pages 25-6 were compiled as a hierarchical scheme, dividing first by material, then by thread size, then by head shape and finally by finish, it would begin to assume the following appearance:

```
Machine bolts
  Stainless steel machine bolts
    0 BA
      Round
        Unfinished
        Chromium plated
        Zinc plated
        Painted
      Flat
        Unfinished
        Chromium plated
        Zinc plated
```

- Painted
- Hexagonal
 - Unfinished
 - Chromium plated
 - Zinc plated
 - Painted
- Square
 - Unfinished
 - Chromium plated
 - Zinc plated
 - Painted
- 1 BA
 - Round
 - Unfinished
 - Chromium plated
 - Zinc plated
 - Painted
 - Flat
 - Unfinished
 - Chromium plated
 - Zinc plated
 - Painted
 - Hexagonal
 - Unfinished
 - Chromium plated
 - Zinc plated
 - Painted
 - Square
 - Unfinished
 - Chromium plated
 - Zinc plated
 - Painted
- 2 BA
 - Round
 - Unfinished
 - Chromium plated
 - Zinc plated
 - Painted
 - Flat
 - Unfinished
 - Chromium plated
 - Zinc plated
 - Painted

and so on.

Several things will be immediately apparent:

1 The length of the schedules

The extract from the scheme shown here has only progressed as far as the subject 'Painted flat 2 BA stainless steel bolts', yet already it is far longer than the faceted scheme.

2 The repetition

As subjects and not concepts are being listed, constituent concepts such as 'unfinished' have to be repeated every time that they occur in a complex subject.

However, it is still only possible to classify a subject if it is listed. For example, the subject 'Chromium plated stainless steel bolts' (that is, omitting the size) cannot be classified nor can the subject 'Chromium plated bolts'.

Because of the problems (1) and (2) noted above, a hierarchical, enumerative scheme tends to become a compromise and enumeration is not carried to its logical conclusion. Therefore the finished scheme is often very complex in nature and the basic principles of construction are difficult to identify.

6 Practical Examples of Hierarchical Enumerative Classification Schemes

Presented here are brief, illustrative descriptions of six classification schemes. They illustrate how widespread and long-lasting these schemes are, ranging from *Dewey* (1876 to date – Example 6.4) to schemes still being devised such as those intended for use in the book trade (Example 6.6). As with the faceted schemes described in the previous chapter they also reinforce the fact that classification is an essential tool for organising knowledge.

Example 6.1 Classification for office organisation (Shaw, 1984)

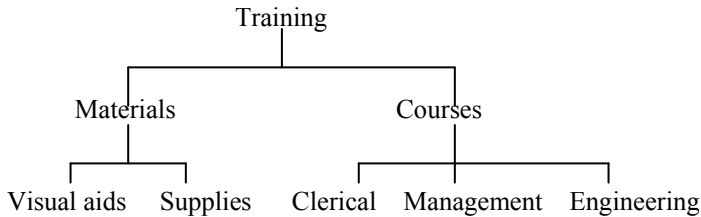
The following are illustrative extracts from a possible enumerative classification for the organisation, by subject, of office files.

16	TRAINING
16.1	MATERIALS
16.1.1	Visual aids
16.1.2	Supplies
16.2	COURSES
16.2.1	Clerical
16.2.2	Management
16.2.3	Engineering

The printed index entries to the above schedule would be:

Clerical courses	16.2.1
Courses	16.2
Engineering courses	16.2.3
Management courses	16.2.2
Materials	16.1
Supplies	16.1.2
Training	16
Visual aids	16.1.1

The scheme uses a very simple, expressive, decimal notation and its hierarchical nature is immediately apparent, that is:



Josephine Shaw, from whose work the above example is taken, suggests that such a system might be suitable for the products, supplies, services and functions of an organisation. She cites as an advantage the flexibility which the system offers, in that it is easy to subdivide and add topics, but, in her view, a disadvantage is the fact that an index is essential. She therefore advocates a combination of systems for storing and retrieving information in the office and recommends 'direct filing', that is, alphabetical sequences by name, subject, and so on, where an index is not required.

A hierarchical, enumerative scheme has the further problem that a subject can only be classified if it is enumerated, or listed. To take a simple example from the above scheme, the subject 'Visual aids for clerical courses', cannot be classified. A choice would have to be made between 'Visual aids' (16.1.1) and 'Clerical courses' (16.2.1).

Example 6.2 Guildhall classification for local material (Smith, 1966)

The following extracts are from *Classification for London Literature based Upon the Collection in the Guildhall Library*, a scheme for the classification of local materials, that is, materials relating to a specific area, which was first published in 1926:

20-59	Social life
...	
40-49	Administration – special subjects
...	
46	Public health and lighting
...	
46.9	Lighting
46.91	Oil lighting
46.92	Gas lighting
46.93	Electric lighting

The main classes have a two-digit notation, with subdivisions extending to a further two, and sometimes three digits after the decimal point.

The above example illustrates the progressive process of division in an enumerative scheme. However, even in a simple scheme, this process,

and especially the consistent application of a particular characteristic of division at one point, is not easy, as the following schedule illustrates:

40	Statistics. Yearbooks
41	Social welfare
42	Crimes. Police
43	Prisons
44	Laws. Courts
45	Markets
47	Medicine. Hospitals
46	Public health and lighting
48	Transport. Traffic
49	Education

Example 6.3 *ACM Computing Classification System (1998)* (1999 – valid through 2009) ¹

This scheme is a revision of the classification system used in the periodical *Computing Reviews*, which itself draws heavily on *Taxonomy of Computer Science and Engineering*.

The heart of the classification is a hierarchical tree, ‘restricted to three levels in order [it is claimed] that the tree be able to reflect accurately the essential structure of the discipline over an extended period’ (*ACM*, Intro., p. 1).

The top level consists of eleven classes identified by capital letters:

A.	General Literature
B.	Hardware
C.	Computer Systems Organisation
D.	Software
E.	Data
F.	Theory of Computation
G.	Mathematics of Computing
H.	Information Systems
I.	Computing Methodologies
J.	Computer Applications
K.	Computing Milieux

The second and third levels make use of a numeric, decimal division, plus a lower case ‘m’ for ‘Miscellaneous’. In H. ‘Information systems’, for example, we find:

H.0	GENERAL
H.1	MODELS AND PRINCIPLES
H.1.0	General
H.1.1	Systems and Information Theory

H.1.2	User/Machine Systems
H.1.m	Miscellaneous
H.2	DATABASE MANAGEMENT
H.2.0	General
H.2.1	Logical Design
H.2.2	Physical Design
H.2.3	Languages
H.2.4	Systems
H.2.5	Heterogeneous Databases
H.2.6	Database Machines
H.2.7	Database Administration
H.2.8	Database Applications
H.2.m	Miscellaneous
H.3	INFORMATION STORAGE AND RETRIEVAL
H.3.0	General
	and so on

In order to provide sufficient detail, there is also a fourth, unnumbered level which consists of alphabetical subject descriptors. Under H.2.1 ‘Logical Design’, for instance:

H.2.1	Logical Design
	<i>Data models</i>
	<i>Normal forms</i>
	<i>Schema and subschema</i>

In addition to these subject descriptors, there are general terms, such as ‘Algorithms’, ‘Design’, and ‘Documentation’, which can apply to any element when relevant. Names of languages of systems may also be used where appropriate, eg: ‘Pascal’ under D.3.2 ‘Language Classification’.

Example 6.4 *Dewey Decimal Classification* (Dewey, 2003)

This is a classification that covers the whole of knowledge. It was devised over a hundred years ago for shelf arrangement in libraries. It has since progressed through twenty-two editions. Its popularity can be gauged by the fact that it is used in over 200,000 libraries in 135 countries and has been translated into over thirty languages.² Because it employs a simple, decimal notation, there are ten main classes:

000	Computer science, information & general works
100	Philosophy & psychology
200	Religion
300	Social sciences
400	Language
500	Science

600	Technology
700	Arts & recreation
800	Literature
900	History & geography

Each main class can be divided into 'ten', in practice nine, subclasses and each subclass into nine more, and so on. Here, for example, are some parts of the Literature class, which the reader will see equate closely to the scheme shown on page 40-1, although the notation is different:

800 Literature

...

810	American literature in English
811	American poetry in English
812	American drama in English
813	American fiction in English
820	English and Old English (Anglo-Saxon) literatures
821	English poetry
822	English drama
823	English fiction
830	Literatures of Germanic languages. German literature
831	German poetry
832	German drama
833	German fiction
840	Literatures of Romance languages. French literature
841	French poetry
842	French drama
843	French fiction

With such a notation, each of the nine subdivisions, eg 821, 822, 823 . . . 829, is theoretically co-equal. As very few subjects divide into nine parts naturally, distortion is bound to occur; nevertheless this is an easily understood system and offers hospitality and expansibility. The *Dewey* scheme utilises a three figure 'minimum' base, as illustrated above; further division continues after the decimal point, for example:

822.3 English drama of Elizabethan period, 1558-1625

where the '3' represents 'Elizabethan period'. A fifth digit may then indicate a particular dramatist within this period, for example:

822.33 Shakespeare

If an item which deals with a work such as ‘a criticism of Shakespeare’s *Macbeth*’ were classified by the above scheme, then classification is by subject, but if the actual *text* of *Macbeth* were to be classified, classification in this case would be by *form*, that is literary form. It will be necessary in a general classification to cater for form classes.

In the above example, the hierarchical method of compilation is easily identifiable, that is division by language, then by literary form, then by period; but this is not always so. Here is a further extract from *Dewey*, taken from class 640 ‘Home economics and family living’:

- 641.52 Breakfasts
- 641.53 Luncheons, lunches, brunches, teas, suppers, snacks
 - 641.532 Brunches
 - 641.534 Box and bag lunches
 - 641.536 Afternoon teas
 - 641.538 Suppers
 - 641.539 Snacks
- 641.54 Dinners
- 641.55 Money-saving and timesaving cookery
 - 641.552 Money-saving cookery
 - 641.555 Timesaving cookery
- 641.56 Cooking for special situations, reasons, ages
 - 641.561 Cooking for one or two persons
 - 641.561 1 Cooking for one
 - 641.561 2 Cooking for two
 - 641.562 Cooking for persons of specific ages
 - 641.562 2 Young people
 - 641.562 22 Cooking for infants
 - 641.562 7 Persons in late adulthood
- 641.56 Cooking for health, appearance, personal reasons
 - 641.563 1 Cooking for persons with medical conditions
 - 641.563 11 Persons with heart diseases
 - 641.563 14 Persons with diabetes
 - 641.563 18 Persons with food allergies
 - 641.563 2 Cooking with specific vitamin and mineral content
 - 641.563 23 Low-salt cooking
 - 641.563 5 Low-calorie cooking
 - 641.563 6 Vegetarian cooking
 - 641.563 7 Health-food cooking
 - 641.563 8 Cooking with respect to carbohydrate, fat, protein content

The above extract shows the great detail of the scheme but it is also somewhat confusing and illustrates the problems of enumeration. A

compromise has had to be reached and the hierarchical process of division has not been continued at each step. For example, division by 'Meal' is not further subdivided by 'Special situations' and then by 'Health', etc. but division by each of these characteristics is carried out separately. Thus it is impossible to classify the subject 'Cooking dinner for one' or 'Cooking for persons in late adulthood who suffer from heart disease'. A choice has to be made in these instances between 'Cooking for dinner' or 'Cooking for one' and 'Cooking for persons in late adulthood' or 'Cooking for persons with heart disease'.

What is known as 'cross-classification' also occurs. This happens when different characteristics are applied at the one point of division. For example, 'For special situations' is divided by the characteristics of 'number of people' and 'age' at the one step.

The printed index to the *Dewey Decimal Classification* runs to well over 1000 pages. It is a very detailed, often praised index and its relative nature, which was 'devised and lauded by Melvil Dewey' (the scheme's creator), has had a considerable influence on many other indexes. Here is an illustrative example:

Muscles	573.75
biology	573.75
drawing	
animals	743.6
humans	743.47
human anatomy	611.73
human biochemistry	612.744
human biophysics	612.741
human physiology	612.74
injuries	617.473 044
medicine	616.74
pharmacokinetics	615.773
surgery	617.473
<i>see also</i> Musculoskeletal system	

This indicates that various aspects of the subject 'Muscles' can be found in many different places in the schedules and the cross reference tells the user that additional aspects may be found by consulting 'Musculoskeletal system'.

Dewey classification numbers can become quite lengthy and the digits after the decimal point are sometimes split into groups of three for greater convenience and ease of use, as shown above for 'injuries', ie:

617.473 044

Dewey also has special tables for subdivision by place, time, and so on. These will be discussed more fully in Chapter 7.

The latest edition of *Dewey* is also available online (*WebDewey*),³ with additional index facilities and regular updates throughout the year. There is also a Web-based version of the abridged 14th edition.

Example 6.5 *Library of Congress Classification* (various dates)

This classification was constructed at the turn of the nineteenth century for the US Library of Congress and it reflects knowledge as represented in the books that are held in this great library's stock. Although designed specifically for the Library of Congress, many libraries, primarily large university and research libraries, have adopted it for their own use, not only in America but in other countries including the United Kingdom.

The scheme consists of some twenty main classes, an alphabetical base enabling each of these to be designated by a capital letter:

A	General works
B	Philosophy, Psychology, Religion
C	Auxiliary sciences of history
D	History: General and Old World (Eastern hemisphere)
E-F	History: America (Western hemisphere)
G	Geography
H	Social sciences
J	Political science
K	Law
L	Education
M	Music
N	Fine arts
P	Language and Literature
Q	Science
R	Medicine
S	Agriculture
T	Technology
U	Military science
V	Naval science
Z	Bibliography. Library science

Some main classes are broken down into sections and a second capital letter used for differentiation purposes, eg:

KD	Law of the United Kingdom and Ireland
KF	Law of the United States

A second upper case letter is also used for major subclasses, eg:

Q	Science
QA	Mathematics
QB	Astronomy
QC	Physics

. . .

Arabic numerals are then used to denote further division, these numbers running, as required, from 1-9999. The following extracts are taken from Class T 'Technology'. Classes are published in separate volumes and the printed Class T contains close on 400 pages. This is only one of upwards of thirty volumes in the complete scheme, which gives some indication of its vast size.

Within Class T, Class TH is 'Building construction', and TH6000 onwards is concerned with 'Building fittings and their installation'. Here is a brief outline of this section of the scheme:

BUILDING FITTINGS AND THEIR INSTALLATION

6010 General works

. . .

6100- PLUMBING AND PIPEFITTING

7000- HEATING AND VENTILATION

7700- ILLUMINATION. LIGHTING

8000- DECORATION AND DECORATIVE FURNISHINGS

9000- PROTECTION OF BUILDINGS

Each of these divisions is further subdivided, for example:

PROTECTION OF BUILDINGS

9025 General works

9031- Against damage from natural causes

9100- Protection from fire

9700- Protection from burglary, sabotage, etc.

The above extracts illustrate the hierarchical nature of the scheme but fail to do the scheme justice in that the printed version of Class TH from division 6000 onwards actually takes up 21 pages of complex schedules and the principles of construction become hidden in a wealth of enumerative detail. Here, for instance, are the divisions of one very small section, 'Plumbing of the bathroom':

Plumbing of ... bathrooms. Toilet rooms. Lavatories

- 6485 General works
- 6486 General special. Specifications, etc.
- 6487 Systems
 - Details. Fixtures
- 6488 General works
- 6489 Catalogs of bathroom fixtures
- 6490 Basins
 - Baths
 - Cf TH6518.B3 Public baths
- 6491 General works
- 6492 Shower baths. Sprays
- 6493 Tubs
- 6494 Foot baths
- 6495 Sitz baths
- 6496 Bidets
- 6497 Soil fixtures
- 6498 Water closets. Toilets
- 6499 Urinals
- 6500 Flush tanks and auxiliary valves, etc.

The subject 'Plumbing of a bidet', for example, would have the class number TH6496. Despite the detail displayed here, the true nature of the Library of Congress scheme is still not fully revealed. Here is a further extract from 'Heating of buildings':

Heating of buildings

- 7201 Periodicals, societies, etc.
- 7205 Congresses
- 7207 Collected works (nonserial)
- 7209 Dictionaries and encyclopedias
 - Directories
- 7212 General works
- 7213 Special localities
 - History
- 7215 General works
- 7216.A-Z Country divisions, A-Z
 - General works
- 7221 Early to 1860
- 7222 1860-
- 7223 Textbooks
- 7224 Popular works. Juvenile works
- 7225 Pocketbooks, tables, rules, etc.
- 7226 General special

7227	Addresses, essays, lectures
7231	Study and teaching
7325	Specifications
7331	Drawings
	Estimates. Measurements. Quantities and costs
7335	General works
7337	Schedules of prices
7338	Superintendence
7339	Inspection
7341	Testing
7355	Catalogs of heaters and fixtures (General)
7391	Special rooms. By name, A-Z
7392	Special classes of building. By name, A-Z
.A6	Apartment houses
.A8	Atomic bomb shelters
.A9	Automobile service stations
.C56	Chemical plants
.C6	Churches
.C65	Commercial buildings
.C7	Creameries
	Dwellings see TH7201-7643
	Factories see TH7392.M6
.F3	Farm buildings
.F6	Foundries
.G37	Garages
.I53	Industrial buildings
.M2	Machine shops
.M6	Mills and factories
.O35	Office buildings
.P33	Packing houses
.P8	Prisons
.P9	Public buildings
.R3	Railroad structures and buildings
.S35	School buildings. University and college buildings
.S65	Sports facilities
.T34	Tall buildings
.T47	Textile factories
.T5	Theaters
	University buildings see TH7392.S35

A number of points of interest are illustrated by the above example. Among these are:

- i The high degree of ‘enumeration’. Even forms such as ‘Dictionaries’ and ‘Encyclopedias’ are given in the schedules, for example:

Encyclopaedia of heating of buildings = **TH7209**

- ii The use of alphabetical order within a systematic arrangement by means of initial letters followed by Arabic numbers. The numbers are used decimally and are assigned from special tables in a manner that preserves the alphabetical order. In the above schedule alphabetical division is applied to both geographical and subject division, for example:

History of the heating of buildings in Japan = **TH7216.J3**

Heating of atomic bomb shelters = **TH7392.A8**

The second example above provides a further illustration of the great detail of this scheme.

- iv The use of references to indicate where a particular topic should be classed or to indicate related classes, eg:

University buildings see TH7392.S35

TH6490 Baths
Cf TH6518.B3 Public baths

- iii The provision of hospitality by leaving gaps in the notation (see also pages 74-5 and 80)

Each class has its own index and the alphabetical authority list *Library of Congress Subject Headings* (see pages 108-9, 117-8 and 125 fn), since many of the headings include the relevant classification numbers, can act to a limited extent as a general index.

Example 6.6 *BIC Subject Categories* (2006) and *BISAC Subject Headings* (2008)

Before leaving this description of practical hierarchical systems, mention should be made of the two recent classification schemes intended for use in the book trade for the electronic transfer of subject information.

In the United Kingdom, the *BIC* (Book Industry Communication) *Subject Categories* system was launched originally in 1998 and Version 2 was released in 2006. The scheme consists of a hierarchical listing of categories and qualifiers. The top level categories are represented by a single letter code, eg: A = Arts; B = Biography and True Stories; C = Language; D = Literature & Literary Studies; E = English Language Teaching; F = Fiction & Related Items; etc. Some categories have a

literal mnemonic value, eg: H = Humanities; L = Law; M = Medicine. Level 2 categories have two letter codes and level 3 categories have three letter codes, and so on. For example:

A	Arts
AS	Dance and other performing arts
ASD	Dance
ASDC	Choreography
ASDL	Ballet
ASDR	Ballroom dancing
ASDT	Contemporary dance
ASDX	Folk dancing

When classifying a book, it need not necessarily be placed in a single category. For instance, examples of book classifications from the *BIC* website include: ‘The effective teaching of mathematics in secondary education’ which is allocated the categories:

JNU	Teaching of a specific subject
YQM	Educational: Mathematics & numeracy
JNLC	Secondary schools

and ‘David Beckham: my son’, a biography by his father would be allocated the categories:

BGS	Biography: Sport
WSJA	Football (Soccer, Association football)

There is an online *BIC Subject Category* Selection Tool, which acts as an index. For example, if ‘physics’ is entered in the search box, all of the subject headings containing the string of letters ‘physics’ will be shown, eg:

HPJ	Philosophy: metaphysics and ontology
PH	Physics
...	
PHM	Atomic and molecular physics
...	
PHVB	Astrophysics
etc	

By clicking on a particular heading, the hierarchical structure of the scheme for that heading will be revealed. Selecting 'Astrophysics', for example, will give:

P	Mathematics and science
PH	Physics
PHV	Applied physics
PHVB	Astrophysics

Like *Dewey*, the *BIC* scheme provides for geographical locations, time periods, etc. This is facilitated by means of special sets of 'qualifiers' (see page 64).

As can be seen from the illustrative examples used here, *BIC* was not designed for shelf arrangement. However, there is a subset of the *BIC* scheme (E4Libraries Category Headings) aimed specifically at the public library market. This is intended to allow the arrangement of a library's stock by broad categories.

Some forty years ago the notion of 'reader interest order' originated in the United States. This involved the arrangement of stock in popular categories based upon the observed interest of readers. This has been used in many general libraries where a high proportion of the stock is recreational. However, most of these schemes are unique to the particular library system. Using the *BIC* scheme involves a similar idea but offers standardisation and the opportunity to automate the categorisation process. Each title is assigned one category, which can incorporate several categories from the full *BIC* scheme, such as CRI for 'Crime' or MED for 'Media studies'. This can be used in combination with the *Dewey* number. For example, an item might be allocated the category 'HIS', which, with the addition of the *Dewey* number, would allow the item to be shelved in the broad category of 'History', with a sub-arrangement by class number.

In the United States, there is the *BISAC* (Book Industry Standards and Communication) *Subject Headings* list. Each *BISAC* subject consists of a nine letter code. The first three characters of this code consist of letters representing the section, for example HIS = History, PHO = Photography, etc. Note the 'literal' nature of the name. The next three numeric characters were initially designed to hierarchically connect subheadings and subsubheadings within a section but, while most codes retain this structure, hierarchy cannot always be inferred from a code.

In classifying, the first step is to decide on the major heading that applies to a book and from within that heading determine the specific term that describes the item. For example, if classifying a book on 'Rose

growing’, the major heading ‘GARDENING’ would be chosen and, from the ensuing list of specific terms ‘GAR004060 GARDENING / Flowers / Roses’ would be selected:

Rose growing = **GAR004060**

As with *BIC* codes, multiple codes can be assigned to a book that deals with more than one topic. A book about ‘agnosticism’ and ‘atheism’, for instance, would be allocated:

REL001000 RELIGION / Agnosticism
and REL004000 RELIGION / Atheism

Codes are intended for EDI (Electronic Data Interchange) and they can also be used as access points for database searching. The long term aim is the unification of the two schemes described above into a single classification system for the English language book market.

Notes

- 1 Copyright by Association of Computing Machinery, Inc.
- 2 www.bl.u/bibliographic/subject.html#ddc
- 3 <http://connexion.oclc.org>

Dewey, Dewey Decimal Classification and WebDewey are registered trademarks of OCLC Online Computer Library Center, Inc.

References

- ACM Computing Classification* (1998, (1999 – valid through 2009), Association for Computing Machinery. Available at: http://www.informatik.uni-stuttgart.de/zd/buecherei/ifibib_hilfe_cr.html and see also www.acm.org/class.
- BISAC Subject Headings* (2008), Book Industry Standards and Communication. Available at: www.bisg.org/standards/bisac_subject/index.html.
- BIC Subject Categories* (2006), Book Industry Communication. Available at: www.bic.org.uk/7/subject-categories.
- Dewey, Melvil (2003), *Dewey Decimal Classification and Relative Index*, edition 22, ed. by Joan S. Mitchell ... [et al]., OCLC, Dublin, Ohio. 4 v.
- Library of Congress (various dates), *Classification*, LC, Washington, 43 v. The examples included here have been checked in 2008 against www.classificationweb.net the online version of the scheme from the Cataloging Distribution Service of LC.
- Shaw, Josephine (1984), *Administration in Business*, 2nd ed., Pitman, London, p. 91.
- Smith, Raymond (1966), *Classification for London Literature Based Upon the Collection in the Guildhall Library*, 3rd ed., The Library Committee, London. Cited in George A. Carter (1973), *J.L. Hobbs's Local History and the Library*, 2nd rev. ed., Deutsch, London.

7 The Use of Synthesis in a Basically Enumerative Scheme

It is possible to introduce an element of synthesis, or ‘number building’, into what is essentially a hierarchical enumerative scheme, in several different ways. This helps to reduce the length of the schedules but improves efficiency.

One method is to provide auxiliary tables of terms that may be applicable throughout the schedules. These tables may include, for instance:

Common subdivisions, such as those of form of presentation, for example ‘Encyclopaedia’ or ‘Serial’.

Geographical tables for specifying ‘place’, for example ‘United States’ or ‘United Kingdom’.

Period tables for specifying ‘time’, for example ‘Twentieth century’.

Another method is to include instructions within the schedules themselves for dividing one classification number in the same manner as another. All plants, for instance, may suffer from ‘injuries’, ‘diseases’ and ‘pests’. Instead of enumerating each of these concepts under every individual plant, they can be listed once and instructions given for the division of each plant in the same way.

PRACTICAL EXAMPLES OF SYNTHESIS IN A BASICALLY ENUMERATIVE SCHEME

Example 7.1 *Bibliographic Classification* (Bliss, 1940-53)

The 1940-53 edition of this general classification scheme for library materials by H.E. Bliss is essentially enumerative but a certain amount of synthesis is provided for by ‘systematic schedules’, some of which may be applied throughout the scheme but others are applicable only in specific classes. Below are shown extracts from two schedules of the former variety:

SCHEDULE 1

NUMERAL DIVISIONS OF ANY CLASS OR SECTION

- . . .
 1 Reference books (including dictionaries,
 glossaries, encyclopedias, indexes, handbooks...
 2 Bibliography...
 3 History...
 4 Biography...

SCHEDULE 2

GEOGRAPHICAL DIVISION

- a America
 aa North America
 . . .
 d Europe
 ea Great Britain

‘Education’, in the *Bibliographic Classification*, is J, and by use of the above schedules:

A history of education in Great Britain = **J3ea**

This number has been derived in a very similar way to the way in which classification is achieved in a faceted scheme. The subject is analysed into its constituent concepts and a notation assigned to each:

History	=	3
Education	=	J
Great Britain	=	ea

The instructions for citation order and method of combination given in the scheme must now be followed. Usually classification would be first by subject, then by ‘place’ and then by ‘form’ (for example: Education – Great Britain – Encyclopaedia) but when ‘form’ is ‘history’, as in this case, ‘place’ is cited last (that is: Birds – History – England). As different symbols are used, a linking device is not required.

Bliss was a firm believer in brevity of notation. He thought that the economic limit should be three or four letters or digits. Perhaps because of this, he did not emphasise sufficiently what he refers to as ‘composite notation’, a general method of number building for complex subjects which he admitted might be necessary for specialist classification. He advocated the possible use of a hyphen to link numbers. Using this

Base number for cats	=	636.80
Add numbers following	636.0	
Number for breeding	=	636.082
Add '82 to '636.80'	=	636.8082

The number for 'feeding cats' would be obtained in a similar way

Add numbers following	636.0	
Number for feeding	=	636.084
Add '84' to '636.80'	=	636.8084
Feeding cats	=	636.8084

Example 7.3 *BIC Subject Categories* (2006)

In the *BIC* scheme there are five sets of qualifiers beginning with a number which can be added to categories: 1 = Geographical; 2 = Language; 3 = Time periods; 4 = Educational purpose; and 5 = Reading level and special interest. Examples are 1D = Europe; 2AB = English; 3JJ = twentieth century; 4KD = UK educational tests; 5AB = For children c0-2 years. These qualifiers are a key feature of the scheme, claimed to provide great flexibility.

For example, a 'History of education in Great Britain during the nineteenth century' would be allocated the category:

YQH	Educational: History
	and to this would be added the geographical qualifier:
1DBK	Great Britain
	and the time period qualifier:
3JH	c1800 to c1900

The work 'Teach yourself German' would be allocated the subject category:

CJBT	Language self-study texts
	and the qualifier for the particular language:
2ACG	German

There are also other, more unusual qualifiers. For instance the book 'Cooking for Christmas parties' would be allocated the category:

WBR	Cooking for parties
	and the 'Holidays and seasonal interest' qualifier:
5HC	Christmas

In practice, a limit of four or five each of categories and qualifiers is stated to be adequate. It is doubtful whether many books would require anything like this level of indexing.

References

BIC Subject Categories (2006), Book Industry Communication.

Available at: www.bic.org.uk/7/subject-categories.

Bliss, H.E. (1940-53), *A Bibliographic Classification*, H.W. Wilson, New York, 4 v.

Dewey, Melvil (2003), *Dewey Decimal Classification and Relative Index*, edition 22, ed. by Joan S. Mitchell ... [et al], OCLC, Dublin, Ohio. 4 v.

8 Synthesis Grafted on to an Enumerative Scheme

There are two major examples of synthesis being grafted on to an enumerative scheme. The first dates back to 1904-7 when the *Universal Decimal Classification* was published. This was an adaptation (originally in French) of the *Dewey Decimal Classification* by Paul Otlet and Henri La Fontaine. The second scheme is of much more recent origin, the ongoing revision of Bliss's *Bibliographic Classification* (the original Bliss scheme was briefly referred to in the previous chapter – pages 61-3).

Example 8.1 *Universal Decimal Classification (UDC)* (various dates – accessed online 2009)¹

UDC is claimed to be ‘the world’s foremost multilingual classification scheme for all fields of knowledge, a sophisticated indexing and retrieval tool’.² Over the past century it has been extensively revised and developed through various editions and has been issued in a number of different languages, to become a ‘highly flexible and effective system’.³

Although based upon the *Dewey Decimal Classification*, signs of combination and abbreviation were introduced to allow for synthesis and number building. This enabled the required, greater detail, as *UDC* was ‘aimed specifically at retrieval rather than at shelf arrangement’ (Maltby, 1975).

Here are some examples of *UDC* classification numbers. As can be seen, they are the same as their counterparts in the *Dewey Decimal Classification* except that the three figure minimum is abolished; therefore 310 in *Dewey* becomes 31 in *UDC*:

31	Statistics
51	Mathematics
53	Physics
54	Chemistry
63	Agriculture

As noted above, synthesis is introduced by the provision of signs of combination and abbreviation (partly described previously, as adopted by *Uniclass*, on pages 29-31) to facilitate number building:

+ and / Signify coordination and extension.

The + sign is used for the sum of the meanings of several *UDC* numbers and the / sign is used to link consecutive numbers.

- : Signifies relation.
The relationship of one number with another.

Applying these signs of combination to the class numbers shown above, in order to build up numbers for composite subjects, would achieve results as illustrated in the following examples:

Mathematics and physics	=	51+53
Physics and chemistry	=	53/54
Statistics relating to agriculture	=	63:31
	or	31:63

There is no stipulated citation order in *UDC*, hence the possible alternative for the last number. This provides for great flexibility (see also pages 31, 75 and 80-1). If necessary a double colon may be used to 'fix' the order of the constituent elements of a compound number, especially when *UDC* is used in a computer-based information system, for example:

Statistics relating to agriculture	=	63::31
------------------------------------	---	---------------

Other examples of *UDC* classification numbers are:

17:7	Ethics in relation to art Where 17 = 'Ethics' and 7 = 'Art'
622+669	Mining and metallurgy Where 622 = 'Mining' and 669 = 'Metallurgy'
624+69+72	Civil engineering, building and architecture. Where 624 = 'Civil engineering', 69 = 'Building' and 72 = 'Architecture'.
57/58	Biology and botany, where 57 = 'Biology' and 58 'Botany'.

In common with other classifications, for further specificity *UDC* provides a number of common auxiliary tables for use throughout the scheme. These include:

<u>Table</u>	<u>Auxiliary</u>	<u>Indicated by:</u>
1c	Language	=
1d	Form	(0/9)

1e	Place	(1/9)
1f	Race	(=...)
1g	Time	"..."
1k	General characteristics	
	Properties	-02
	Materials	-03
	Persons	-05

For example, if 339.5 is 'Trade relations' then:

339.5(410/44) = Trade relations between the United Kingdom and France, where 410 = 'United Kingdom' and 44 = 'France'.
The 'place' numbers are derived from Table 1e and the / is used to indicate relationship between the two countries.

and:

339.5(410/44)"17" = Trade relations between Great Britain and France in the eighteenth century, where "17" = 'Eighteenth century' from Table 1g.

Where form is concerned, an example would be:

636.8(031) = Encyclopaedia of cats, where 636.8 = 'Cats' and (031) = 'Encyclopaedias' from Table 1d.

An example of the use of the 'General characteristics' table is provided by the subject 'Life expectancy of men', which would be allocated the number:

314.118-055.1 where 314.118 = 'Life expectancy' and -055.1 = 'Men' from Table 1k.

Some auxiliaries might not always be required in certain situations. For example a book on ornithology written in English could be allocated the number:

598.2(02)=111 where 598.2 = 'Ornithology', 02 = 'book' (from Table 1d) and =111 is 'English' (from Table 1c).

Clearly, if a collection consists only of books, and mainly books in the English language, there is no point in classifying each item as a book or as a book in English. However, where various types of media or various

languages are included in an information system, it may become necessary to distinguish between them.

The fact that many of these auxiliaries have biterminal signs such as brackets or inverted commas allows for great flexibility. Signs can be affixed to any part of a *UDC* number. Thus 'Coal mining in Great Britain', where 'Coal mining' = 622.33 and (410) = 'Great Britain' could be classified as:

	622.33(410)	Coal mining – Great Britain
	622(410).33	Mining – Great Britain – Coal
or	(410)622.33	Great Britain – Coal mining

For many years, right up to the last decade of the twentieth century, certain aspects of enumeration in *UDC* could conflict with the synthetic facility offered by the auxiliary tables. For example, the subject 'Women as criminals' was actually listed in the schedules as 343.914. Fortunately, in the latest version of *UDC*, this has been rectified and the number would now be built up by means of Table 1k:

343.91-055.2 = Women as criminals
 where 343.91 = 'Criminals' and -055.2 = 'Women'.

There are also special auxiliaries which occur at various places in the scheme and which express concepts that occur in a limited subject range. For example, in 'Physics':

53.05 = 'Observation and recording of physical phenomena'

The 05 can be detached and added to any direct subdivision of the main number, thus:

531 = 'Mechanics'
531.05 = 'Observation and recording of mechanical phenomena'
531.5 = 'Gravity'
531.5 05 = 'Observation and recording of gravitational phenomena'

UDC is used by a large number of European special libraries and in a great variety of both manual and automated information systems. Many collections of different kinds of media, including maps, motion pictures, coins and model soldiers have been organized by it.⁴

A *UDC* Consortium (UDCC) was formed in 1991, with FID (International Federation for Information and Documentation), BSI (British Standards Institution) and four other publishers as founder

members. The UDCC Master Reference File is continually updated and the standard edition in English (BS1000) is available online.⁵

Example 8.2 *Bibliographic Classification (Second Edition)*. (Mills, 1977-)⁶

Henry Evelyn Bliss devoted a large part of his life to the study of classification and his thoughts and ideas are expounded in a number of articles and books beginning in 1910 and culminating in 1953 with the publication of his *Bibliographic Classification*. A Bliss Classification Association was formed in the United Kingdom 1967 and it was suggested that a new and completely revised edition of the full *BC* should be made available. ‘However, the revision has been so radical that it is more accurately described as a completely new system, using only the broad outline developed by Bliss’.⁷ One of the main features of the original scheme was the carefully designed main class order. This order in fact reflects the considerable attention that Bliss gave to the development of this part of his scheme, in an attempt to make it adhere to what he called ‘the scientific and educational consensus’, the way in which ‘the majority of subject specialists expect their work to be organized’ (Maltby, p. 208).

The Association began to publish this much-improved, revised version of the original Bliss scheme in 1977 – the *Bliss Bibliographic Classification (Second Edition)* or *BC2*. It is based on the principles of faceted classification and some 15 of approximately 28 volumes of schedules have so far been published. The latest was Class W, The Arts, published in 2007 (Mills). The schedules are constantly updated and amendments are published in the *Bliss Classification* annual *Bulletin*.

In the original enumerative scheme, many subjects are not listed and therefore cannot be classified. The revised scheme is fully faceted and synthetic and therefore allows for much greater detail. Any class may be qualified by all the classes following it in citation order (and therefore filing before it). This may be illustrated by the following example taken from class Q ‘Social welfare’ (1977).

Q	Social welfare
QEL	Residential care
QLV	Old people

After analysing the subject, classification is achieved by arranging the elements in the reverse order of the *BC2* schedule going from the most specific to the most general. The classmark is then formed by combining

the elements but the repeated initial class letter is dropped from all but the first component. Therefore:

Residential care for the elderly = **QLV EL**

The 'Q' from 'QEL' is dropped in the final classmark.

Here is a further example of the detail that *BC2* provides.

KMU R	Rural communities
...	
KOM	Unemployed persons
...	
KVQ E	Indian society

Therefore:

Unemployment in rural communities in India = **KVQ EOM MUR**

The initial 'K' is dropped from 'KOM' and 'KMU R' before combining the elements.

The notation is primarily alphabetical, although some use is made of the digits 1-9, and purely ordinal (see page 82). A space is inserted after every three letters for clarity and as an aid to understanding.

Detailed alphabetical indexes to each class are provided. These are created using the terms in the schedules but chain procedure (see pages 115-6) is used in order to bring together scattered items. Where the examples 'Residential care for the elderly' and 'Unemployment in rural communities in India' cited above are concerned, the index entries would be:

Old people : Residential care : Social welfare QLV EL

Rural communities : Unemployment : Indian society KVQ EOM MUR

Although the scheme is 'designed to give the most helpful and logical order', *BC2* recognises that the rigid application of its citation order may not best suit the needs of readers in all contexts'.⁸

The new *Bliss* is an ambitious project 'using and developing the work of the Classification Research Group towards the elusive goal of a completely new general classification scheme' (Rowley, 2000). Although it has its adherents, particularly in education libraries (Foskett, 2000), it is not widely used. Foskett asserts, that because of finance and other factors, its future seems less assured than *UDC* (*ibid.*).

Notes

- 1 www.udconline.net.
BSI British Standards (British Standards Institution) is the distributor of English Language UDC products including UDC Online.
- 2 www.udcc.org/about.htm
- 3 *ibid.*
- 4 www.udcc.org/BSI_brochure_2005.pdf.
- 5 www.udconline.net.
- 6 Examples used here have been taken from the *Bliss Classification* website at:
www.blissclassification.org.uk/index.htm.
- 7 www.blissclassification.org.uk/bchist.htm.
- 8 www.blissclassification.org.uk/bcclass.htm.

References

- Foskett, A.C. (2000), 'The Future of Faceted Classification', in Rita Marcella and Arthur Maltby (eds.), *The Future of Classification*, Gower, Aldershot, Hants; Burlington, Vt., p.78.
- Maltby, Arthur (1975), *Sayers' Manual of Classification for Librarians*, 5th ed., Deutsch, London, p. 208. Marcella (1994) is essentially a revised edition of this manual.
- Mills, J. (1976), 'Bibliographic Classification', in A. Maltby (ed.), *Classification in the 1970s: a Second Look*, rev. ed., Bingley, London, pp. 25-50.
- Mills, J. *et al*, (1977-), *Bliss Bibliographic Classification (Second Edition)*, Butterworths, London, and Saur, München, vols in progress.
- (1977), Class Q Social Welfare, by J. Mills and V. Broughton, with the assistance of Valerie Lang, Butterworths, London. A revised edition by C. Peddle was published in 1994.
- (2007), Class W The Arts, by J. Mills and C. Ball. Saur, München.
- Rowley, Jennifer and Farrow, John (2000), *Organizing Knowledge: an Introduction to Managing Access to Information*, 3rd ed., Gower, Aldershot, Hants; Burlington, Vt., p. 233. A 4th ed. of this work, by Rowley and Hartley, was published in 2008.
- Universal Decimal Classification* (various dates), English text, (BS 1000).
Available at: www.udconline.net. Examples shown here were checked via this site in 2009. BSI British Standards (British Standards Institution) is the distributor of English Language UDC products including UDC Online.

9 More about Notation

A common error is to think that choice of notation is the first step in the compilation of a classification scheme. It should now be apparent that this is not so; the allocation of a notation is one of the final steps.

Notation is, however, a very important feature. 'Poor notation can reduce the value of even the best constructed scheme' (Aitchison, 1982). Some of the qualities which might be looked for are listed below and it is also 'desirable for a notation to be easy to read, write, type, say and remember' (Palmer and Wells, 1951).

Uniqueness

Care should be taken that one concept or subject cannot be mistaken for another. In a machine-based system, for example, it could be disastrous if the coding for a component, say a ball-bearing, were the same as that allocated to a completed product such as a gear box. A distinction must however be made between unique identification of a subject and identification of an actual item. For example, in a library using the *Library of Congress Classification*, the subject 'Use of artificial satellites in telecommunication' might have the number TK5104 but there could well be a great number of books dealing with this subject, all with the same class number. For unique identification of each book something must be added to this number, so that in effect it becomes a class+book number. This addition could consist of letters and digits that could represent author, copy number, volume, and so on. A similar situation exists in a computer-based system where a code might be used as the sole key for identification and location of a data record. If several records have the same classification number then some further coding must be added.

Simplicity

Notation should be easily comprehensible and must have ordinal value as it reflects and 'fixes' chosen orders, therefore:

1		A
2	or	B
3		C

not

\$
&
*

Confusion should be avoided, for example the use of both the upper case letter 'O' and the digit '0', or the lower case letter 'l' and the digit '1'.

For classification and coding in industry, 'in order to assist the many functions of manufacture by providing a logical and meaningful system of identification for information, data and components . . . the best type of symbolisation is generally considered to be purely numerical and of uniform length' (McConnell, 1971, p. 1).

Brevity

Notation should be as brief as possible. Note that the longer the base, the shorter the classification numbers. For example, using letters, with a base of 26, it is possible to allocate class numbers no longer than two letters to 676 subjects (that is, 26×26). Using numerals, with a base of 10, only 100 subjects (that is, 10×10) can be allocated numbers no longer than two digits.

Note also that synthesis does not lend itself to brevity in notation.

Hospitality

Notation should have the ability to accommodate new concepts or subjects as necessary. It should allow for the insertion of both subordinate and coordinate subjects. The former is catered for quite easily by the decimal use of numbers (or the use of letters on the same principle). If 1 represents 'Fiction' and 2 'Poetry' then further expansion is possible in this manner:

1	Fiction
11	Historical fiction
12	Romantic fiction
13	Science fiction
2	Poetry
21	Lyric poetry

The insertion of new coordinate subjects is more difficult but it could be provided for by leaving gaps, for example:

1	Construction
4	Testing
7	Maintenance

but the gaps may not be in appropriate places and therefore could interfere with the required order.

Flexibility

It may be useful to be able to alter citation order according to the needs of a particular set of users. Facet analysis lends itself admirably to variations in preferred citation order (see also pages 30 and 80-1).

Expressiveness

In a narrow sense any notation which reflects and fixes a chosen order could be said to be 'expressive' but authors such as Maltby (1975) equate expressiveness with hierarchical classification where the notation reflects the structure of the scheme, for example:

1	Military ordnance
11	Artillery
111	Field artillery

Although sometimes considered helpful (the user can understand the systematic order more easily and it can be an effective tool in hierarchical computer searching), expressiveness is not essential. It results in longer classification numbers and limits hospitality.

Although the above features: Uniqueness; Simplicity; Brevity; Hospitality; Flexibility and Expressiveness may be desirable, it is obviously going to be difficult to incorporate them all into a single scheme. It is particularly difficult in a faceted scheme, for synthesis does not lend itself to brevity, simplicity, or expressiveness.

Retroactive notation

One device which is sometimes used in faceted schemes in an attempt to shorten numbers is 'retroactive' notation. 'Retroactive' simply means 'backwards'. In notational terms a change of facet is indicated by a reversal of alphabetic or numeric direction from forwards to backwards. A classification number such as:

ABCD

shows no change of direction and therefore is derived from one facet only.

However the classification number:

CDEBCDABC

shows a change of direction, and therefore a change of facet, between the E and the B, and the D and the A. It could have been written as:

CDE/BCD/ABC

but the ‘/’ is superfluous because of the retroactive nature of the number.

In allocating such a notation, it will be seen that the first letters of the alphabet are used to introduce those facets which are to be cited *last* and not for the primary facets. The letter ‘A’, having been used once, must not be used again in any earlier facet. Similarly ‘B’, having been used once, must not be used again. Thus the notational base that is available is progressively reduced. Care must therefore be taken that when the primary facets are reached there are still sufficient notational symbols remaining. It is usual to ensure that an adequate amount of the notational base is reserved for the primary facets before allocating a notation to other facets.

Here is a retroactive notation applied to the estate agent’s scheme described on pages 12-23.

DE Atwell	CD Apartment	BC 1 bed.	AB £50-100,000
DF Blanford	CE Semi-det.	BD 2 bed.	AC £100-15,000
	bungalow		
DG Crosswood	CF Det.	BE 3 bed.	AD £150-200,000
	bungalow		
DH Denby	CG Terraced	BF 4 bed.	AE £200-250,000
	house		
	CH Semi-det.	BG 5 bed	AF £250-300,000
	house		AG £300-350,000
	CJ Det. house		

Sample classification numbers would be:

1 bedroomed apartment. Crosswood. £95,000 = **DGCDBCAB**

Apartments in Blanford = **DFCD**

PRACTICAL EXAMPLES OF NOTATION

Example 9.1 Uniqueness – from the *NATO Codification System (NCS)*
(NATO Group of National Directors on Codification, 1958 to date)

This is a commodity classification and coding plan used by the member countries of NATO (North Atlantic Treaty Organisation). Its ‘primary goal is to ensure that military personnel deployed in an operational

scenario can be assured of getting the right item to successfully complete their mission' (NATO...2008, p. [1]). It is based on the Federal Supply Classification, part of the Federal Catalog System developed in the United States during the period 1946-52. It was adopted by NATO in 1958 and has 'evolved over the years from a paper system into a sophisticated computer based system' (Pergolesi, 2006). It is operated by the National Codification Bureau of each member country. It enables a unique stock number to be allocated to each item of supply and 'facilitates the interchange of equipment and stores between NATO forces' (MacConnell, 1971).

The first section of four digits of the *NATO* number (the NSC number) is a Group/class prefix which indicates the type of equipment, commodity or component. The first two digits identify a particular group and the next two digits identify a particular subdivision or class within that group. In this section the classification is hierarchical, for example: group 10 = 'Weapons' and 1040 = 'Chemical weapons'; group 34 = 'Metalworking machinery' and 3416 = 'Lathes'; group 59 = 'Electrical and electronic equipment' and 5905 = 'Resistors'.

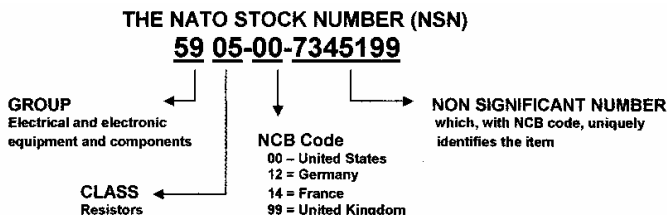
5905 identifies a component as a resistor

The second section of the code number is a two digit nation code (NCB) used to identify the country which originated the equipment or commodity, for example 00 = USA and 99 = Great Britain.

5905 – 00 a resistor originating in the United States

The third section of the code number is the item identification number that contains seven digits. These digits give uniqueness to the complete NATO number. They are allocated at random by the particular country and have no significant meaning.

Thus the complete 13 digit NATO Stock Number would be (NATO ..., 2008, p. [7]):



To enable user-friendly reading, it is common for a dash to be inserted at strategic points in the NSN, eg: **5905-00-734-5199** (NATO...2003, p. [15]).

With only two stages of division in the first hierarchical section and because the Group/class prefix is limited to four digits, two for the main group and two for the sub-group, the maximum number of groups and sub-groups that can be accommodated is $100 \times 100 = 10,000$. However, this relates only to the subject classification; the further subdivision by country of origin and then by random number in the range 0 to 9,999,999 means that actual number of components that can be handled is almost infinite.

The last nine digits of the number, comprising the country number and the random number, constitute a *NATO* item identification that is unique world-wide.

Several points are illustrated by the above example:

- 1 The use of a purely numeric notation of uniform length with a fixed number of digits.
- 2 The identification of the country of origin.
- 3 The addition of a unique random numeric identifier that will distinguish, for instance, the particular component in the above example from other resistors.

Example 9.2 Simplicity – from the *London Education Classification* (Foskett, 1974)

As we have seen, to be simple, ideally a notation must convey order, should be brief, easily comprehensible, memorised, written and spoken. Clearly, it is difficult to accommodate all of these attributes into one system.

A scheme that reflects many of these features is considered to be the *London Education Classification*. The pure alphabetic notation conveys order; most of the class numbers are just three letters long; they are easily comprehensible and written; they are also, in most instances, pronounceable. Examples are:

Bux	Documentation
Fab	Teaching profession
Hab	Management of education
Lob	Audio visual aid
Pil	Sex education
Ror	Choir schools

The pronounceable notation led to some interesting mnemonic classification numbers as illustrated by 'Bux', 'Pil' and 'Ror' above.

Unfortunately, in the second edition some four letter notations had to be introduced and the pronounceable element was therefore reduced.

Example 9.3 Brevity – from the *Bibliographic Classification* (Bliss, 1940-53)

As noted in the example from the *London Education Classification* above, the fact that the notation is just three letters long makes class marks reasonably short. Another scheme where one of the underlying aims was a concise, brief notation is the original *Bliss Classification*. For example:

History of Europe = **M**

whereas, in *Dewey*:

History of Europe = **940**

The basic notation in *Bliss* consists of capital letters and Bliss considered that a main class number need never exceed four letters. A base of three letters can accommodate 17,576 topics and four letters 456,976 topics. This allows specific subjects to be given fairly brief classification numbers, e.g.

GER= Biochemistry of muscles in animals

UCJF = Milk production

Class marks longer than three letters are rare, but the use of systematic schedules and synthesis (see pages 61-3) makes the notation more complex.

Example 9.4 Brevity from the *Bliss Bibliographic Classification, revised edition (BC2)* (Mills, 1977-)

It is claimed that the wide notational base and the ordinal nature of the notation in *BC2* 'produce classmarks which are exceptionally brief in relation to their specificity. For example, the class of a work on the nurse as a caregiver for terminal patients and their families is exactly represented by the classmark **HPK PEY FBG K**'.¹

Example 9.5 Hospitality – from the *NATO Codification System*

In this system, as explained in Example 9.1, the first two digits identify a particular group and the next two a particular class within that group.

Some of these numbers are unassigned, which leaves room for expansion:

11	Nuclear Ordnance
1105	Nuclear Bombs
1110	Nuclear Projectiles
1115	Nuclear Warheads and Warhead Sections
...	
20	Ship and Marine Equipment
22	Railway Equipment
23	Ground Effect Vehicles, Motor Vehicles, Trailers and Cycles

Example 9.6 Hospitality – from the *Library of Congress Classification*

In the fifth edition of Class Q Science (Library of Congress, 1951) will be found:

QD	CHEMISTRY
	ORGANIC CHEMISTRY
241	Periodicals
243	Yearbooks
245	Collected works
248	History
...	

‘Encyclopaedias’ are not enumerated or listed, and therefore ‘An encyclopaedia of organic chemistry’ cannot be classified. In the sixth edition of Class Q (Library of Congress, 1973), however, appears:

246	Dictionaries and encyclopedias
-----	--------------------------------

This class number has been neatly slotted into the appropriate place in the scheme by making use of the gap left, for hospitality purposes, in the notation between 245 and 248.

‘An encyclopaedia of organic chemistry’ can now be classified, the number being:

QD 246

In this scheme the letters I, O, W, X and Y are not used in the main divisions (see also page 146) and there also some gaps left at the second letter position allowing for insertions at a later stage.

Example 9.7 Flexibility – from the *London Education Classification* (Foskett, 1974)

In this scheme the facets are arranged in an order of increasing concreteness, for example:

Lab	Teaching methods
...	
Mim	Mathematics
...	
Ref	Secondary schools (11-18 years)

The recommended or preferred citation order is therefore the reverse of this, for example:

Ref Mim Lab = Teaching mathematics in secondary schools

It is pointed out, however, by Foskett, the compiler, that individual libraries or information services may wish to vary this order: 'One of the greatest advantages of a faceted classification . . . is that . . . the facets can be arranged in any sequence, to suit the particular user' (*ibid.* p. 9). If it were desired to bring together all items dealing with 'Teaching methods' and to sub-arrange first by subject and then by environment, then the citation order could be altered to conform to the schedule order and the classification number would become:

Lab Mim Ref = Teaching mathematics in secondary schools

Note also that a change of facet is consistently indicated by a change from a lower to an upper case letter. The upper case letter therefore acts as a facet indicator.

A further example of flexibility (from *UDC*) is described on pages 68-9.

Example 9.8 Expressiveness – from *Uniclass* (Crawford, 1997)

P	Materials
P4	Metal
P41	Steel
P413	Alloy steel
P4131	Stainless steel

Note the way in which the structure of the scheme is reflected in the notation.

Example 9.9 Non-expressiveness – from the *British Catalogue of Music Classification* (Coates, 1960)

PW	Keyboard instruments
...	

Q	Piano
· · · ·	
R	Organ
· · · ·	
RW	String instruments
· · · ·	
S	Violin

The structure of the scheme is not revealed by the notation. The 'Piano' and the 'Organ' are both 'Keyboard instruments' but it is not clear that Q and R are subordinate to PW. A similar situation exists with the 'Violin' and 'String instruments'.

A purely ordinal notation such as that shown here, in which no attempt is made to be expressive, permits new concepts to be added anywhere in the schedules and is therefore more hospitable.

Example 9.10 Retroactive notation – from the *British Catalogue of Music Classification* (Coates, 1960)

G	Suites
P	Solos
Q	Piano

As the notation can be combined retroactively, the classification number for:

Suites for solo piano = **QPG**

Note the fact that the facet changes each time that the alphabetical direction reverses.

Example 9.11 Retroactive notation from the *Bliss Bibliographic Classification, revised edition (BC2)* (Mills, 1977-)

The subject 'Library provision for elderly people in residential care' would be classified as:

QLV EPX L

This number is derived from the classification schedule below as described on page 70.

Q	Social welfare
QEL	Residential care
QEP X	Library provision
QLV	Old people

A change of facet is indicated each time that the alphabetic order reverses, ie: between the ‘V’ and the ‘E’ and between the ‘X’ and the ‘L’. Thus the notation is retroactive

Example 9.12 Retroactive notation – from the *Dewey Decimal Classification* (Dewey, 2003)

Perhaps without realizing it, Dewey included a ‘flavour’ of retroactive notation in his ‘History’ class:

History of England

942.05 History of England in the Tudor period

942.1 History of London

942.105 History of London in the Tudor period

Note that whenever the notation returns to ‘0’, that is goes backwards to ‘0’, then the characteristic of division changes from ‘Place’ to ‘Time’.

Note

1 www.blissclassification.org.uk/bchist.htm.

References

- Aitchison, Jean (1982), ‘Indexing Languages, Classification Schemes and Thesauri’, in L.J. Anthony (ed.), *Handbook of Special Librarianship and Information Work*, 5th ed, Aslib, London, pp. 207-61.
- Coates, E.J. (1960), *The British Catalogue of Music Classification*, British National Bibliography, London.
- Crawford, Marshall, Cann, John and O’Leary, Ruth (eds) (1997), *Uniclass: Unified Classification for the Construction Industry*, RIBA, London.
- Dewey, Melvil (2003), *Dewey Decimal Classification and Relative Index*, edition 22, ed. by Joan S. Mitchell ... [et al], OCLC, Dublin, Ohio. 4 v.
- Foskett, D.J. and Foskett, Joy (1974), *The London Education Classification: a Thesaurus/classification of British Educational Terms*, 2nd ed., University of London, Institute of Education Library. Currently, (2009) the Institute continues to use this scheme but see note on pages 106-7.
- Library of Congress (various dates), *Classification*, LC, Washington, 43 v. *Class Q : Science*, 5th ed., 1951, and 6th ed., 1973.
- MacConnell, W. (1971), *Classification and Coding: an Introduction and Review of Classification and Coding Systems*, British Institute of Management, London, p.2.
- Mills, J., et al. (1977-), *Bliss Bibliographic Classification (Second Edition)*, Butterworths, London, and Saur., Munchen, vols in progress.

- Palmer, Bernard I. and Wells, A.G. (1951), *The Fundamentals of Library Classification*, Allen & Unwin, London, p. 62.
- NATO Group of National Directors on Codification (2008), *Brochure on the NATO Codification System (NCS)*. Available at: www.nato.int/codification.
- NATO Group of National Directors on Codification (2003). *Guide to the NATO Codification System (NCS)*. Available at www.nato.int/codification.
- Pergolesi, Mauro (2006), *50 years of NATO : Codification Plays a Vital Role*, p. [5]. Available at: www.nato.int/codification.

10 More about Schedule and Citation Order

In the estate agent's scheme with a retroactive notation shown on page 76, the facets are shown in the order of combination, but, in practice, it would be necessary and obviously more useful to set it down in the order of the alphabetical notation:

AB	£50-100,000	BC	1 bed.	CD	Apartment	DE	Atwell
AC	£100-150,000	BD	2 bed.	CE	Semi-det. bungalow	DF	Blanford
						DG	Crosswood

etc.

In fact, this is one of those occasions mentioned on pages 17-18 when the schedule order would differ from the citation order. In this instance citation order would be the *reverse* of schedule order, for example:

1 bedroomed apartment. Crosswood. £95,000 = **DGCDBCAB**

The citation order may be the reverse of the schedule order not only when retroactive notation is used but whenever it is desired to adhere to the commonly accepted principle that 'general' should precede, or file before 'special'.

If, for instance, in a particular scheme 'Bridge' has the class number A1, 'Construction' the class number B1, and the citation order is 'type of structure' followed by 'operation', then the class number for 'Construction' is B1 and the class number for 'Bridge construction' is A1B1. These numbers would file as:

A1B1
B1

This is not a 'general' before 'special' order. A work on civil engineering construction should file before a work on the construction of a specific structure.

What happens if the same citation order is retained but this is made the reverse of the schedule order? In order to do this the notation must be changed so that the first letter in the alphabetical sequence introduces the first facet to be set down in the schedules, that is:

<u>Operation</u>	<u>Type of construction</u>
A1 Construction	B1 Bridge

The classification numbers now become A1 for 'Construction' and B1A1 for 'Bridge construction'. These would file as:

A1	Construction
B1A1	Bridge construction

and a 'general' before 'special' order has been achieved. This reflects the 'principle of inversion', which states that filing order should be the reverse of citation order.

Remember, however, that it is not essential for citation order and schedule order to differ.

When various types of symbol introduce different facets a filing order for symbols can be laid down so that a 'general' before 'special' sequence will be obtained. For example, if in a classification scheme for 'Literature', B is 'English literature', 2 is 'Drama' and e is 'Essays', and if facets are combined in this order, then sample class numbers would be:

B2	English drama
B2e	Essays on English drama
Be	Essays on English literature

'Essays on English literature' is more 'general' than 'Essays on English drama' and should come first in a filing arrangement. It may be stipulated therefore that lower case letters file before numbers, giving this order:

Be	Essays on English literature
B2	English drama
B2e	Essays on English drama

which is the 'general' before 'special' order that is required.

Facet formulae

On page 37 reference was made to Ranganathan's PMEST formula. Difficulty can be experienced in the understanding of this formula. What precisely is 'Personality', for example? It is 'remarkably elusive when succinct unambiguous definition is demanded' (Maltby, 1975).

However, Ranganathan was the first classification theorist to develop an explicit and comprehensive theory for citation order. It was based on the notion of ‘decreasing concreteness’ – ‘those concepts most readily received by the mind (most ‘concrete’) should be cited before concepts less readily received’ (Bliss, 1977-). Other theorists have subsequently attempted to produce a generalized facet formula that could be applied more easily. Having identified the primary facet, the ‘thing’, the objective of study, the end product, then this is cited first. The other facets ‘are then cited in an order of dependence; e.g., an operation has to be performed on something and the agent of an operation must have an action to be an agent of’ (*ibid.*). This standard citation order can be expressed thus:

**Things – Kinds – Parts – Materials – Properties – Processes –
Operations – Agents**

In certain areas this formula appears to work well. A subject such as ‘Machinery for the X-ray testing of stainless steel wires on suspension bridges’, for instance, could be analysed thus:

Thing (Bridge) – Kind (Suspension) – Part (Wires) – Material (Steel) –
Property (Stainless) – Process (X-ray) – Operation (Testing) – Agent
(Machinery)

However, as Foskett (1996) points out, how could such a formula be applied to literature, for example?

‘Place’ and ‘Time’ do not appear in the formula given above, although they are included in PMEST. The facet analytical theory developed by members of the United Kingdom Classification Research Group, whilst basically agreeing with the above formula, recognizes thirteen categories which include ‘Space’ and ‘Time’.¹

‘Place’ and ‘Time’ are, of course, facets common to many subjects and, even in a faceted scheme, they are often given in ‘auxiliary’ tables preceding or following the main schedules.

Regardless of generalized facet formulae, the important thing is to ensure that the citation order in a particular scheme caters for user requirement.

The first part of the formula, ‘Thing – Kind’, implies that it may well be necessary to introduce sub-facets, or ‘facets within facets’. As an example, if a classification scheme for ‘Food’ were being compiled it would be possible to group according to ‘source’, for example:

<u>Flesh foods</u>	<u>Grown foods</u>	etc.
Fish	Fruit	
Meat	Vegetables etc.	
Poultry etc.		

A sub-facet could obviously be established for 'Meat', for instance, according to the animal from which it was derived, for example:

Meat

From cattle
Beef

From pigs
Bacon
Ham
Pork

From sheep
Lamb
Mutton

Notes

1 www.blissclassification.org.uk/bchist.htm.

References

- Foskett, A.C. (1996), *The Subject Approach to Information*, 5th ed., Library Association, London, p. 154.
- Maltby, Arthur (1975), *Sayers' Manual of Classification for Librarians*, 5th ed., Deutsch, London, p. 61.
- Mills, J., et al. (1977-), *Bliss Bibliographic Classification (Second Edition)*, Butterworths, London, and Saur, Munchen, vols in progress.
- Introduction and auxiliary schedules, by J. Mills and Vanda Broughton. p.40.

11 Other Features of Classification Schemes – Summary of advantages and disadvantages of faceted and enumerative classification

The generalia class

In a scheme which attempts to cater for library materials, there must be provision for documents which do not fall into a particular subject area but which are all pervasive. Examples are general encyclopaedias and general newspapers or periodicals such as *The Columbia encyclopedia*, *The Times* or the *Spectator*. What is required, therefore, is a ‘general works’ or ‘generalia’ class, which will enable such documents to be classified.

The *Dewey Decimal Classification*, for example, uses Class 000 and, within this class, 030 is used for general encyclopaedic works. *Everyman’s Encyclopaedia*, being an encyclopaedia in English, would be classified as 032, with the ‘2’ representing English.

Care must be taken to distinguish between a general encyclopaedia or periodical and a *subject* encyclopaedia or periodical. Where the latter is concerned classification is first by subject and then by form. An ‘Encyclopaedia of cats’, for example, would be classified by *Dewey* at 636.8003, as indicated on page 63, and the periodical *Cats magazine* would be classified at 636.8005.

A further example of an all-pervasive topic is in fact ‘Librarianship’, which is concerned with information handling over the whole range of knowledge. *Dewey* also places ‘Library and information sciences’ in the generalia class at 020. Within this class 025.4 is ‘Subject analysis and control’ and the subject of this text, ‘Classification’, would be allocated the number 025.42.

Literary warrant

The ‘literary warrant’ principle upon which the *Library of Congress Classification* is based, that is, that the classification reflects the subjects

of items actually in stock (see page 53), permits the classification of subjects which often cannot be specified adequately in other schemes, for example:

Shakespeare and Anne Hathaway	=	PR 2906
Heat, light and sound	=	QC 220

Many books have been written about the subject 'Heat, light and sound' but if, for instance, an attempt is made to classify this subject by the *Dewey Decimal Classification*, it is found to be impossible; the only available choices being:

530 Physics in general
 534 Sound
 535 Light
or
 536 Heat

Literary warrant is however catered for in faceted schemes because the concepts included in such schemes are obtained from the literature of the subject. These concepts may then be combined to form the composite subjects of actual documents. The principle may be illustrated using the *Universal Decimal Classification*. The scheme is based upon the *Dewey Decimal Classification* but, as indicated on pages 66-9, the introduction of signs and symbols permits general number building, therefore:

Heat, light and sound	=	534/536
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Main class order

One problem that has exercised the minds of classification theorists is that of the order of the so-called 'main classes' in a general scheme. The *Dewey Decimal Classification*, for instance, is often criticised for the separation of 'Language' (400) from 'Literature' (800) by 'Science' (500), 'Technology' (600) and 'Arts' (700). But what constitutes a 'main' class? *Dewey* divides the whole of knowledge into ten such classes but is this feasible? It has no realistic logical base but is primarily governed by the nature of the chosen notation.

It is generally recognised that the order *within* classes is more important than main class order. Is a study of the latter of any significance therefore? Maltby (1975) maintains that, among other things, 'it sets the student to study the structure of the field of knowledge itself'.

A full discussion of main class order is outside the scope of this text. However one point needs to be made and that is that a major criterion is, of course, the need of the user, which must be paramount. If one main class order is of more help to the user than another then that is the one that should be chosen. This perhaps refers not so much to the overall order but to ensuring that the more closely related classes, for example Language and Literature, Science and Technology, or Sociology and History, are placed in close proximity.

One system that obviates the need for a main class order is what is known as ‘free’ classification.

Free classification

As we’ve seen, in a general scheme, covering the whole of knowledge, whether hierarchical or faceted, the tendency is to initially divide knowledge into a number of disciplines. This means that a single concept may have several different notations dependent upon the discipline. For example, the concept ‘Gold’ may have one notation under mining in technology, another under finance in economics, another under jewellery in business, and so on. In ‘free classification’ there is just one notation for a concept no matter in which context it is being viewed and main class order would therefore be irrelevant.

In such a classification, the concept ‘Gold’ would have a single fixed notation and a search for that notation would retrieve all documents relevant to ‘Gold’ regardless of the related discipline. A document on ‘Gold mining’ could be retrieved by a search for a combination of the notations for ‘Gold’ and for ‘Mining’. This is akin to post-coordinate indexing (see pages 113-4). The *Brisch* classification (see page 26) could be searched in this way using punched cards. A card would be punched with holes that related to the number of a particular concept and information retrieved by passing a needle through to select those cards indexing the required concept or combination of concepts.

Such a system lends itself admirably to computerised database searches and, for this reason, free classification has been advocated as a possible useful method for online searching on the Internet (Gnoli and Mei, 2006).

Gnoli asks ‘why not join the merits of free classification with those of faceted classification?’ (*ibid.*). The result is then a ‘freely faceted classification’ where the notation can include facets expressing the kind of relations holding between the concepts (*ibid.*). This takes us back to research undertaken by Austin and other members of the Classification Research Group in the United Kingdom.¹ Austin put theory into practice in PRECIS (see page 117).

Summary of advantages and disadvantages of faceted and enumerative classification

There follows a brief summary of some of the major advantages and disadvantages of the two approaches to classification.

Faceted

A faceted scheme lists concepts only – classification numbers for complex subjects are built up by means of synthesis.

Advantages

Because complex subjects are not listed, such schemes are easier to compile.

The schedules are shorter for the same reason but, despite their brevity, they permit the classification of both very simple and very complex subjects.

New subjects can very often be catered for by the combination of already existing concepts.

A faceted scheme is said to be the best way to provide detail and accuracy.

A faceted scheme enables the classification of subjects which are not listed in an enumerative scheme.

Disadvantages

The notation can become long and complex and may be unsuitable for the arrangement of documents on shelves as in a library.

The problem of citation order can cause difficulty.

Enumerative

An enumerative scheme attempts to list all possible subjects, both simple and complex, within the defined subject field or fields.

Advantages

Such schemes are well-established; they have been generally accepted and widely used with considerable success throughout the world for a long period of time.

It is possible to use a fairly short and uncomplicated notation and 'notationally' it is easier to display the structure of the scheme.

In relation to the use of general enumerative schemes in libraries, standardisation is facilitated because classification numbers may be included in centralised records. For example, *Dewey Decimal Classification* and *Library of Congress Classification* numbers are included in eye-readable and machine-readable records supplied by the Library of Congress and the British Library.

With schemes such as the *Dewey Decimal Classification* there is also the possibility of 'integrity of numbers', that is, the stabilizing of existing numbers, a policy initiated by Dewey.

Disadvantages

It is impossible to list every conceivable subject.

There can be a lack of accommodation for even simple subjects.

New subjects cannot be accommodated and regular revision may be required. An enumerative scheme has virtually a built-in obsolescence.

General enumerative schemes may be unsuitable for certain purposes. For example, in libraries and information services which specialise in particular subject fields, they are not detailed enough in specific areas. Special classification schemes, restricted by subject, are therefore required. These may be enumerative or faceted. Special schemes are also useful for the manual filing of office papers and for the coding of entities in computerised systems.

Note

- 1 See, for instance, Austin, Derek (1976) 'The CRG Research into a Freely Faceted Scheme', in *Classification in the 1970s: a Second Look*, ed. by Arthur Maltby, Bingley, London, 1976, pp. 158-194.

References

- Gnoli, Claudio and Mei, Hong (2006), 'Freely Faceted Classification for Web-based Information Retrieval', *New Review of Hypermedia and Multimedia*, 12 (1), pp. 63-81.
- Maltby, Arthur (1975), *Sayer's Manual of Classification for Librarians*, 5th ed., Deutsch, London, p. 59.

12 The Relationship between Classification and Alphabetical Authority Lists of Indexing Terms — the Compilation of Thesauri

The printed book has been a prime information source for some hundreds of years. A book's alphabetical index unlocks its contents and, without it, a specific piece of information may be very difficult to find. Here are some abbreviated extracts from such an index (Pictorial knowledge, 1970):

aeronautics *see* aviation

aviation

• • •

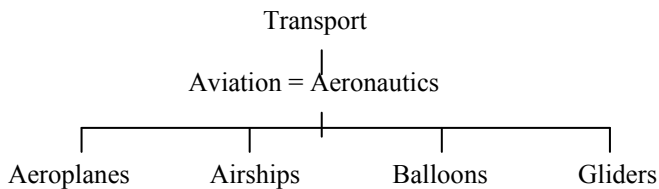
see also aeroplanes; airships; balloons; gliders

transport

• • •

see also aviation; . . .

A close examination of these entries will reveal a hierarchical structure:



These entries therefore provide a simple example of the use of classification in alphabetical indexing.

Just as an alphabetical index must be provided for a book, so any information system, including a computerised information system, will need to cater for an alphabetical subject approach.

In order to ensure that the selection of indexing terms is consistent, a record or 'authority list' of the terms that have been used will need to be

compiled. For example, the entry for 'Aviation', in the authority list, might appear as follows:

Aviation

<i>Use for</i>	Aeronautics
Broader terms:	Transport
Narrower terms:	Aeroplanes; Airships; Balloons; Gliders

This indicates that the lead term 'Aviation' is the preferred term that must be used and that 'Aeronautics' is not to be used. 'Aviation' is subordinate to the broader term 'Transport' and superordinate to the narrower, co-ordinate terms: 'Aeroplanes'; 'Airships'; 'Balloons'; and 'Gliders'.

A further entry in the book index is:

flight
 . . .
see also aviation

This additional related term 'flight', which could, of course, be concerned with 'bird flight', would also need to be included in the authority list:

Aviation

<i>Use for</i>	Aeronautics
Broader terms:	Transport
Narrower terms:	Aeroplanes; Airships; Balloons; Gliders
Related terms:	Flight

As a further illustration of how such an authority list might be used, let us take, for example, a library catalogue. In this type of system, when the subject of a document, or some other record, has been ascertained, instead of, or as well as, translating this subject into a classification number, alphabetic indexing terms that succinctly and specifically describe the subject will need to be selected. An examination of a particular document, for instance, might reveal that it is concerned with the subject 'Wages of post office workers'. Using 'natural' language, that is, terms taken directly from the document, the terms that describe the subject are:

Wages
 Post office
 Workers

If a natural language index is all that is required, then these are the terms that would be used but if consistency is the aim, then an appropriate

‘authority list’ must be consulted. This may well instruct that ‘Post office’ is acceptable but that ‘Remuneration’ is to be preferred to ‘Wages’, and ‘Personnel’ is to be preferred to ‘Workers’. The terms that must be used are therefore:

Remuneration
Post office
Personnel

The indexing language is being ‘controlled’ so that the same subjects are consistently indexed under the same terms.

Today, such a structured, controlled vocabulary or indexing language is more often referred to as a ‘thesaurus’.

Here is a typical entry from a published thesaurus (Thesaurus, 1997):

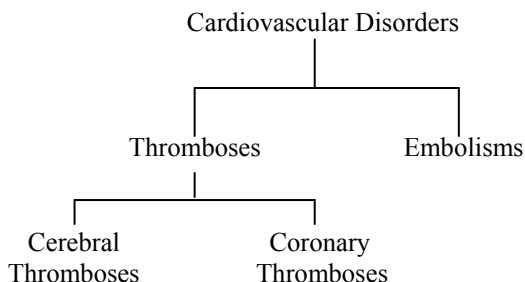
Thromboses

Broader Cardiovascular Disorders

Narrower Cerebral Thromboses
Coronary Thromboses

Related Embolisms

Clearly, in a similar manner to the illustrative entry for ‘Aviation’, hierarchical and coordinate relationships or groupings are being established, that is:



Because the grouping of concepts in this way very obviously embodies a process of classification, can a classification scheme possibly provide a base from which to compile an alphabetically arranged thesaurus? The answer is a very definite ‘Yes’ and a faceted classification lends itself admirably to this process. The method helps to keep inconsistencies to a minimum and ensures that the thesaurus is grounded upon firm principles.

If, for example, a classification scheme were being compiled for the subject ‘People’, then the facets would probably include:

People		
<u>By age</u>	<u>By sex</u>	<u>By race</u>
Children	Male	European
Adults	Female	Asiatic
etc.	etc.	etc.

The first of these facets will be used as an example to illustrate how a thesaurus can be compiled from such a scheme. A more fully developed facet might appear as follows:

People

By age

Babies

Infants

Children

Teenagers

Adults

Young adults

Middle aged

Old aged

Nb Sub-facet within Adults

Having chosen the concepts as shown they should be defined, as certain terms could be interpreted in different ways.

People

By age

Babies

0-2 years

Infants

3-5 years

Children

6-12 years

Teenagers

13-17 years

Adults

over 18 years

Young adults

18-39 years

Middle aged

40-59 years

Old aged

over 60 years

In addition alternative 'non-preferred' forms, if any, of each concept should be indicated at this stage, for example:

People

Use for Persons

By age

Babies

0-2 years

Infants

3-5 years

Children

6-12 years

Teenagers

13-17 years

Use for Youth

Adults	over 18 years
Use for Grown up persons	
Young adults	18-39 years
Middle aged	40-59 years
Old aged	over 60 years
Use for Aged	

A thesaurus entry must now be formulated for each concept. The following abbreviations, which have become standard in many thesauri, will be used:

SN	Scope note
UF	Use for
BT	Broader term
NT	Narrower term
RT	Related term

The thesaurus entry for 'Old aged', for example, would be:

Old aged

SN	Over 60 years
UF	Aged
BT	Adults
RT	Young adults
	Middle aged

Note how this entry has been compiled using the already existing hierarchical and coordinate relationships indicated by the classification scheme.

A further entry would have to be made for the alternative non-preferred form of the term 'Old aged', that is:

Aged use **Old aged**

The reader, if he or she so wishes, may now attempt to produce, in a similar manner, a thesaurus entry for the term 'Adults'. The answer will be found on the opposite page.

The thesaurus entry for the term 'Adults' would be:

Adults

- SN Over 18 years
- UF Grown up persons
- BT People
- NT Young adults
 - Middle aged
 - Old aged
- RT Babies
 - Infants
 - Children
 - Teenagers

Check this result against your own entry and note that an entry would also be required for the non-preferred term 'Grown up persons', that is:

Grown up persons use **Adults**

A further point is that, in practice, it is advantageous to alphabetize terms within a particular category of relationship:

Adults

- SN Over 18 years
- UF Grown up persons
- BT People
- NT Middle aged
 - Old aged
 - Young adults
- RT Babies
 - Children
 - Infants
 - Teenagers

Entries for all preferred and non-preferred terms are built up in this way. Completed entries are filed in an alphabetical sequence.

The end result is a systematically developed alphabetically arranged list of terms recommended for use as 'index' terms in retrieval systems, together with an indication of related terms and alternative or synonymous terms.

Such thesauri as those described may be produced as separate entities or they may be published together with the classification scheme from which they were evolved.

Where the classification scheme is not combined with the thesaurus, and is not intended for use in its own right, but is used merely as a

basis for thesaurus construction, then compilation is considerably simplified. There is no need to pay attention to such things as order in array (a simple alphabetical order is all that is required), schedule order, or citation order.

When the thesaurus is published together with the classification scheme, it can then perform a dual function in that it also acts as an index to the scheme. If, for instance, a notation based on upper case letters were added to the 'age' facet, for example:

P People

Use for Persons

By age

PA	Babies	0-2 years
PB	Infants	3-5 years
PC	Children	6-12 years
PD	Teenagers	13-17 years

Use for Youth

PE	Adults	over 18 years
----	--------	---------------

Use for Grown up persons

PEA	Young adults	18-39 years
PEB	Middle aged	40-59 years
PEC	Old aged	over 60 years

Use for Aged

then a typical index/thesaurus entry would be:

Middle aged

PEB

SN 40-59 years
BT Adults
RT Old aged
Young adults

However, the methods of presentation vary and in some schemes it is necessary to consult the classification schedules themselves in order to use the scheme as a full thesaural facility.

The sample thesaurus is shown in full on the next page.

On subsequent pages are descriptions of three examples of classification schemes combined with thesauri, each of which adopts a different method of presentation.

Adults

SN Over 18 years
 UF Grown up persons
 BT People
 NT Middle aged
 Old aged
 Young adults
 RT Babies
 Children
 Infants
 Teenagers

Aged *use* **Old aged**

Babies

SN 0-2 years
 BT People
 RT Adults
 Children
 Infants
 Teenagers

Children

SN 6-12 years
 BT People
 RT Adults
 Babies
 Infants
 Teenagers

Grown up persons
use **Adults**

Infants

SN 3-5 years
 BT People
 RT Adults
 Babies
 Children
 Teenagers

Middle aged

SN 40-59 years
 BT Adults
 RT Old aged
 Young adults

Old aged

SN Over 60 years
 UF Aged
 BT Adults
 RT Middle aged
 Young adults

People

UF Persons
 NT Adults
 Babies
 Children
 Infants
 Teenagers

Persons *use* **People**

Teenagers

SN 3-17 years
 BT People
 RT Adults
 Babies
 Children
 Infants

Young adults

SN 18-39 years
 BT Adults
 RT Middle aged
 Old aged

Youth *use* **Teenagers**

Sample Thesaurus

Preferred terms are in bold type

PRACTICAL EXAMPLES OF FACETED CLASSIFICATION SCHEMES COMBINED WITH THESAURI

Example 12.1 *London Education Classification* (Foskett, 1974)

The *London Education Classification* (see also pages 78 and 80-1) was originally compiled for, and is still used by, the University of London Institute of Education Library.¹ The second edition, influenced by the publication of *Thesaurofacet* (Example 12.2), contains an alphabetical section in thesaural form.

Here is an extract from the classification schedules followed by examples of entries from the index/thesaurus that have been derived from this extract:

Bus	Research in Education
. . .	
But	Research Method . . .
Butj	Data Collection
Butk	Research Technique
Butl	Interview
Butm	Questionnaire
Butn	Survey
Butp	Longitudinal Survey
Butr	Analysis
Buts	Computing
Butt	Statistical Analysis
Butv	Testing Results
Buty	Presentation of results

Sample index/thesaurus entries derived from the above extract:

Analysis		Butr
SN	Examination of research data; distinguish from philosophical analysis	
BT	Research Technique	
NT	Computing	
	Statistical Analysis	
RT	Interview	
	Questionnaire	
	Survey	
Computing		Buts
SN	As a research technique	
BT	Analysis	
RT	Statistical Analysis	

Research Technique	Butk
SN	Tactics of research
BT	Research in Education
NT	Analysis
	Interview
	Questionnaire
	Survey
RT	Data Collection
	Presentation of results
	Research Method
	Testing Results

Examine the above entries in the light of the principles explained in the 'People' example. Note how they have been obtained directly from the classification schedules and how they perform the additional function of an alphabetical index to the scheme. In the case of the *London Education Classification*, concepts listed in the classification schedules are repeated in the thesaurus. Therefore the alphabetical thesaurus can be used as an entity in its own right, without reference to the schedules.

Example 12.2 *Thesaurofacet* (Aitchison, 1969)

This is a faceted classification and thesaurus for engineering and related subjects, which was developed for use in the English Electric Company (no longer extant as a separate company). Although now somewhat dated, it is included here because of its importance as the scheme that began the trend to classification-thesaurus integrated systems.

In *Thesaurofacet* related terms indicated by the classification hierarchy are *not* repeated in the index to the scheme. The index includes *additional* related terms only. In consequence the scheme must be used, whether for classification or alphabetical subject indexing, as an integrated whole.

Here is a brief extract from the schedules:

SV	Sewage engineering
SV2	Sewage
SV4	Domestic sewage
SV6	Industrial sewage
SV7	Soil sewage
SV9	Surface water sewage
SVB	Sewers
SVC	Outfall sewers
SVD	Storm sewers
SVE	Manholes

SVG	Sewage pumping
SVH	Syphons (sewage)
SVJ	Sewage treatment
...	
SVS	Sewage disposal

and here is the entry for 'Sewage' from the index/thesaurus:

Sewage	SV2
UF	Sewerage
RT	Corrosion atmospheres
BT(A)	Wastes

The point to note is that where the term 'Sewage' is concerned, the broader term 'Sewage engineering', the narrower terms 'Domestic sewage', 'Industrial sewage', and so on, and the related terms 'Sewers', 'Sewage pumping' and so on, are indicated in the classification schedules and are therefore not repeated in the index/thesaurus. The thesaurus entry does however contain related terms that are not apparent from the schedules. In this case, for instance, there is the related term 'Corrosion atmospheres' and there is the broader term 'Wastes' (Class SU6) additional to the broader term 'Sewage engineering' revealed by the classification hierarchy. (The 'A' in BT(A) stands for 'Additional'.) Clearly this is an economy device but it does mean that both the schedules and the thesaurus have to be used in order to obtain all the terms related to 'Sewage'.

Thesaurofacet 'had no direct successor' but the *BSI Root Thesaurus* (see page 106) was based upon it (Foskett, 1996).

Example 12.3 *London Classification of Business Studies* (2005)

This classification, already referred to (pages 31-3), also acts as a thesaurus. In earlier versions it was the schedules that performed this function rather than the alphabetical index, which was a conventional location device, but, in the latest revision, it is the alphabetical sequence which indicates hierarchical relationships most clearly.

Here, for example, is a brief extract from the alphabetical sequence:

Insurance	FGC
UF	Life insurance
BT	Financial benefits
NT	Health insurance
	Unemployment benefits
RT	Annuities
	Insurance companies

Referring to the schedules at FGC reveals the following:

FGC Insurance

- UF Life insurance
- RT Annuities
- Insurance companies

FGCB Health insurance

- UF Accident insurance
- Ill-health insurance
- Injury compensation
- Long term disability insurance
- RT Health service
- Industrial health

In the alphabetical sequence, broader and narrower terms are indicated in the normal thesaural manner. In the classified sequence, the notation and the degree of indentation do not necessarily reveal such relationships. Whilst it seems clear that FGCB 'Health insurance' is subordinate to 'Insurance', the broader/narrower relationship is not explicit.

One of the related terms to 'Health insurance' is listed as 'Health service'. This would not be indicated by the classification schedule, as it comes from class LWL. The inclusion of such terms is necessary but they can be difficult to pinpoint as the relationship is not revealed automatically by the scheme.

Abbreviations in thesauri

The abbreviations BT, for 'Broader term/s', and NT for 'Narrower term/s', although fairly standard, are not necessarily used in all thesauri. One example of a thesaurus that varies slightly in this respect is the *NASA Thesaurus*, which contains the authorised subject terms by which the documents in the NASA Aeronautics and Space Database are indexed and retrieved. Terms are organised within a hierarchical structure but instead of BT and NT, the abbreviation GS, for 'Generic Structure', is used. Here is a simple example:

acoustic fatigue

- UF sonic fatigue
- GS fatigue (materials)
- . **acoustic fatigue**
- RT acoustics

The GS (generic structure) in this example indicates that 'acoustic fatigue' is narrower than the broader term 'fatigue (materials)'. The abbreviations UF (for Use For) and RT (for Related Terms) are used in

the normal manner. Note the use of a parenthetical qualifier, ie: ‘fatigue (materials)’.

The abbreviations GS, BT, NT, RT, and so on, being abbreviations of English terms, are language dependent. A more neutral or language-independent system of symbols might be preferred for foreign language or multi-lingual thesauri. A system has been proposed (British Standards Institution, 1979) that could well form part of a future international standard, that is:

- precedes the preferred term
- = precedes the non-preferred term
- < precedes the broader term
- > precedes the narrower term
- precedes the related term

These symbols have been used in the *BSI* (British Standards Institution) *Root Thesaurus* (1985).²

Here are the entries for ‘Adults’ and ‘Grown up persons’ from the sample thesaurus on page 101 reproduced in this manner:

Adults

- Over 18 years
- = Grown up persons
- < People
- > Middle aged
- Old aged
- Young adults
- Babies
- Children
- Infants
- Teenagers

Grown up persons

- **Adults**

Notes

- 1 Illustrative examples used in this text are taken from the 1974 edition, which is still the last published version. Currently (2009), the Institute of Education Library continues to use the scheme but in a form that is substantially the original edition, as the degree of change required to move to the later edition was prohibitive. However, some class numbers remain the same as those cited here and searches can be made in the web-based online catalogue (<http://ioe.sirsidynix.net.uk/uhtbin/cgiisr/f1733GAeBe/SIRSI/62510058/60/502/x>) via numbers such as: Bux (‘Documentation’); Mim Lab (‘Teaching mathematics’) or Mim Ref (‘Mathematics in secondary schools’). The thesaurus did not form part of the original

scheme and, in the Institute, remains a separate entity. It is now far larger than the original and is more frequently updated than the classification schedule, new terms being added as required. The scheme used in-house has been re-keyed over recent years, in order to provide a machine-based version. Institute staff and time restrictions have precluded the adequate revision of the scheme as a whole.

Although use of *LEC* outside of the Institute has been limited, Aitchison (1982) maintained that this 'belies its importance, since its influence is seen in the categories of the *EUDISED Thesaurus*, in Class J Education of *BC2* [the *Bliss Bibliographic Classification*, second ed.] and in the educational section of the *UNESCO Thesaurus*'.

- 2 The *BSI Root Thesaurus*, based, as noted on page 106, on *Thesaurofacet*, is therefore another example of a scheme that consists of a classified subject display with a complementary alphabetical list. The classified section, which uses facet analysis to assist in the arrangement of the subject fields, gives full information about each term. This information is repeated in the alphabetical list but to one level of hierarchy only.

References

- Aitchison, Jean (1982), 'Indexing Languages, Classification Schemes and Thesauri', in L.J. Anthony (ed.), *Handbook of Special Librarianship and Information Work*, 5th ed., Aslib, London, pp. 205-261.
- Aitchison, Jean, Gomersall, Alan and Ireland, Ralph (1969), *Thesaurofacet: a Thesaurus and Classification for Engineering and Related Subjects*, 4th ed., English Electric, Leicester.
- British Standards Institution (1979), *Guidelines for the Establishment and Development of Monolingual Thesauri*, BSI, Milton Keynes. (BS 5723). Based upon the 'guidelines' previously issued by the International Standards Organisation (ISO 2788).
- BSI Root Thesaurus* (1988), 3rd ed., British Standards Institution, Milton Keynes.
- Foskett, A.C. (1996), *The Subject Approach to Information*, 5th ed., Library Association, London, p. 368.
- Foskett, D.J. and Foskett, Joy (1974), *The London Education Classification: a Thesaurus/Classification of British Educational Terms*, 2nd ed., University of London, Institute of Education Library.
- London Business School Library (2005), *London Classification of Business Studies: Thesaurus*, The Library, 2 v. [Vol.1] Classified Sequence, [Vol. 2] Alphabetical Sequence. This is a new version of the scheme first devised by K.D.C. Vernon and Valerie Lang in 1970, revised by K.G.B. Bakewell and David Cotton in 1979.
- Pictorial Knowledge* (1970), International Learning Systems, London, 8 v.
- Thesaurus of Psychological Index Terms* (2007), 11th ed., edited by Lisa Gallagher Tuleya, American Psychological Association, Washington. The entry cited in the text is taken from the 1974 edition and varies slightly in this latest edition.
- Unesco Thesaurus: a Structured List of Descriptors for Indexing and Retrieving Literature in the Fields of Education, Science, Culture, Communication and Information* (1994), New ed., Unesco, Paris. Previous edition compiled by Jean Aitchison, 1977.
- Viet, Jean and Slype, Georges van (1984), *EUDISED Multilingual Thesaurus for Information Processing in the Field of Education*, New ed., English version, Mouton, Berlin. At head of title: Council of Europe. Commission of the European Communities.

13 Classification as a Search Tool

Imagine that you are working as a library assistant, or as a bookshop assistant, when a client or customer asks whether you have anything on the ‘stegosaurus’. You look on the shelves and, as you might expect, there is nothing specifically on the required subject. Being an efficient assistant you will check to see if there is anything on ‘dinosaurs’ or perhaps ‘prehistoric animals’ or ‘prehistoric life’, to see if there are any items which contain relevant information. When you carry out a search process in this way, you are making use of classification in that you have identified the superordinate classes to which the required subject belongs.

Use of authority lists and thesauri

The ‘stegosaurus’ example used above is a simple one. Most people would know that a stegosaurus is a dinosaur. However, one may not always be aware of the context of a particular subject term or of the related terms that might be used as alternative approaches. Alphabetical lists of subject headings, or thesauri, showing related terms, as described in the previous chapter, are useful in this respect and therefore these tools may be used not only for indexing but also for *searching*. An entry such as the following (from the *RCN Library Thesaurus of Nursing Terms*, 2007):

FERTILISATION
UF Human fertilisation
BT Reproduction
NT Assisted conception
RT Fertility

indicates that ‘Fertilisation’ is a term which could have been used as an indexing term and also that the search may be broadened (via ‘Reproduction’), narrowed (via ‘Assisted conception’), or ‘widened’, coordinately (via ‘Fertility’).

The widely used *Library of Congress Subject Headings* (2008)¹ works in a similar fashion, as the following entries illustrate:

Rubber

. . .	
UF	Caoutchouc Ebonite Gum elastic India rubber Vulcanite
BT	Latex Non-timber forest products
RT	Elastomers Gutta-percha
NT	Elastic fabrics Electric insulators and insulation--Rubber Foam rubber Guayule rubber Vulcanization
. . .	
	Vulcanite
	USE Rubber

When indexing using this list, whether the document being indexed uses the term 'Vulcanite' or the term 'Rubber' in the constituent text makes no difference. The document *must* be indexed under the term 'Rubber'. In this way, as was explained in the previous chapter, the indexing language is *controlled*. A search for the term 'Vulcanite' would be unproductive but the searcher could, by referring to the list, ascertain that the preferred term is 'Rubber'. In addition, by referring to the term 'Rubber', a range of related terms that could be used in searching will be revealed.

If natural language, where terms are taken directly from the text of a document, is employed, a thesaurus, or an alphabetical subject heading list such as the Library of Congress list, may still assist in the search process.

Imagine that an enquirer is searching a *natural* language system for information on the possible use of parachutes to slow down ocean-going ships. A search is made under 'Parachutes AND Ships' but nothing is found. There is, in fact, a relevant document in the information system entitled 'Putting a halt to super tankers' but the textual content does not include the word 'ship' and, as this is a natural language system, the document has not been retrieved. A check in *Thesaurofacet* (see page 103) indicates that among the possible alternative terms to 'Ships' is the narrower term 'Tankers'. Amending the search to 'Parachutes AND Tankers' will retrieve the document. Without this aid, the searcher would either (a) have assumed, wrongly, that there is nothing relevant in the

system, or (b) have used some other source or even guesswork to come up with possible alternative search terms.

Boolean searching and full text databases

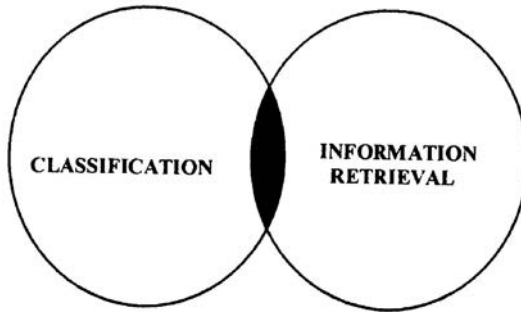
The search for 'Parachutes AND Tankers' utilises the 'operator' AND to link the two required terms. 'AND' is only one of three operators, the others being 'OR' and 'NOT'. The use of these operators in searching is now very common and widely accepted. AND, OR and NOT are known as *Boolean* operators and this type of searching is referred to as Boolean searching. A search for 'Venice AND Climate' would locate items that had been indexed under both of these terms. A search for 'Venice AND (Climate OR Weather)' would find items which had been indexed under or contained 'Venice' and *either* 'Climate' or 'Weather'. A search for 'Venice AND (Glass NOT Crystal)' would yield those items indexed under 'Venice' and 'Glass' but *not* those indexed under 'Venice' and 'Crystal'.

Today, many online information systems contain abstracts or the full text of articles, documents, books, etc. It is now possible to search a complete text, even a large work such as an entire encyclopaedia, for a single word or term, that is 'full text' searching. Because of this, some writers argue that a knowledge of classification is no longer necessary for the information worker. Burton (1997), for instance, maintains, perhaps tongue in cheek, that 'Internet search engines can rapidly find the growing volumes of information created and stored electronically using full text retrieval techniques. These search engines ... are capable of handling complex search strategies with Boolean operators'. What this statement fails to recognise is that even Boolean searching involves elements of classification. To take a practical example, let us imagine that an enquirer searches for the subject of this book: 'Classification' and the search results in thousands, or even millions, of 'matches' or 'hits'. This is far too great a number of items to browse and many of the items found will be irrelevant as classification can be concerned with a number of different subject areas: 'Biology'; 'Diseases'; 'Plants'; and so on. The enquirer will then add a second term, such as 'Information retrieval', in order to narrow the search to the particular subject area required and thereby reduce the number of items retrieved and increase their relevance.

If we examine what has happened here, this type of searching, although Boolean in that the user is searching for 'Classification AND Information retrieval', can be seen also as a type of classification, in that the user is, in effect, identifying characteristics of a subject in order

to bring like items together and separate unlike, the basic principle of classification.

The above Boolean search could be represented diagrammatically thus, the shaded area representing documents that have been indexed under both terms:



Despite the extensive use of Boolean in online searching, not everyone is convinced that it is the best search methodology. Hildreth (1989), for example, states that 'Much research and experience with Boolean retrieval systems ... indicates clearly and repeatedly that Boolean search formulation syntax and retrieval techniques are not very effective in search performance and not very usable or efficient search methods for end-users'. 'Determined explorers and the just plain curious need a flexible, rich, contextual subject search and browsing mode which offers plenty of navigation and trail blazing options'. Schneiderman (1997) maintains that 'until recently, computer scientists argued that the best way to search for information on the Web was by using keyword searching ... But keyword searching often fails miserably'.

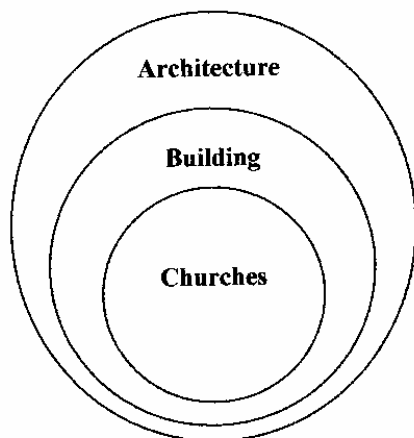
There is one word in the first of the above quotes that implies that classification must have a more significant and direct role to play over and above its inherent link with Boolean searching, and that word is 'contextual'. Imagine that a user searches for the term 'Churches' and gets the response

Your search did not match any documents:

Try to broaden your search using more general keywords

If the user is interested in the church as a physical entity, then, using a similar strategy as in the manual search for 'Stegosaurus' already described, he or she might enter the more general, broader term 'Buildings' or perhaps 'Architecture'. Clearly, this type of searching is making use of classification and, where a hierarchical relationship such as

this is concerned, the diagrammatic representation would be very different to the Boolean diagram shown above. It would appear thus:



This sort of relationship is therefore not conducive to Boolean type searches but requires some more explicit form of hierarchical classificatory facility.

The distinction between the two kinds of inter-term relationship described here equates with the syntactical (or posteriori) and thesaural (or priori) relationships described in the international standard ISO 2788-1986. The thesaural, or priori, relationship, this standard states, 'adds a second dimension to an indexing language' and 'the effectiveness of a subject index as a means of identifying and retrieving documents' in any system (including 'those systems in which a computer is used to store and manipulate terms or to identify documents associated with terms') 'depends upon a well-constructed indexing language' (International Standard for Organization, 1986). The use of classification is essential for efficient subject access.

Use of the classification schedule

So far in this chapter we have been concerned with searching under alphabetical terms but it is, of course, also possible to make use of the classification *schedules*. Notation can help solve terminological problems and allow complex subjects to be represented by simpler coding. For example, using the *Dewey Decimal Classification*, a search for 598.2 would find 'Birds', 'Ornithology' and 'Aves'. Also using *Dewey*, a search for 670.427 would equal a search for 'Mechanisation and automation of factory operations' and, using the *British Catalogue of Music*

Classification, a search for QPG would equal a search for ‘Suites for solo piano’.

When searching a system arranged according to an enumerative classification, the search will usually be made for a classification number that represents the *complete* subject. For example a search for the *Dewey Decimal Classification* number 629.4753, or the *Library of Congress Classification* number TL783.5, is a search for the subject ‘Nuclear propulsion systems for spacecraft’.

When using a faceted scheme, the search *may* be for the complete subject, for example using *Thesaurofacet* for:

RKM/SBH

where RKM is ‘Spacecraft’ and SBH is ‘Nuclear propulsion’, or a ‘string’ search (that is, a search for a specified string of characters contained within a larger string) could be made for any class number which includes a particular element, for example a search for:

RKM

The idea of searching for a particular element rather than a complete subject can be extended further. In order to understand how this can be done the reader must be aware of the difference between *pre-coordinate* and *post-coordinate* indexing. If the indexer devises an alphabetical subject entry or classification number for the complete subject, for example:

Spacecraft : Nuclear propulsion

or 629.4753

or RKM/SBH

then this is referred to as *pre-coordinate* indexing; the significant point being that the concepts, or elements, which together make up the complete subject, are coordinated, or combined, by the *indexer*.

If the indexer merely indexes the constituent components of the subject, for example:

Spacecraft

or Nuclear propulsion

or RKM

or SBH

then this is referred to as post-coordinate indexing. The components of the subject description or class number are left separate and they must be coordinated by the *searcher*. Boolean searching is a post-coordinate method, the terms 'Spacecraft' and 'Nuclear propulsion' could be coordinated, or linked, by the searcher as: 'Spacecraft AND Nuclear propulsion'. Because a faceted classification scheme provides class numbers for separate concepts rather than complete subjects, such a scheme can also be used post-coordinately and searches may be made for subjects such as 'RKM AND SBH'.

Note that a hierarchical enumerative scheme must always be used pre-coordinately but a faceted scheme may be used in either a pre- or post-coordinate manner. When using a faceted scheme post-coordinately no attempt is made to combine class numbers for constituent concepts into a composite number. Citation order and facet linking devices are therefore irrelevant (see also page 91).

Searches can be broadened or narrowed by reducing or increasing the number of elements. For example, using the *London Classification of Business Studies* (see pages 31 and 104), a search for 'Safety measures for materials handling in the explosives industry' would be for:

CMG AND JZRD AND KDQ

where CMG is 'Materials handling', JZRD is 'Safety measures' and KDQ is 'Explosives industry'.

It is then possible to broaden the search by reducing the number of elements. If a search is begun using three elements, as in the above example, each of these could be discarded in turn by searching for:

CMG AND JZRD i.e. 'Safety measures for materials handling'

CMG AND KDQ i.e. 'Materials handling in the explosives industry'

or JZRD AND KDQ i.e. 'Safety measures in the explosives industry'

Alternatively, the three searches could be combined as:

(CMG AND JZRD) OR (CMG AND KDQ) OR (JZRD AND KDQ)

The search could be broadened even further by searching for a single element, for example:

KDQ i.e. 'Explosives industry'

This illustrates the degree of flexibility that can be achieved when searching post-coordinately with a faceted scheme.

Some of the other special facilities introduced to improve alphabetical term searching in computerised systems, can be adapted for use with classification numbers. An example is the 'truncation' device, that allows searching on word stems, e.g. a search for 'Comput*' would find 'Computer', 'Computers', 'Computing', 'Computerization' and 'Computerisation'. This same device can also be used on classification numbers as a means of broadening a search. Where the previous example of 'Churches' is concerned, in the *Dewey Decimal Classification* this topic would be classified at 726.5. Truncation would allow the search to be widened progressively. Searching for:

726.5	would equal a search for	'Churches' (Buildings associated with Christianity)
726	"	'Religious buildings'
72	"	'Architecture'

Thus, if a classification scheme is hierarchical and the notation is expressive, the search can obviously be broadened, narrowed or widened in a similar manner to that explained above for a thesaurus.

Chain procedure

The above type of search relates to what Ranganathan referred to as a 'chain', that is, the classification hierarchy for a subject, working through from the general to the more specific. Such a chain can play a significant part in the search process, not only by the use of classification numbers but also by means of alphabetical entries derived from the classification number by the 'chain procedure' method. This method can be applied not only to hierarchical classification with expressive notation but also to faceted classification and non-expressive notation. For example, using the *Universal Decimal Classification*, the classification number for the subject 'Public health aspects of petroleum pollution of sea water' would be:

614.777(26):665.6

This number clearly is not expressive but a hierarchical 'chain' can still be constructed, that is:

6	Technology
61	Medicine
614	Public health
614.7	Pollution

614.777	Water pollution
614.777(26)	Seas
614.777(26):665.6	Oil. Petroleum

Alphabetical subject index entries can be produced from this chain by beginning with the last, or most specific 'link' and proceeding step by step back through the chain, qualifying where necessary by a more general term or terms to indicate the subject context:

Petroleum: Sea water pollution: Public health	614.777(26):665.6
Oil: Sea water pollution: Public health	614.777(26):665.6
Sea water pollution: Public health	614.777(26)
Water pollution: Public health	614.777
Pollution: Public health	614.7
Public health	614
Medicine	61
Technology	6

The process of qualification would lead to the production of a relative index (see pages 42-3) to the classified arrangement, for example:

Petroleum: Economic geology	553.982
Petroleum: Mining	622.323
Petroleum: Sea water pollution: Public health	614.777(26):665.6

Because of problems relating to terminology and 'missing' or 'false' links, this method is not purely mechanical but semi-mechanical in that some adjustment to the chain, as derived from the classification, may be required. Nevertheless it does provide a means of producing a specific alphabetical entry for a subject, based upon the classification schedule in use, which will indicate the context in which the subject is treated. Entries for related aspects of the same subject can also be pinpointed.

Chain procedure has been applied to a number of information systems, especially in libraries and information services, and it may also be used, either consciously or unconsciously, for book indexing.

The prime example of the successful use of chain is probably the *British National Bibliography (BNB)*. The *BNB* is a weekly list which aims to include all new and forthcoming books newly published in the United Kingdom and Republic of Ireland. Works are arranged by the *Dewey Decimal Classification* and name, title and subject indexes to this classified sequence are provided. Chain procedure was used to produce the printed subject index between 1950 and 1970.

PRECIS and COMPASS

When the British Library decided to produce the *BNB* by computerized methods, chain was replaced by PRECIS in 1971. PRECIS was independent of any particular classification scheme but, nevertheless, it had its roots in classification research and was founded on classification principles.

PRECIS is an acronym for PREserved Context Index System and this conveys the intention of allowing a user to enter an alphabetical subject index at any one of the significant terms which together make up a compound subject statement and establish at that point the full context of the subject which contains the selected term:

Aircraft

Engines. Design	629.1343532
-----------------	-------------

Similar entries would be made under 'Engines' and 'Design'. In addition, the system provides for the automatic production of references to link synonymous or related terms.

Due to cost-effectiveness issues and the need for an interactive mode facility (PRECIS had as its main aim the production of a printed output), a simplified system, COMPASS, was developed and introduced to replace PRECIS in 1990. Although simplified, COMPASS entries retained the essential appearance of PRECIS entries.

Library of Congress Subject Headings

In 1995 the British Library reinstated *Library of Congress Subject Headings* (see also pages 108-9 and 125) to its *BNB* records as the means of controlled language subject access and COMPASS was dropped in 1996. The printed subject index to the *BNB* also uses Library of Congress headings and *BNB* software arranges the entries producing a lead term followed by the strings in which it occurs, for example:

Minorities

Minorities. Civil rights	305.488
Minorities. Education	370.117
Minorities. Legal status, laws, etc.	364.089
Minorities. United States. Political activity	327.73

This arrangement, however, decrees that the classified section may need to be consulted in order to ascertain the context of the various distributed relatives as the following entries illustrate:

Business ethics

Business ethics	174.4
Business ethics	303.34
Business ethics	657.45
Business ethics	658.408

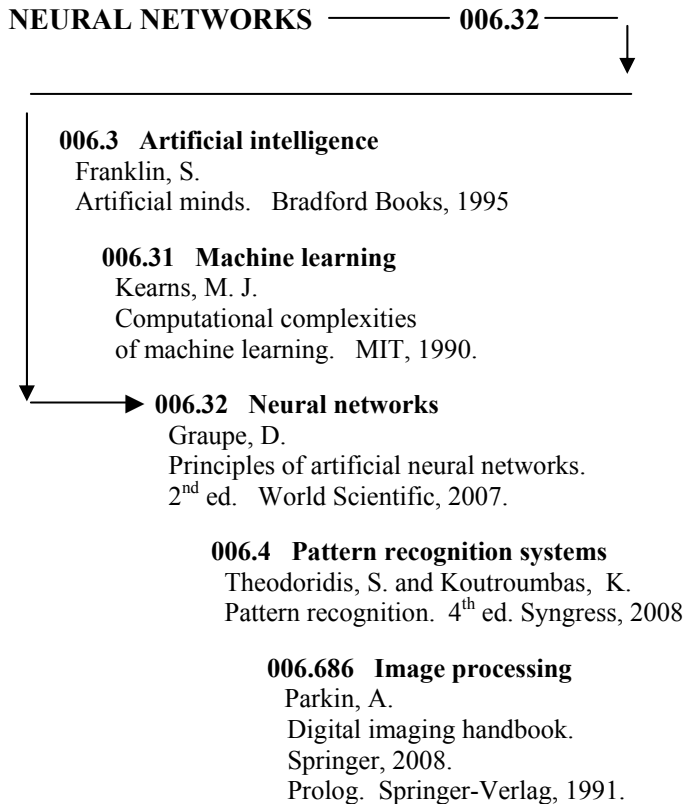
The document specificity allowed by PRECIS compares favourably with the broad subject heading approach of the Library of Congress list (see, for instance, Wellisch, 1977). However, the prohibitive economic and administrative implications, and the understandable, entrenched attitude of the American library community, precluded any movement away from what is the most comprehensive file of subject headings in existence.

Library of Congress Subject Headings, which reflect Cutter's rules (1904), has been the bedrock of United States practice for over one hundred years. The list consists of an alphabetical listing of subject headings, cross-references and subdivisions (a total of 280,000 in the 30th edition 2007). Although the Library of Congress maintains and edits its authority file, the list has become an increasingly international standard in recent years due to the contribution of new subject heading proposals from libraries across the world'.² The printed edition is in five volumes (known as the 'red books') and the list is available online via 'Classification Web'.³ Also available online is Library of Congress Authorities,⁴ which enables one to search, browse and view authorities for subject and other headings from the Library of Congress online catalog.

Systems with multiple search options

The arrangement of the *BNB*, that is a classified sequence with alphabetical indexes, is the sort of arrangement that was used for many years with great success in the catalogues of the majority of British libraries. Such a system is designed to cater for both an alphabetical subject approach or a classified approach.

As can be seen from the diagram on the next page, one of the basic features which the traditional, manual, classified catalogue offers is the ability to look in an alphabetical subject index for the classification number for a particular subject and then to go to that number in a classified sequence to find relevant items. Not only is this an efficient way to find all of the items dealing with a specific subject but it also facilitates browsing for coordinate, subordinate and superordinate related subjects.



Computerised catalogues are now the norm, giving the user online access (OPAC = Online Public Access Catalogue). However, in the early stages, many researchers concluded that the online catalogue, despite its numerous virtues, had not improved *subject* access. Some OPACs offered nothing more than a ‘keyword from title’ facility. Searching under keyword from title obviously requires the title words to reflect the subject clearly. Too often this does not happen; the title *Starting from scratch* is, for example, concerned with teaching methods. Such a title requires ‘enrichment’ by the addition of appropriate subject terms. Other subject methods tried were little better and, eventually, a number of libraries, including, for instance, those of the Liverpool City Libraries, the University of Warwick and this writer’s own Liverpool John Moores University, discovered that the original classified catalogue arrangement could offer a useful methodology for the online catalogue. Unfortunately (in this writer’s view), this is no longer the case, the ‘classified catalogue’

approach has 'been almost forgotten by contemporary librarians' (Bland, 2008). Searches by author, title and keyword are now commonplace, with more advanced searches by subject, class number, or other access points possible in many cases. However, the ability to browse 'up and down' a classified list of items is rarely offered, although 'recent developments indicate that this is changing as libraries seek to enhance the capabilities of their online catalogs' (*ibid*). One library catalogue that does allow such browsing is the University of Liverpool Library.⁵

The University of Liverpool Library uses the *Library of Congress Classification* and *Library of Congress Subject Headings*. Searches can be carried out using either of these tools. When searching for a class number, the user enters as much of the class number that he/she knows and the system will display the section of the catalogue around that number. A search for TH6000, for example, will result in a section of the catalogue being displayed as shown below (simplified for illustration purposes, the actual result would also give site locations and enable required records to be marked and saved):

TH5606.P85	
Carpentry and joinery : work activities /	
Brian Porter, Reg Rose	2000
TH5608.7.B87	
The genius of Japanese carpentry : the secrets	
of a craft	1989
TH5667.B63	
Stairs / Alan and Sylvia Blanc	2001
TH5667.H11	
Stairs / design and construction / Karl J. Habermann	
2003	
Your entry TH6000 would be here	
TH6010.G7.E51	
Co-ordination and components in housing : the	
dimensional framework and component size,	
addendum	1970
TH6010.H17	
Essential building services and equipment / F. Hall	1995
etc.	

The user can browse through the displayed records and can scroll up and down the classified listing using a 'Prev / Next' page facility.

This example demonstrates forcibly the value of using a classified approach to improve access. In the United States the Western North Carolina Library Network (WNCLN) has also developed a 'classification browse' feature.⁶ This again makes use of the *Library of Congress*

Classification. One might choose ‘U Military science’ from the top level display of LC classes, then ‘UD1 Infantry’ from the second level and ‘UD 480-485 Airborne troops. Parachute troops’ from the third. A list of the most used titles in UD 480-485 would then be displayed:

- 1 Assault from the sky / John Weekes ...
- 2 Airborne : the history of American parachute forces / Edwin P. Hoyt ...
etc.

This is not to say that the classified approach should necessarily be *the* approach in an online catalogue. The computer has given us a wonderful, flexible capability that offers search possibilities that were impossible previously and we must make full use of all of these ‘added value’ facilities. However, to ignore classificatory techniques is to ignore one of the most powerful access tools that we possess.

The University of Liverpool catalogue also provides access by *Library of Congress Subject Headings*. A search for ‘Nuclear reactors’, for instance, would result in a display of relevant subject headings (again simplified for illustration purposes; the number of items indexed under each heading is included together with other information):

Nuclear reactors
Nuclear reactors Accidents
Nuclear reactors Congresses
Nuclear reactors Containment Painting Handbooks Manuals etc
Nuclear reactors control
Nuclear reactors cool
... and so on

Using a thesaurus as an aid to searching online catalogues

A further useful illustration of the way on which a thesaurus may be used in online catalogue searching is provided by the *African Studies Thesaurus*.⁷ This is a structured vocabulary of 12,100 (2008) English terms in the field of African studies, developed and maintained by staff at the Library of the African Studies Centre Leiden. It is used for indexing and retrieving material in the library collection and is directly linked to the catalogue. If, for instance, one searches in this thesaurus for ‘dairy industry’, the following entry appears:

dairy industry
Search catalogue
Used for
dairy products industry
Broader terms

food industry
Narrower terms
cheese industry
Related terms
dairy farms
milk

If 'dairy industry' satisfies the user requirement, then a search of the catalogue could be done immediately but, if one of the other terms is more appropriate, then the examination of the thesaurus can continue. 'Milk' might be more suitable and selecting that term will produce:

milk
Search catalogue
Broader terms
beverages
Related terms
dairy industry

Selecting Search catalogue at this point will reveal a list of items of dealing with the subject 'milk' which may be of relevance. The first of these, for example, is:

Stationarity tests in geographic markets : an application
to South African milk markets. William H. Boshoff /
In: The South African journal of economics / 2007

Clicking on this catalogue entry will give full details of the item and an abstract.

Using the complete classification scheme for searching

When searching via a computerised online catalogue system, options are often provided via a 'menu':

Enter your search in the appropriate box:

Author:

e.g: Underdahl Brian

Title:

e.g: Teach yourself Microsoft Office 2000

Keyword:

e.g: Office

When the record which describes an item is retrieved, it will include details such as author, title, publisher, date, series (if any), and a class number. But what if the record contents were expanded to include the subject terms which are shown alongside the class number in the classification schedules and the relevant subject index entry from the scheme's index? It then becomes possible to make use of the complete classification scheme for online searching.

Using the *Dewey Decimal Classification* as an example, a record for a bibliographic item might contain the following fields:

Class number	623.89
Dewey subject	Nautical engineering and seamanship – Navigation – Selection and determination of course
Dewey index	Navigation – Technology – Seamanship
Author	Taylor
Title	The geometrical seaman

The user would be able to conduct a search in various ways:

- 1 By class number, for example:
623.89
- 2 By pre-coordinate phrases from the alphabetical subject fields, for example: Nautical engineering
- 3 By 'keyword' from the alphabetical subject fields, for example:
Navigation
or
Seamanship

The latter facility would permit 'users to search systematically and receive direction from the system as to fruitful areas of the classification where there are items matching the entered keywords' (Markey, 1986).

In addition the user could browse backwards and forwards through the index to the system, for example:

Navigation – aquatic sports	797.1
Navigation – maritime transportation	623.89
Navigation – maritime transportation – law	343.096 6
Navigation – space flight	623.89
Navigation aids	387.155
Navigators	629.045092

This would assist in the identification of related topics that might also be of use.

Conclusion

Clearly, classification can play an important role in searching, some aspects of which have been described here. There is obviously considerable potential and, where online retrieval systems are concerned, Svenonius (1983) envisaged eight uses of classification:

In areas of knowledge admitting of natural taxonomies, classification can be used to improve recall and precision and to save the time of the user in keying in search terms. In other areas of knowledge, perspective hierarchies can be used to contextualise the meaning of vague search terms, enabling the computer to simulate in part the negotiation of a search request carried out by reference librarians. An important use of traditional classification in online systems is to provide a structure for meaningful browsing. Classification can be used to provide a framework for the representation and retrieval of non-bibliographic information, e.g. statistical data. Automatic classification can be used to collocate citations in ways not possible in manual systems, e.g. by similarity of linguistic features. Finally, classification can be used to achieve compatibility of retrieval languages by serving as a mediating or switching language.

A full explanation of certain possibilities mentioned, that is switching languages and automatic classification, is outside the scope of an elementary introductory text. However it may be of benefit to the reader to provide some brief information.

The major objective of a switching language is to enable indexing decisions recorded in one indexing language to be 'translated' into equivalent decisions in another. This is of particular concern when the information systems to be searched are multilingual. Classification schemes such as the *Dewey Decimal Classification* or the *Universal Decimal Classification*, which have been translated into many languages and which make use of an 'international' language, that is, numeric digits, as notational symbols, immediately spring to mind as possible candidates. 'User's search terms entered into the system in one language can be switched through DDC [or UDC] numbers to retrieve documents in several different languages' (*ibid.*). Since this statement was made, machine-readable versions of these schemes have become available, thereby increasing their possible usefulness in a switching role. There is one general classification scheme, namely the *Broad System of Ordering* (see pages 33-5) which has been specifically designed for information exchange and switching and the *BSI Root Thesaurus* (see page 106) 'has shown that that a multilingual thesaurus is feasible — if the relationships are shown by symbols rather than letters' (Foskett, 2000).

By removing the human intellectual effort from indexing and allowing the computer to select suitable terms, a form of automatic indexing is obtained. Sparck-Jones (1976) has stated her belief that large files need more sophisticated tools and, since automatic classification has not been conclusively shown to be unhelpful compared with human classification, its longer-term prospects should be of interest to all concerned with information systems. The Internet has acted as a catalyst to revitalize this interest (see pages 135-6).

To conclude, the search methodology chosen should be one that is most appropriate for the particular purpose and which is of the greatest assistance to the user. In current online systems subject searching is usually done via alphabetical keywords or subject terms. However, it is now becoming more widely accepted that a greater use of classification in searching could lead to an improvement in search technique and in system efficiency.

Notes

- 1 Where, historically, the United Kingdom tended to adopt a classified approach to the catalogue, in the United States the 'dictionary' catalogue, with author, title, subject and other entries filed in one alphabetical sequence, has been pre-eminent, influenced by the pioneering work of Charles A Cutter well over 100 years ago. The dictionary catalogue makes use of *Library of Congress Subject Headings* and, from the 13th edition, 1990, this list adopted a revised format using the accepted thesaural abbreviations of BT, RT, etc. The list was criticised severely for this move; 'the fact of the matter is that the LCSH list is not a thesaurus, not any more now than it ever was' (Dykstra, 1988). A thesaurus differs from a conventional list such as the LC list in that a listed term need not necessarily be used alone but may be coordinated with other terms. LC headings such as 'Cosmetics for men' or 'Cosmetics containers' or 'Video games and children' would be listed as separate concepts in a true thesaurus. The thesaurus also defines relationships between terms more clearly. Dykstra claims that 'North America has never bothered to grasp the significance of the research on analytico-synthetic classification that has been undertaken in the rest of the world' (*ibid.*)
- 2 <http://www.bl.uk/services/bsds/nbs/subject.html>.
- 3 <http://classificationweb.net>.
- 4 <http://authorities.loc.gov>.
- 5 <http://library.liv.ac.uk>.
- 6 bullpup.lib.unca.edu/scripts/lcclass/p2..htm.
- 7 www.ascleiden.nl/libraryAfrica.

References

- Bland, Robert N. and Stoffan, Mark A. (2008), 'Returning Classification to the Catalog', *Information Technology and Libraries*, (Sep), pp. 55-60.
- Burton, Paul (1997), 'The Decline and Fall of "Cat. and Class."', *Catalogue a Index*, 124, p. 9.

- Cutter, Charles A. (1904), *Rules for a Dictionary Catalog*, 4th ed., GPO, Washington.
- Dykstra, Mary (1988), 'LC Subject Headings Disguised as a Thesaurus', *Library Journal*, 113 (4), pp. 42-7.
- Fosket, A.C. (2000), 'The Future of Faceted Classification', in Rita Marcella and Arthur Maltby (eds.), *The Future of Classification*, Gower, Aldershot, Hants; Brookfield, Vt., pp. 69-80.
- Hildreth, Charles R. (1989), *The Online Catalogue: Developments and Directions*, Library Association, London, p. 19.
- International Organization for Standardization (1986), *Documentation: Guidelines for the Establishment and Development of Monolingual Thesauri*, 2nd ed., The Organization, Geneva, (ISO 2788), p. 1.
- Library of Congress (2008), *Subject Headings*, 31st ed., LC, Washington, 5 v. The examples included here have been checked in 2008 against 'Classification Web', available from the Cataloging Distribution Service of LC at <http://classificationweb.net>.
- RCN Library Nursing Thesaurus: a Thesaurus of Terms Used in Nursing, Midwifery, Health Visiting and Related Subject Areas* (2007), 5th ed., RCN [Royal College of Nursing] Library and Information Services, London.
- Svenonius, Elaine (1983), 'Use of Classification in Online Retrieval', *Library Resources and Technical Services*, 27 (1), pp. 76-80.
- Schneiderman, R. Anders (1997), 'A Non-librarian Explains Why Librarians Rule the 'Net'', *Information Outlook*, 1 (4), 34-5.
- Sparck-Jones, Karen (1976), 'Automatic Classification', in Arthur Maltby (ed.), *Classification in the 1970s: a Second Look*, Bingley, London, pp. 209-225.
- Wellisch, Hans H. (ed.) (1977), *The PRECIS Index System: Principles, Applications and Prospects*, H.W. Wilson, New York.

14 Classification and the Internet¹

In 1996, the two major trends for the accessing of Internet resources were identified by Dodd as (i) search engines that look for matches on keywords and (ii) the subject-oriented, hierarchical, classification system. 'The explosion of the use of the Internet, particularly via the World Wide Web, has given rise to an interesting phenomenon: the proliferation of semi-professional attempts to give some subject-based access to Internet resources via hierarchical guides'. More recently, Saumare (2008) notes that 'there is a sense that faceted classification is [also] becoming a key means of organising information on the Web'.

Use of classification by search engines – Google² and Yahoo³

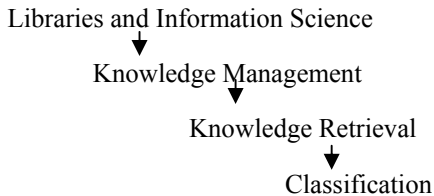
From the beginning Internet search engines recognised the fact that some form of classification could be used for searching. Google and Yahoo are now, of course, far more than just search engines; these portals provide a wide array of products. Google is 'a giant of the Web' and 'straddles everything from search to e-mail, video to social networking' (Beamont, 2009). As a search mechanism, at the present time Google is probably by far the most popular of such services but how does Google make use of classification? At first glance one might not think that it does, as Google's entry screen requires the input of alphabetical search terms. By default, Google usually returns pages that contain all of the words one searches for. A search for 'plastic bottles', for example, will return pages that contain the words 'plastic' AND 'bottles'. This is an implicit Boolean 'AND' search. As demonstrated in Chapter 13 (see pages 110-1), in this type of searching the characteristics or context of the required subject are being identified and this involves quite clearly elements of classification. If, for instance, one is looking for information on the god 'Vulcan', and a search is made on Google for that name, many of the upwards of 350 thousand sites found will be irrelevant as they are concerned with the Vulcan bomber, Vulcan Engineering, Vulcan fireworks, the Vulcan Riders Association, and so on. However, if the search is amended to 'Vulcan mythology', then the search now relates to 'Vulcan' solely in the context of the class 'mythology'; irrelevant sites will be avoided and all of the sites found should be of interest. 'OR' searches can be undertaken on Google by including the word OR between search terms. 'AND', 'OR' and 'NOT' searches can also be done via Google's Advanced Search.

More directly, Google makes use of classification on the Google Directory⁴ which organises the Web by categories:

Arts Movies Music Television	Home Consumers Homeowners Family	Regional Asia Europe North America
Business Industries Finance Jobs ...	Kids and Teens Computers Entertainment School	Science Biology Psychology Physics
Computers Hardware Internet Software	News Media Newspapers Current Events	Shopping Autos Clothing Gifts
Games Board Roleplaying Video	Recreation Food Outdoors Travel	Society Issues People Religion
Health Alternative Fitness Medicine	Reference Education Libraries Maps	Sports Basketball Football Soccer

One could, for example, select 'Reference' from the above list. This section contains a number of subordinate categories, one of which is 'Library and Information Sciences'. When this is selected, a further list is presented which includes 'Knowledge Management'. From the list under 'Knowledge Management', 'Knowledge Retrieval' might be chosen and, from the list under 'Knowledge Retrieval', 'Classification'.

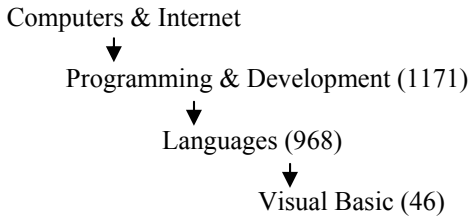
An examination of this search shows that the searcher is being guided through a hierarchical structure (indentation indicating the subordinate topics):



This categorisation can be extremely useful for its 'classificatory' nature concentrates the search on the appropriate subject area. Compare this with a search on the keyword 'Classification' in the main Google home page which, currently (2009), produces over 100 million hits, many of which are irrelevant to the required topic!

Other search engines provide a similar categorisation facility. Yahoo,⁵ for instance, has a subject-based directory listing Websites in a wide range of topics, from arts, entertainment, society and culture, to science, education and health.

If one was looking on Yahoo for Websites covering the programming language Visual Basic, 'Computers & Internet' would be selected from the top level. As with Google, the search could then progress through the hierarchical structure via subsidiary lists:



The numbers in parentheses indicate the number of relevant sites. One of the sites listed, for example, under Visual Basic is VB Web (www.vbweb.co.uk), that presents downloads, tutorials, source codes and news relating to this programming language

Dodd (1996) is perhaps right to describe some of these attempts at categorisation as ‘semi-professional’. Although the hierarchical structures do support subject browsing, the nature of the ‘classification’ does not appear to be as systematic as can be found in the more traditional, established schemes; frequently cross-classification is apparent and a further possible disadvantage is that they do not have a notation. Whilst, as has been pointed out earlier in this text, a notation is not absolutely necessary for a system of classification to work, it can provide added value.

Use of conventional classification schemes

Is it possible, then, to utilise conventional schemes such as the *Dewey Decimal Classification* or the *Library of Congress Classification* for searching the Web? The answer, of course, is a qualified ‘Yes’, although services which make use of these schemes are usually non-commercial, much more restrictive than the search engines referred to above, and possibly experimental. In the early days of the Web, classification schemes such as *UDC* were used as organisers for many subject gateways but, by 1904, Broughton was claiming that they had ‘largely been superseded in this role’. Various reasons can account for this: lack of funding; difficulty of keeping up with the growth of the Web; change to another form of indexing; collaboration and merging of subject gateways and portals; and so on. In the previous edition of this present work, for instance, NISS (National Information Services and Systems) was cited as an example of a service which used *UDC* as a subject gateway to its Directory of Networked Resources but NISS became part of HERO (Higher Education & Resource Opportunities) and ceased using *UDC* in 2003.

However, there are still a number of services that use traditional schemes. The *Dewey Decimal Classification* is used by ‘Webrary’ and

‘BUBL’. The *Library of Congress Classification* is used by ‘Cyberstacks’ and the ‘Scout Report Archives’ uses *Library of Congress Subject Headings*.

Webrary⁶ is a service provided by the Morton Grove [US] Public Library. The Webrary Links Menu consists of links to what are claimed to be the most useful reference and informational Web sites, organised by *Dewey* class numbers. Selecting one of the ten main classes of *Dewey* provides a breakdown for that class, e.g. for class 000:

000-	010-	020-	etc
Computers	Bibliography	Library Science	

Upon selecting 020, two of the first relevant sites named are those of the ‘American Library Association’ and the ‘Library of Congress Information System’.

The **BUBL**⁷ Information Service for the library and higher education community also uses the *Dewey Decimal Classification* system as the primary organisation structure for its catalogue of internet resources. It is run by the Centre for Digital Library Research at the University of Strathclyde. The name was originally short for the Bulletin Board for Libraries. The search method is similar to that described for Webrary. The home page shows the main ten divisions of *Dewey* from which you select the class that you require. This will give you a breakdown of the divisions in that class from which you can make a further selection and so on. For example one might select 200 Religion and then 220 Bible and then 220.3 Encyclopedias and topical dictionaries. One of the resources given at this number is the internet version of Easton’s 1897 *Bible dictionary*, which defines unusual or religious terms from the scripture.

Cyberstacks (sm)⁸ is a collection of significant World Wide Web and other Internet resources categorised using the *Library of Congress Classification*. As with Webrary, resources are categorised firstly within a broad classification, e.g.

G	Geography, Anthropology and Recreation
H	Social Sciences
J	Political Sciences
. . .	

then within narrower subclasses and finally under a specific classification range and associated subject description. For example, ‘T Technology’ might be selected from the main menu, then ‘TL Motor vehicles, Aeronautics and Astronautics’ from the sub-classes, and finally ‘TL 787-4050 Astronautics’ from the specific classification range. A relevant

resource listed here is 'NASA Astronaut Biographies'. For each resource, a brief summary is provided and, when necessary, instructions on using the resource. At present, CyberStacks(sm) is a prototype demonstration service and is limited to significant Internet resources in selected fields of science and technology.

The **Scout Report Archives**,⁹ located in the Computer Sciences Department of the University of Wisconsin, is a searchable and browsable database of nearly nine years of the *Scout Report*. Searches may be made under keyword or a more advanced search may be made in other fields such as title, author, publisher or *Library of Congress Subject Headings*. If, for instance, 'D' is selected from the 'Browse by Library of Congress Subject Headings' alphabetical array, a list of headings beginning with that letter is displayed. Among these will be found 'Dinosaurs (67)'. This indicates that there are 67 resources with subject headings that start with the term 'Dinosaurs'. Selecting this entry will reveal a list of 'Resources (19)' entered directly under 'Dinosaurs' and further 'Classifications', or subheadings, such as 'Bibliographies (4) ... Databases (1) ... and Study and Teaching (15)', relating to other resources. One subheading listed here is 'Cardiovascular system (2)' and selecting this will, for example, reveal the site 'Willo: the dinosaur with a heart', that refers to the discovery of a fossilised dinosaur heart. At the time of writing, the Scout Report Archives contain over 25,000 critical annotations of carefully selected Internet sites.

In certain subject areas, the use of special rather than general classification schemes or lists of subject headings may be more appropriate. Among the options offered by, **Intute: Social Sciences**,¹⁰ for instance, is an online thesaurus. This is a service which aims to provide details of the very best Web resources for education and research in the social sciences. The thesaurus engine helps to indicate alternative search terms to the one selected, organised into a hierarchy of relationships. For example, searching for 'Offences' will produce the following suggested matches:

drinking offences [0]
driving offences [4]
offences [78]
political offences [2]
sexual offences [19]

At this point, one may examine the items indexed under a particular term (the number being indicated in parentheses) or continue to consult the thesaurus. Clicking on 'sexual offences', for instance, will give the following:

sexual offences [16]		
Broader terms	Narrower terms	Related terms
<u>offences</u> [65]	<u>pornography</u> [15]	<u>prostitution</u> [38]
	<u>rape</u> [44]	<u>sexual behaviour</u> [21]
	<u>sexual harassment</u> [29]	<u>gender</u> [730]
	<u>child sexual abuse</u> [30]	<u>behavioural disorders</u> [5]

Again, one has the option of examining the items indexed or continuing to consult the thesaurus. If, at this point, it is decided that ‘behavioural disorders’ is the term required, selecting this will reveal the five items. These include, for instance, details of the Website of the Association for the Advancement of Behavioural Therapy.

It should be noted that services such as Webrary, BUBL and the Scout Report Archives offer other facilities such as searching by keywords. However, as stated above, these services are far less exhaustive than commercial search engines. They are more properly described as search *directories* or *catalogues* rather than search *engines*. The latter attempts to seek out as many as possible of the Web pages that match a particular search criteria. The former is more selective in that it is limited to a particular directory’s database; sites are usually categorised and therefore this type of directory is more likely to utilise hierarchically organised and cross-referenced structures. The difference in coverage can be illustrated by the fact that the 38 ‘hits’ for ‘prostitution’ on the Intute: Social sciences service compares with over 16 million on Google!

Online Public Access Catalogues (OPACs)

As described in the previous chapter, most libraries now have online public access to their catalogues. Many of these catalogues are available on the Internet. The University of Liverpool catalogue¹² (see pages 119-120) and the catalogue of the Library of the African Studies Centre Leiden¹³ (see pages 121-2) are two examples. Thus is it possible to sit at home and use one’s computer to search a great number of systems for citations and availability of required documents. In many instances, classification plays a central role in this search process. Some institutions provide merely a version of the in-house system. Other institutions are more innovative. The London Business School, for example, provides ‘Concept space’¹³, a visual search tool for business concepts linked to a wide range of information sources. This is a ‘point and click’ system based upon the *London Classification of Business Studies* (see pages 31-3, 104-5 and 114). It is possible to navigate through the system using a ‘graphic’ view or a ‘text’ view. Colour coding is used in both views, e.g.:

red = entry term; green = broader term; blue = narrower term; purple = related term.

Having found a relevant term in either view, the user can click on the 'Search related sites' link in order to obtain information on where to find books, articles, academic information, related companies and more.

Another institution, the North Carolina State University¹⁴ uses a 'Guided navigation' system, designed by Endeca, which it is claimed 'lets you search its libraries' collections using faceted classification'.¹⁵ However, this is not faceted classification as described in this text. Endeca, itself states that it only 'resembles' what librarians call faceted classification.¹⁶ The 'facets' include 'topic', 'author', 'genre', 'language', 'format', 'material type' and 'availability'. The 'topic' facet does allow a search to be narrowed to a more relevant subject area but use is then made of the far from 'faceted' *Library of Congress Subject Headings*. For example, the topics revealed by a search for 'Roses' include: 'African Americans', 'Architecture', 'Awards', 'Catholic Church', 'Children', 'Children's poetry', and so on.

Nevertheless, this is a sophisticated system which is said to provide 'the speed and flexibility of popular online search engines while capitalizing on existing catalog records'.¹⁷

Classification of electronic documents

MacLennan (2000) considers that if Internet resources were adequately classified there seems every probability that schemes such as *Dewey* and *Library of Congress* could provide adequate access. We have seen how these schemes can be used for directories and catalogues but is it feasible that *all* electronic documents carried on the Internet could be classified in the same way as the items in a conventional library are classified, thus permitting a particular class number to be used by a search engine as the search criteria? In order to achieve this, the relevant classification information would need to be carried within the document itself. A metadata (data about data) standard that could be used for this purpose is the Dublin core.

The Dublin Core

The Dublin Core Metadata Element Set (DCMES) is a system that facilitates the inclusion of tagged description and identification within Web documents. It provides a simple and standardised set of conventions for describing documents stored online in a way that people can understand, in order to make them easier to find. The basic Dublin Core Metadata (data about data) consists of fifteen 'properties' which comprise

a standard data set for describing electronic documents. The name “Dublin” is due to its origin at a 1995 invitational workshop in Dublin, Ohio. It is maintained by the Dublin Core Metadata Initiative (DCMI). The term ‘core’ is used because its elements are broad and generic, usable for describing a wide range of resources. These elements include the ‘creator’, ‘title’, ‘date’, ‘format’ and so on. One of the fifteen elements is the ‘subject’ and, typically, this will be represented using keywords, key phrases, or classification codes. It is the latter that is of interest here as it would enable the inclusion of classification numbers from a scheme such as *Dewey* or *Library of Congress* to be added to a document, which could then be retrieved with a classification number search. Whilst this is therefore theoretically possible, the problem is one of number, there are many, many millions of electronic documents on the Internet and the logistics are daunting.

In relation to the Dublin Core, readers might well come across a variety of terms and abbreviations. For example, implementations of the Dublin Core typically make use of XML (eXtensible Markup Language) and also RDF (Resource Description Framework). XML is a specification for computer based documents; it provides a syntax for describing *how* data is presented rather than *what* data is presented. RDF is used to present data; it provides a standard model to describe Web resources. More information is provided about XML and RDF in the section on the Semantic Web (page 140).

Classification schemes as aids to searching

Despite the difficulty of including classification numbers in documents, there is the possibility of using a general classification scheme such as *Dewey* or *Library of Congress* as an *aid* to searching. There are machine-readable versions of *Dewey* and *Library of Congress* which could assist here and, if the complete scheme was considered to be too detailed, outlines of these schemes are available.

Schemes are also available online on the Web. ‘*WebDewey*’¹⁸ is provided by OCLC; ‘*UDCOnline*’¹⁹ by the British Standards Institution; and ‘*Classification Web*’²⁰ (giving access to the *Library of Congress Classification* and *Library of Congress Subject Headings*) by the Library of Congress Cataloging Distribution Service. Special systems such as the *ACM Computing Classification*²¹ (see pages 48-9) are also available on the Web.

Both *Dewey* and *Library of Congress*, as the reader now knows, are *enumerative* schemes. One *faceted* classification based on modern theory, which has been advocated for use as an aid to searching is the *Broad*

System of Ordering (BSO) (see pages 33-5). The following description attempts to explain how the scheme might be used:

For an Internet searcher who has exhausted keyword and hypertext link search methods and needs to search further the *Broad System of Ordering (BSO)* offers a subject contextual reference framework against which to plot a comprehensive and systematic search strategy. It is particularly useful in clarifying decision-making in searching for subjects which can be named only by phrases rather than single words.

When your search question dives straight into the haystack and instantly retrieves the sought needle, then you don't need *BSO*. At other times searching in a large information store or network may all too readily bring the needle-in-haystack problem to mind. You may draw a blank or be presented with an offering which is not quite what you were looking for. At such times *BSO* could often help in suggesting alternative search approaches. Similarly, if you are not quite sure of the appropriate subject word to begin your subject search with, a glance at the index and systematic schedules of *BSO* could help to set you on your way.²²

BSO can also be consulted online.²³

Automatic classification

As noted above, for a classifier to sit down and classify all of the electronic documents that appear on the Internet in the same way as this might be done in a library or information service clearly is not feasible. However, is it possible to remove the human intellectual effort and allow the computer to assign documents to appropriate subject categories *automatically*? Is there a role for data obtained automatically from source documents to be used in the classification process? Is it possible to automatically identify characteristics that a document should have in order to place it in a particular category or class. Whilst there do not appear to be any practical, 'real-life' examples of this at present, research into automatic classification has been going on for a considerable length of time. This research has assumed a greater importance as the need to improve access to Internet resources has intensified. Internet users can often be irritated and frustrated by a search engine's tendency to produce a vast plethora of results from which relevant information has to be sifted.

A number of projects have attempted to examine whether automatic classification can be used to improve the above situation. Various algorithms and sampling techniques have been used in investigations aimed at discovering whether automatic classification is a practical proposition.

One method, for instance, entails the statistical analysis of the way in which terms co-occur in documents. 'Documents that share the same frequently occurring keywords and concepts are usually relevant to the

same queries. Clustering such documents together enables them to be retrieved more easily and helps to avoid the retrieval of irrelevant information' (Jenkins, 2001).

There is also the possibility of a statistical comparison of citations. Documents that cite a similar set of citations or documents that are cited by a similar set of documents are clearly subject related.

In the United States, the OCLC Office of Research has undertaken research into whether standard library classification schemes can be adapted to automatically classify materials, especially Web resources and other digitised electronic documents.²⁴ OCLC has also investigated whether there is a role for indexes and topic maps (see also page 141) that are obtained directly from source documents. The OCLC Scorpion project²⁵ offers software that implements a system for automatically classifying Web-accessible text documents. Scorpion is intended for use by investigators who have a machine-readable subject classification scheme (such as the *Library of Congress Classification*) or thesaurus and wish to incorporate it into an automatic classification system. There is also the OCLC RDF Topicmaps Project,²⁶ which explores subject navigation of Web sites using semi-automatically generated finding aids.

Joanna Yi-Hang Pong and others (2008) describe a comparative study of automatic document classification methods using two well-known machine learning algorithms (the *k*-nearest neighbour (KNN) and the naïve Bayes) using the *Library of Congress Classification*.

A full explanation of these and the other research projects in this area is outside the scope of an elementary text of this nature but there seems little doubt that automatic classification can have a part to play in the provision of Internet access. Nevertheless, it is interesting to note that, in an ISKO UK (International Society for Knowledge Classification) conference in 2008, in a linked presentation on the implementation of auto-categorisation by three British Broadcasting Corporation (BBC) information architects, one of the conclusions was that 'you still need humans'! There is also the view that no single classificatory technique, whether manual or automatic, outperforms other classificatory methods in all situations and perhaps there is a need to provide different approaches which can be used simultaneously and the results combined so that users can achieve an optimum, satisfactory outcome (Blumberg and Atre, 2003).

E-commerce

The buying and selling of products and services over the Internet and other computer networks has become known as 'electronic commerce' or 'e-commerce' or 'e-business'. We have already seen that classification

has a significant part to play in this activity. Examples can be found in Chapter 6 in the classifications used by the book trade and in Chapter 9 in the codification system used by NATO.

Hierarchical classification

We saw in Chapter 1 how classification must be used in a supermarket in order to aid the shopper in the selection of goods. Although not quite so essential for the Internet shopper, nevertheless some form of categorisation is very helpful.

Ebay²⁷

Currently one of the most used buying and selling websites is Ebay. Searching can be done via a straight keyword search or by categories. If one is interested in, for example, 'Antique ceramic boxes', a search directly under those terms will reveal the number of items in all categories. Alternatively, one can select an appropriate category from the many listed on the Ebay home page:

- Antiques
- Art
- Baby
- Books
- etc

Selecting 'Antiques' will reveal a number of further categories, one of which is:

- Decorative Arts

Within 'Decorative Arts', divisions include:

- Ceramics, Porcelain
- Clocks
- Glass
- Lamps
- etc

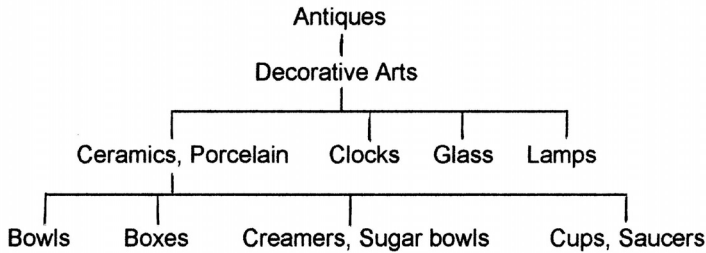
and within 'Ceramics, Porcelain' we find:

- Bowls (638)
- Boxes (176)
- Creamers, Sugar Bowls (565)
- Cups, Saucers (1933)
- etc

The numbers (which are constantly changing as items are offered or sold) refer to the number of items available.

Clicking on 'Boxes' will enable the searcher to browse through the large number of boxes.

The searcher is being guided through the hierarchical structure:



A great many retail websites offer a similar facility.

If one is interested in computing, for example, the website of the particular retail dealer chosen may well list various categories, i.e.

Computing equipment

- Desktop PCs
- Laptops
- Monitors
- Peripherals
- Printers
- etc

As with any hierarchical methodology, the selection of a particular category will reveal a further set of options.

UNSPSC

Today, 'increasingly business is e-business' and 'fast, simple, accurate classification of goods and services is imperative in the marketplace' (UNSPSC, 2009). In order to achieve this aim, the United Nations Development Programme and the Dun & Bradstreet Corporation jointly developed *UNSPSC (United Nations Standard Products and Services Code)* which can be used to classify all types of products and services. It is claimed to be 'the most efficient, accurate and flexible classification system available today for achieving company-wide visibility of spend analysis, enabling procurement to deliver on cost-effectiveness demands and allowing full exploitation of electronic commerce capabilities' (UNSPSC, 2009). Microsoft has selected *UNSPSC* as its standard commodity classification system (Turner, 2007?).

UNSPSC is a hierarchical classification of five levels. The first four levels totalling eight digits comprise: Segment; Family; Class; and Commodity. The hierarchical structure for the code 44103103, for example, would be:

Segment	44	Office Equipment and Accessories and Supplies
Family	10	Office machines and their supplies and accessories
Class	31	Printer and facsimile and photocopier supplies
Commodity	03	Toner

There is also a two digit code for 'Business function', e.g. **14** = 'Retail'.

The management and development of *UNSPSC* is coordinated by ECCMA,²⁸ the Electronic Commerce Code Management Association. A link has been created between this system and the *NATO Codification System* described on pages 76-8 and 79 (Pergolesi, 2006).

Faceted classification

Heidi P Adkisson (2003) maintains that 'unlike a single hierarchical scheme, faceted classification gives the users the ability to find items based on more than one dimension'. Adkisson claims that 69% of sites that she examined made at least some use of faceted classification. She cites jewellery as an example. Some people may be interested in browsing by particular types of jewellery (earrings, necklaces, etc.), some in a particular material and others in a particular occasion.

Below is part of the home page of the jeweller Ernest Jones:²⁹

Categories

Bangles	Bracelets
Charms	Earrings
Rings	
etc	

Material

Gold	Platignum
Silver	White gold
etc	

Recipient

For baby	For bride
etc	

Price

£50-£149	£150-£499
£500-£999	
etc	

Stone type

Cubic zirconia	Diamond
Pearl	Ruby
etc	

Brand

D&G Jewels	Emporia Armani
etc	

Occasion

Anniversary	Birthday
etc	

The user has the option of searching by any one of these categories. If ‘Anniversary’ is chosen from ‘Occasion’, then a selection of appropriate jewellery, including earrings, bangles, lockets, bracelets, etc. will be presented. A choice can then be made, if necessary, from a number of further categories. One might select ‘Bracelets’ alone, or ‘Heart’, from the ‘Style’ category, which appears at this second level, if that was the interest. The prospective purchaser can begin his/her search in any one of a number of categories and then refine the result by further categorisation.

If, for example, one was looking for ‘Gold earrings priced at £50-£149’, then one could start from ‘£50-£149’ in the ‘Price’ category and progress through ‘Earrings’ in ‘Categories’ and ‘Gold’ in ‘Material’. Alternatively, the progression could be ‘Earrings’, ‘Gold’, ‘Price’, or ‘Gold’, ‘Earrings’, ‘Price’. Any one of six different approaches could be used.

This methodology is ‘faceted’ in the sense that one can choose a focus or value from one category and then refine the search by a focus from another category. It is not a fully-fledged faceted classification scheme but it does offer a very flexible shopping experience. It has something in common with post-coordinate indexing (see pages 115-6), in that the qualities that make up a sought item are combined by the ‘searcher’ and not the ‘indexer’.

The Semantic Web

The increasing diversity of use which the Internet has fostered has led to a re-evaluation of the ways in which it might be developed.

The original concept of the World Wide Web was devised by Sir Tim Berners-Lee. The Web would not exist if Berners-Lee had not made his original Web software freely available.

HTML (Hypertext Markup Language) became the predominant language for Web pages. No doubt many readers will have come across files with an ‘HTM’ (a throwback to the time when file extensions were

limited to three characters) or ‘HTML’ extension. HTML is a means of giving a structure to text based information in a document, indicating features and links. ‘Markup’ is achieved by elements, or ‘tags’, which tell the Internet browser how to present a page. Tags are words or letters typed between two angular brackets. <html> for instance indicates the start of an html document and </html> indicates the end. classification would be the code or instruction for printing the word ‘classification’ in bold.

In 1999, ten years after the Web was conceived, Berners-Lee postulated a new concept, the ‘Semantic Web’, where computers would become capable of analysing all of the data on the Web, the contents, links and transactions between people and computers. One important requirement of the Semantic Web is the availability of languages for representing data and knowledge.

For the Semantic Web to become a reality, something more advanced than HTML was required and other markup languages were developed. One of these is XML (eXtensible Markup Language) which looks very much like HTML but is ‘extensible’ because it allows users to define their own elements and its main purpose is to allow different information systems to share structured data. A classification number written in XML might appear as: <classification>TH6010</classification> There is also XHTML, a revised version of HTML, that conforms to XML syntax.

Where XML provides a syntax to encode data, RDF (Resource Description Framework), another of the Berners-Lee’s World Wide Web Consortium specifications, can be used to model information through a variety of syntax formats. In simple terms, XML can be said to be a way of encoding data and RDF can be said to be a means of giving an interpretation of that data.

As noted above, XML allows users to create their own markup but it does not explain meanings. One way to add meanings is by a link to terminological resources such as glossaries, dictionaries, or thesauri. Because XML is tree-based it supports hierarchical collections (thesauri, etc). For non-hierarchical relationships an additional structure can be added. In addition, ‘hyper glossaries’ can be developed as an ‘umbrella’ to link various resources.

Another Semantic Web tool is the ‘Topic Map’, a standard for the representation and interchange of data, with the emphasis on how efficiently information can be found. A topic map represents informational concepts, countries, organisations, software, etc, and also relationships between topics and relevant information resources. The Topic Maps data model is covered in ISO/IEC 13250:2007. This standard

defines an XML markup language vocabulary for interchanging topic maps.

A detailed study of these recent developments in digital information retrieval is outside the scope of this introductory work. However, any student of classification must be aware of their existence so that future innovations will not come as a complete surprise.

Sir Tim Berners-Lee is currently Director of the World Wide Web Consortium (W3C). A great deal of information about W3C activity can be obtained at its website.³⁰

Web 2.0

The term Web 2.0 has been used since 2004 to ‘describe the changing trends in the use of World Wide Web technology and Web design that aim to enhance creativity, communications, secure information sharing, collaboration and functionality of the Web’ (Web 2.0, 2009). Web 2.0 ‘makes it easier for everyone to publish online, using tools such as blogs, wikis and social networking sites’ (Harrison, 2009).

‘There’s still a huge amount of disagreement about just what Web 2.0 means, some people decrying it as a meaningless marketing buzzword and others accepting it as the new conventional wisdom’ (O’Reilly, 2008). Tim Berners-Lee ‘has questioned whether the term can be used in any meaningful way since many of the technology components of Web 2.0 have existed since the early days of the Web’ (Web 2.0, 2009).

Does Web 2.0 have any relevance to the subject of this present work? Some would argue perhaps that it does. Steve Bailey, for instance, suggests that IT advances have created a new information environment that renders classical record management methods impractical (Poulter, 2008). This author, however, tends to agree with Alan Poulter when he states that there is no easy way to manage information via Web 2.0 or anything else (*ibid.*).

Conclusion

Newton (2000) tells us that the deficiency of search engines has been noted by a number of authors. It appears that current engines are not efficient enough tools to provide adequate access to such an extended resource base as the Internet. ‘The tendency of automated search engines to inundate users with irrelevant results has prompted reconsideration of the merits of classification.’ As we have seen, classification also has a significant part to play in e-commerce.

It seems clear that there is an important role for classification in all aspects of the Internet and, of course, as demonstrated in the previous

chapter, *all* types of searching must involve classification to a certain degree.

Notes

- 1 Examples of Internet searching included in this chapter were valid at the time of writing in 2008/9.
- 2 www.google.com
- 3 www.yahoo.com.
- 4 www.google.com/dirhp.
- 5 <http://dir.yahoo.com>.
- 6 www.webrary.org/ref/weblinksmenu.html.
- 7 www.bubl.ac.uk.
- 8 www.public.iastate.edu/~cyberstacks/homepage.html.
- 9 www.scout.cs.wisc.edu/archives/
- 10 www.intute.ac.uk/socialsciences.
- 11 <http://library.liv.ac.uk>.
- 12 www.ascleiden.nl/library.
- 13 <http://conceptspace.london.edu/>
- 14 www.connexion.oclc.org
- 15 http://endeca.com/corporate-info/press-room/pr/pr_2006-0113.html
- 16 www.lib.ncsu.edu/endeca.
- 17 <http://connexion.oclc.org>.
- 18 www.lib.ncsu.edu/searchcollection.
- 19 www.udconline.net.
- 20 <http://classificationweb.net>.
- 21 www.acm.org/class/1998.
- 22 Taken from Coates' original *BSO* website at www.classbso.demon.co.uk.
This website has now been reduced in scale since UCL took over the management of *BSO*.
- 23 www.ucl.ac.uk/fatks.bso.
- 24 www.oclc.org/research/projects/auto_class/default.htm.
- 25 www.purl.oclc.org/scorpion.
- 26 www.oclc.org/research/software/rdf_topicmaps/default.htm.
- 27 www.ebay.com.
- 28 www.eccma.org.
- 29 www.ernestjones.co.uk
- 30 www.w3.org.

References

- Adkisson, Heidi P. (2005), *Use of Faceted Classification*. Available at: www.webdesignpractices.com/navigation/facets.html.
- Beaumont, Claudine and Warman, Matt (2009), 'The Digital Life', *Daily Telegraph*, (12 Mar).
- Blumberg, Robert and Atre, Shaju (2003), 'Automatic Classification : Moving to the Mainstream'. reprinted from *DM Review*, (Apr) 2003 and available at: www.soquelgroup.com/articles/dmreview_0403_classification.pdf.
- Dodd, David G. (1996), 'Grass Roots Cataloguing and Classification : Food for Thought from World Wide Web Subject-oriented Lists', *Library Resources and Technical*

- Services*, 40 (3), pp. 275-286.
- Harrison, Nicola and Place, Emma (2009), 'The Best of the Web', *Library & Information Update* (Jan-Feb), p.48-50.
- ISOK UK Conference (4th : 2008 : University College, London), *Semantic Analysis Technology in Search of Categories, Concepts & Context*, ISOK UK, London.
- Jenkins, Charlotte, et al. (2001), *Automatic Classification of Web Resources Using Java and Dewey Decimal Classification*. Available at: www.scu.edu.au/programme/posters/1846/com1846.htm
- MacLennan Alan (2000), 'Classification and the Internet', in Rita Marcella and Arthur Maltby (eds.), *The Future of Classification*, Gower, Aldershot, Hants; Brookfield, Vt., pp. 59-68.
- Neely, Mark (1999), *All About Searching the Internet*, rev. ed., Net.Works, Harrogate, Yorks., p. 17.
- Newton, Robert (2000), 'Information Technology and New Directions' in Rita Marcella and Arthur Maltby (eds.), *The Future of Classification*, Gower, Aldershot, Hants; Brookfield, Vt., pp. 59-68.
- O'Reilly, Tim (2008), *What Is Web 2.0 : Design Patterns and Business Models for the Next Generation of Software*, Available at: www.oreillynet.com/pub/a/oreilly/tim/news/2005/09/30/what_is_web_2.0.html.
- Pergolesi, Mauro (2006), *50 years of NATO : Codification Plays a Vital Role*, p. [5]. Available at: www.nato.int/codification.
- Poulter, Alan (2008), 'Is Web 2.0 the Answer to Records Management?': [review of Bailey, Steve, *Managing the Crowd: Rethinking Records Management for the Web 2.0 World*, Facet, London, 2008], *Library and Information Update*, (Dec), p. 23.
- Turner, David (200?), *UNSPSC: Introduction and Overview*. Available at: www.unspsc.org/documents/UNSPSCBasicSlides.pdf.
- UNSPSC (2009), What is UNSPSC and What Are the Benefits of Coding Products and Services. Available at: www.unspsc.org/FAQs.asp.
- Web 2.0* (2009), Available at: http://en.wikipedia.org/wiki/web_2.0.

15 Conclusion

This text has attempted to outline some of the ways in which classification can be used in information storage and retrieval. Hopefully it has succeeded in showing that classification must be considered a very useful tool for the organization and location of documents, records or data in an information system. Langridge (1973) states that in information work, 'classification is the basis for all index languages' and he emphasizes that 'there is no substitute'. Clifton (1994) maintains that 'coding' is an essential feature in computer-based business systems because 'it is impossible to identify, uniquely and unerringly, even as few as a thousand different entities if only their descriptions are available'. Batley (2007) tells us that 'the principles of classification and thesaurus construction enhance findability in the Web environment'.

No one classification scheme will be suitable for all purposes and the choice, or design, of a classification scheme will be governed by factors such as the type of information system, the objectives of the system, and user requirement. A public or university library, for example, will require a general scheme, covering the whole of recorded knowledge, and with special features such as a generalia class. For shelf arrangement, in a library of this nature, notational complexity should be avoided; not only is this confusing but lengthy numbers are inconvenient to inscribe on the spines of books or pamphlets.

A school library's requirement will be similar to that of the public library but there may well be a need for the classification to reflect the curriculum, that is the way that subjects are taught. Teachers sometimes criticize certain schemes from this point of view, the *Dewey Decimal Classification*, for instance, which distributes the subject 'Geography' through various classes: 380, 550, 910, and so on. The school library may therefore prefer some other scheme that, in its opinion, more closely adheres to the needs of library users. However, the *Library of Congress Classification* and the *Dewey Decimal Classification* are in widespread use in public libraries in the United States, the United Kingdom and elsewhere. A school library using one scheme and an adjacent public library using another is itself a situation which creates a further problem, namely a lack of standardization, which is hardly conducive to ease of use and customer satisfaction.

The smaller school library could of course consider a scheme such as *Dewey* too detailed and decide to look for something simpler. In fact, for this reason, abridged versions of schemes such as *Dewey* and the *Bliss*

Bibliographic Classification have been compiled for use in school libraries. There are also some schemes which target school libraries specifically; the United Kingdom School Library Association's *Primary School Classification Scheme* (2004), for example, which provides a way of classifying non-fiction materials using a simplified *Dewey* system. This is claimed to be 'easy for everyone' and 'what teachers and school library services have been asking for'.¹

Conversely, a special library, or information service, could find a general scheme not detailed enough. A perfume and cosmetics firm would discover that the *Dewey Decimal Classification* provides only four divisions for their particular subject area, that is:

668.5	Perfumes and cosmetics
668.54	Perfumes
668.542	Natural perfumes
	Including floral oils and waters
668.544	Synthetic perfumes
668.55	Cosmetics

This would be completely inadequate; much greater specificity would be needed. A special scheme would have to be adopted or designed, although a general scheme might be used for peripheral subject areas. Table Q of *Uniclass* (see page 29), for example, consists of an outline of the *Universal Decimal Classification*. This 'indicates how *UDC* can be used to classify subjects not covered elsewhere in the *Uniclass* system' (Crawford, 1997, p. 13). The United States *National Library of Medicine Classification* (latest edition 2008) makes use of the *Library of Congress Classification* for subjects bordering on medicine and for general reference materials. The NLM scheme uses a mixed notation (letters and numbers) resembling that of LC and utilises the letters QS-QZ and W-WZ, which are permanently excluded from the Library of Congress scheme.²

It may well be that choice is not possible because a classification scheme is already in use in a particular system and the cost of reclassification, even if warranted, would be prohibitive. If *choice* is possible, it could be that a certain scheme may be adopted because it is already in use elsewhere in a similar information environment and therefore is tried and tested. There are many existing classification schemes, both general and in a wide range of subject areas. Some such schemes have been mentioned in this text; other examples are the American Mathematical Society's *Mathematics Subject Classification*³ and the American Institute of Physics *Physics and Astronomy Classification* (PACS).⁴ If the use of classification is being considered

for the first time, it could be advantageous to attempt to ascertain whether an appropriate scheme is already available.

It is also possible to incorporate the features of one scheme in another. The new edition of the *Bibliographic Classification* makes such use of other schemes, for example the *British Catalogue of Music Classification*, where appropriate. In the *London Classification of Business Studies*, Class P Law owes a great deal to *A Classification Scheme for Law Books* by Elizabeth Moyes and Class K Industries is adapted from the Central Statistical Office's *Standard Industrial Classification*. In turn, class C Management in *Uniclass* owes a debt to the *London Classification of Business Studies*.

Whether an existing classification is being considered or an original scheme is being compiled, attention must be paid to a number of the points that have been referred to throughout this text. For instance, decisions may need to be taken with regard to structure, citation order and notation.

A hierarchical structure might be chosen where it was desired to build in an expressive notation, so that superordinate, coordinate and subordinate relationships can be easily recognised from class numbers. Alternatively, where expressiveness was not so important, but flexibility and the ability to classify complex subjects were, the choice would probably be a faceted scheme. It is possible to make the notation expressive in a faceted scheme (see page 81) but only to a limited extent. The very nature of this type of scheme militates against expressiveness, especially when classification numbers for complex subjects can be built up by the combination of notations from various classes. It is easy to visualise 12 being a division of 1, or AB being a division of A, but how is the structure of the *CI/SfB Construction Indexing Manual* revealed by a number such as 81(24)Xf(J)? 'The brackets and the combinations of upper case and lower case letters made the codes difficult to understand. This complex notation also caused problems with computerisation' (Crawford, 1997, p. 9).

A citation order should be selected which will ensure that the scheme adheres as closely as possible to user needs. In a manual system the arrangement resulting from the citation order must be carefully considered. A certain filing order may suit the users of one information service but not another, and for this reason schemes such as *Uniclass*, the *London Education Classification* and the *London Classification of Business Studies* allow for flexibility and permit the adoption of a citation order which most closely caters for an information service's particular requirement. The latter scheme, for example, places the subject 'Organisation and methods and work study' at X, alphabetically one of the last classes. If a particular information system desires to collect everything on 'Organisation and work study' together, then class X could be cited first rather than last. Thus

‘Work study in the food industry’ would be XE/KBA (where KBA = Food industry) rather than KBA/XE, and ‘Work study in banks’ would be XE/ECB (where ECB = Banks) rather than ECB/ XE.

Notation should be as brief and simple as possible. If specificity, that is the degree to which an exact subject may be specified, is required then lengthy notations may have to be accepted. If the notation is to be handled solely by a computer, length and complexity will not matter; only human beings become confused by complexity, not machines. However for product coding in industry a fixed length numeric notation may well be preferred, for example 2310, which, in the *NATO Codification System*, indicates ‘Passenger motor vehicles’ (see also pages 76-8 and 79-80).

Searching is obviously facilitated in computerised systems if concepts are identified consistently by the same notation. In fixed length class numbers, identification need not be unique but, if not unique, there must be positional consistency. For example, in the classification for machine bolts (page 25-6), a search for ‘2’ in position one would be for ‘brass’ but in position four would be for ‘chromium plated’. In other types of notation, uniqueness would be necessary. In *Uniclass*, the concept ‘Women’ is N5331 and this is the only number for that concept. In the *Library of Congress Classification*, the concept ‘Women’ is to be found in a great many enumerated subjects, in various classes and with differing notations and it would be impossible to search for all the various aspects of this concept by class number. Clearly consistency is easier to achieve in a faceted scheme, although it does not always happen in practice. *Uniclass* itself often uses different notations for the one concept as is demonstrated by the extract from the index on page 31.

Decisions relating to structure, citation order and notation may be the only decisions required where a simple scheme for the coding of entities in a machine-based system is concerned. For more complex computer systems, and for other purposes, other factors must be taken into account. For detailed schemes an alphabetical index will be necessary and an explanatory introduction should be supplied.

Classification can aid the search procedure and, as explained in Chapters 12-14, alphabetical indexing techniques should be based upon classificatory principles. Computerised information systems are now the norm. The Internet has revolutionised the accessing of information. In the 1980s, it was forecast that there would be a move towards ‘the refinement of the use of classification in machine systems’ (Gorman, 1983) and that it was likely that classification would become mandatory in online searching’ (Richmond, 1983). Whilst these forecasts have materialised only in part, ‘increasingly, local online systems and search engines are using hierarchical or classification-based browsers to

organise and navigate Internet resources. This is a logical development, after all classification was devised in the beginning as a response to the need for organising large amounts of knowledge and information' (Chan, 2000). Mills (2004) sees faceted classification as 'the only viable form enabling the locating and relating of information to be optimally predictable'. As noted on page 139, the use of faceted classification is being used increasingly for organising information on the Internet. In 2004, La Barre reported that a Google search for 'faceted classification' returned over 8,000 results whereas five years previously it was only 100. A similar search today (2009) produces something over 80,000, which appears to indicate a greatly increased interest in the subject. Nevertheless, currently, keywords remain the preferred and predominant option for Internet searching. However, as explained earlier in this work, keywords may not provide the best solution. A study by Gross and Taylor (2005) found that if no *Library of Congress Subject Headings* were assigned to records and searchers had to rely on keyword searches alone, more than one third of the records could not be retrieved. Admittedly this study related to library catalogues but, nevertheless, it illustrates the possible greater efficiency of a controlled language, which can result in a reduction of irrelevant 'hits'.

Classification can also be used on the Web to facilitate e-commerce and e-business, which consists, essentially, of the exchange of data and the conducting of financial and other transactions over the Internet. Systems such as *BIC Subject Categories* (see pages 57-9), the *NATO Codification System* (see pages 76-8 and 79-80) and *UNSBSC* (see page 139) assist in the exchange of electronic data and the conduct of business. General retail systems such as Amazon and eBay (see pages 137-8) categorise products in order to help customers make an informed choice, as do retailers who specialise in a particular type of goods, such as jewellers (see pages 139-140). Some of these systems may be perceived as simple when compared with complex systems such as the *Library of Congress Classification* but they rely on similar basic principles and satisfy a perceived need.

It is clear from the evidence as outlined in this work that classification is, as it always has been, essential and indispensable in mankind's efforts to organise knowledge and retrieve relevant information.

To sum up, 'one of the major objectives of an information retrieval system is to allow the user to discover with the minimum of effort any items not relevant to an enquiry. Classification schemes set out to achieve this by grouping items according to specified characteristics'. Foskett (1982), however, from whose work this quote is taken, adds an important warning: 'provided that our specification meets the user's needs, this grouping will be helpful, but if it does not, our organisation of information may prove to be a positive hindrance instead of a help'. This is certainly a

message to be heeded, classification is a means to an end, not an end in itself. Cutter (1904) had it right way back in the nineteenth century when he maintained that the ease of the user is to be preferred to that of the indexer.

Notes

- 1 www.sla.org.uk/primary-school-classification-scheme.php.
- 2 wwwcf.nlm.nih.gov/class/nlmclassintro.html.
- 3 www.ams.org.msc/.
- 4 www.aip.org/pacs/.

References

- Batley, Sue (2007), 'Going Nowhere?', *Catalogue & Index*, 155 (Spring), p. 6-7.
- Chan, Lois Mai and Hodges, Theodora L. (2000), 'The Library of Congress Classification', in Rita Marcella and Arthur Maltby (eds), *The Future of Classification*, Gower, Aldershot, Hants; Brookfield, Vt., pp. 105-127.
- Clifton, H.O. and Sutcliffe, A.G. (1994), *Business Information Systems*, 5th ed., Prentice Hall, New York; London, p. 320.
- Crawford, Marshall, Cann, John and O'Leary, Ruth (eds.) (1997), *Uniclass : Unified Classification for the Construction Industry*, RIBA, London.
- Cutter, Charles A. (1904), *Rules for a Dictionary Catalog*, 4th ed., Government Printing Office, Washington, p. 6.
- Foskett, A.C. (1982), *The Subject Approach to Information*, 4th ed., Bingley, London, p. 158.
- Gorman, Michael (1983), 'Technical Services, 1984-2001 (and before)', *Technical Services Quarterly*, 1 (1/2), pp. 65-71.
- Gross, Tina and Taylor, Arlene (2005), 'What Have We Got to Lose? The Effect of Controlled Vocabulary on Keyword Searching Results', *College and Research Libraries*, 66 (3), pp. 212-230.
- La Barre, Kathryn, (2004), 'Adventures in Faceted Classification : a Brave New World', in I.C. McIlwaine (ed.), *Advances in Knowledge Organisation ...* vol. 9, pp. 79-84. Available at: <http://netfiles.uiuc.edu/klabarre/www/isko2004/sysweb/2blabarre.ppt>
- Langridge, Derek (1973), *Approach to Classification for Students of Librarianship*, Bingley, London, p. 112.
- Mills, Jack (2004), 'Faceted Classification and Logical Division in Information Retrieval', *Library Trends* (Winter). Available at: http://findarticles.com/p/articles/mi_m1387/is_3_52/ai_n6080403.
- Richmond, Phyllis A. (1983), 'Futuristic Aspects of Subject Access', *Library Resources and Technical Services*, 27 (1), pp. 88-93.

Bibliography

This list contains entries for classification schemes and other monographic items referred to in the text but a selection of other works on classification and related topics, which might be useful for further reading, have been added.

Mention should also be made of serials relevant to the subject. Notable examples are *Cataloging and Classification Quarterly* and *Knowledge Organisation* (formerly *International Classification*). Other, less subject specific serial publications such as *Library Resources and Technical Services* may also contain articles and information that are of interest

In addition, the reader should note that the Internet is a veritable mine of information on the subject.

ACM Computing Classification (1998) (1999 – valid through 2009), Association for Computing Machinery. Available at: www.acm.org/class.

Aitchison, Jean (1982), 'Indexing Languages, Classification Schemes and Thesauri', in L.J. Anthony, (ed.), *Handbook of Special Librarianship and Information Work*, 5th ed., Aslib, London.

Aitchison, Jean, Gomersall, Alan and Ireland, Ralph (1969), *Thesaurofacet: Thesaurus and Classification for Engineering and Related Subjects*, 4th ed., English Electric, Leicester.

Aitchison, Jean, Gilchrist, Alan and Bawden, Alan (1997), *Thesaurus Construction and Use: a Practical Manual*, 3rd ed., Aslib, London.

Austin, Derek and Dykstra, Mary (1984), *PRECIS : a Manual of Concept Analysis and Subject Indexing*, 2nd ed., British Library, London.

Batley, Susan (2005), *Classification in Theory and Practice*, Chandos, Oxford.

Bengsten, Betty G. and Hill, Janet-Swan (eds) (1990), *Classification of Library Materials: Current and Future Potential for Providing Access*, Neal-Schuman, London, New York.

BISAC Subject Headings (2008), Book Industry Standards and Communication. Available at www.bisg.org/standards/bisac_subject/index.html.

BIC Subject Categories (2006), Book Industry Communication. Available at www.bic.org.uk/7/subject-categories.

Bliss, H.E. (1940-53), *A Bibliographic Classification*, H.W. Wilson, New York, 4 v.

Blumberg, Robert and Atre, Shaju (2003), 'Automatic Classification: Moving to the Mainstream'. reprinted from *DM Review*, Apr 2003 and available at: www.soquelgroup.com/articles/dmreview_0403_classification.pdf.

Bowman, John H. (2004), *Essential Dewey*, Facet, London.

British Standards Institution (1979), *Guidelines for the Establishment and Development of Monolingual Thesauri*, BSI, Milton Keynes. (BS 5723).

- Based upon the 'guidelines' previously issued by the International Standards Organisation (ISO 2788).
- Broughton, Vanda (2004), *Essential Classification*, Facet, London.
- Broughton, Vanda (2006), *Essential Thesaurus Construction*, Facet, London.
- BSI Root Thesaurus (1988), 3rd ed., British Standards Institution, Milton Keynes.
- BSO : *Broad System of Ordering* (1991), FID/BSO Panel (Eric Coates, et al.), rev. ed. Available, including 1994 update, at www.ucl.ac.uk.uk/fatks/bsc. The copyright of BSO is now held by University College London and the responsibility for its future management now lies with this institution.
- Caplan, Priscilla (2003), *Metadata Fundamentals*, Rev ed., ALA Editions, Chicago.
- Chan, Lois Mai (2007), *Cataloging and Classification: an Introduction*, 3rd ed., Scarecrow, Lanham, Md.
- Chan, Lois Mai, et al. (1996), *Dewey Decimal Classification: a Practical Guide*, 2nd ed., Forest Press, a Division of OCLC Online Computer Library Center, Albany, NY.
- Chan, Lois Mai and Hodges, Theodora L. (2000), 'The Library of Congress Classification', in Rita Marcella and Arthur Maltby (eds), *The Future of Classification*, Gower, Aldershot, Hants; Brookfield, Vt., pp. 105-127.
- Chan, Lois Mai (2005), *Library of Congress Subject Headings : Principles and Practice*, 4th ed., Libraries Unlimited, Englewood, Co.
- Chowdry, G.C. (2003), *Introduction to Modern Information Retrieval*, 2nd rev.ed., Facet, London.
- Clifton, H.O. and Sutcliffe, A.G. (1994), *Business Information Systems*, 5th ed., Prentice Hall, London; New York.
- Coates, E.J. (1960), *The British Catalogue of Music Classification*, British National Bibliography, London.
- Coates, E.J. (1960), *Subject Catalogues: Headings and Structure*, Library Association, London.
- Crawford, Marshall, Cann, John and O'Leary, Ruth (eds) (1997), *Uniclass: Unified Classified Classification for the Construction Industry*, RIBA, London.
- Cutter, Charles A. (1904), *Rules for a Dictionary Catalog*, 4th ed., Government Printing Office, Washington.
- Dewey, Melvil (2003), *Dewey Decimal Classification and Relative Index*, edition 22, ed. by Joan S. Mitchell ... [et al.], OCLC, Dublin, Ohio. 4 v.
- Dittman, Helena (2000), *Learn Library of Congress Classification*, Scarecrow Press, London.
- Downing, M.H. and Downing, D.H. (1992), *Introduction to Cataloguing and Classification*, 6th ed., McFarland, Jefferson, N.C.
- Drabenstott, K.M. and Vizine-Goetz, D. (1994), *Using Subject Headings for Online Retrieval: Theory, Practice and Potential*, Academic Press, San Diego.
- E.G. Brisch & Partners [n.d.] *The Brisch Building Classification : prepared for the British Building Documentation Committee*, E.G. Brisch & Partners, [S.I.].

- Ellis, David (1996), *Progress and Problems in Information Retrieval*, 2nd ed., Library Association, London.
- Foskett, A.C. (2000), 'The Future of Faceted Classification', in Rita Marcella and Arthur Maltby (eds), *The Future of Classification*, Gower, Aldershot, Hants, Brookfield, Vt., pp. 69-80.
- Foskett, A.C. (1996), *The Subject Approach to Information*, 5th ed., Library Association, London..
- Foskett, D.J. and Foskett, Joy (1974), *The London Education Classification: a Thesaurus/classification of British Educational Terms*, 2nd ed., University of London, Institute of Education Library.
- Harry, Mike (1994) *Information Systems in Business*, Pitman, London.
- Hildreth, Charles R. (1989), *The Online Catalogue: Developments and Directions*, Library Association, London.
- Immoth, John Phillip and Chan, Lois Mai (1999), *A Guide to the Library of Congress Classification*, 5th rev. ed., Greenwood, London.
- International Organization for Standardization (1986), *Documentation: Guidelines for the Establishment and Development of Monolingual Thesauri*, 2nd ed., The Organization, Geneva, (ISO 2788).
- Intner, Sheila, [et al.] (2005), *Metadata and its impact on libraries*, Libraries Unlimited, Englewood, Colo.
- ISOK UK Conference (4th: 2008: University College, London), *Semantic Analysis Technology in Search of Categories, Concepts & Context*, ISOK UK, London.
- La Barre, Kathryn, (2004), 'Adventures in Faceted Classification: a Brave New World', in I.C. McIlwaine (ed.), *Advances in Knowledge Organisation ...* vol. 9, pp 79-84. Available at: <http://netfiles.uiuc.edu/klabarre/www/isko2004/sysweb/2blabarre.ppt>
- Lancaster, F.W. (1998), *Indexing and Abstracting in Theory and Practice*, 2nd ed., Library Association, London.
- Langridge, Derek (1973), *Approach to Classification for Students of Librarianship*, Bingley, London.
- Langridge, Derek (1992), *Classification: its Kinds, Systems, Elements and Applications*, Bowker-Saur, London.
- Library of Congress (various dates), *Classification*, LC, Washington, 43 v. The examples included here have been checked in 2008 against 'Classification Web', available from the Cataloging Distribution Service of LC at <http://classificationweb.net>.
- Library of Congress (2008) *Subject Headings*, 31st ed., LC, Washington, 5 v. The examples included here have been checked in 2008 against 'Classification Web', available from the Cataloging Distribution Service of LC at <http://classificationweb.net>.
- Library of Congress (2008), *Subject Headings Manual*, LC, Washington.
- London Business School Library (2005), *London Classification of Business Studies : Thesaurus*, The Library, 2 v. [Vol.1] Classified Sequence, [Vol. 2] Alphabetical Sequence. This is a new version of the scheme first devised by K.D.C. Vernon and Valerie Lang (see below).

- MacConnell, W. (1971), *Classification and Coding: an Introduction and Review of Classification and Coding Systems*, British Institute of Management, London.
- McIlaine, I.C., with participation from A.Buxton (1993), *Guide to the Use of the UDC*, International Federation for Information and Documentation, The Hague. FID occasional paper, 5.
- MacLennan Alan (2000), 'Classification and the Internet', in Rita Marcella and Arthur Maltby (eds), *The Future of Classification*, Gower, Aldershot, Hants; Brookfield, Vt., pp. 59-68.
- Maltby, Arthur (1975), *Sayers' Manual of Classification for Librarians*, 5th ed., Deutsch, London. Marcella (1994) is essentially a revised edition of this manual.
- Marcella, Rita and Maltby, Arthur (eds. (2000), *The Future of Classification*, Gower, Aldershot, Hants; Burlington, Vt.
- Marcella, Rita and Newton, Robert (1994), *A New Manual of Classification*, Gower, Aldershot, Hants; Burlington, Vt. Essentially a revised edition of Maltby (1975).
- Mills, J. (1976), 'Bibliographic Classification', in A. Maltby (ed.), *Classification in the 1970s: a Second Look*, rev. ed., Bingley, London, pp. 25-50.
- Mills, J., et al. (1977-), *Bliss Bibliographic Classification (Second Edition)*, Butterworths, London, and Saur., Munchen, vols in progress.
- Neely, Mark (1999), *All About Searching the Internet*, rev ed., Net.Works, Harrogate, Yorks.
- Newton, Robert (2000), 'Information Technology and New Directions', in Rita Marcella and Arthur Maltby (ed.), *The Future of Classification*, Gower, Aldershot, Hants; Brookfield, Vt., pp. 59-68.
- Palmer, Bernard I. and Wells, A.G (1951), *The Fundamentals of Library Classification*, Allen & Unwin, London.
- Pictorial Knowledge* (1970), International Learning Systems, London, 8 v.
- Ranganathan, S.R. (1987), *Colon Classification*, 7th ed., edited by M.A. Gopinath, Sarada Ranganathan Endowment for Library Science, Bangalore.
- Ranganathan, S.R. (1963), *Colon Classification*, 6th ed., with annexure, Asia Publishing House, Bombay; London.
- Ray-Jones, Alan and Clegg, David (1991), *CI/SfB Construction Indexing Manual*, RIBA, London. Abridged reprint of 1976 revision.
- RCN Library Nursing Thesaurus: a Thesaurus of Terms Used in Nursing, Midwifery, Health Visiting and Related Subject Areas* (2007), 5th ed., RCN [Royal College of Nursing] Library and Information Services, London.
- Rowley, Jennifer (1996), *The Basics of Information Systems*, 2nd ed., Library Association, London.
- Rowley, Jennifer and Hartley, Richard (2008), *Organizing Knowledge: an Introduction to Managing Access to Information*, 4th ed., Ashgate, Farnham, Surrey.
- Scott, Mona L. (1998), *Dewey Decimal Classification: a Study Manual and Number Building Guide*, Libraries Unlimited, Englewood, Colo.
- Shaw, Josephine (1984), *Administration for Business*, 2nd ed., Pitman, London.

- Smith, Raymond (1966), *Classification for London Literature Based Upon the Collection in the Guildhall Library*, 3rd ed., The Library Committee, London. Cited in George A. Carter (1973), *J.L. Hobbs's Local History and the Library*, 2nd rev. ed., Deutsch, London.
- Soergel, Dagobert (1985), *Organising Information: Principles of Database and Retrieval Systems*, Academic Press, Orlando, Florida; London.
- Sparck-Jones, Karen (1976), 'Automatic Classification', in Arthur Maltby (ed.), *Classification in the 1970s: a Second Look*, Bingley, London, pp. 209-225.
- Taylor, Arlene G. and Joudrey, Daniel N. (2008), *The Organisation of Information*, Libraries Unlimited, Englewood, Colo.
- Taylor, Arlene G. (2004), *Wynar's Introduction to Cataloging and Classification*, 9th rev. ed, Libraries Unlimited, Englewood, Colo.
- Thesaurus of Psychological Index Terms* (2007), 11th ed., edited by Lisa Gallagher Tuleya, American Psychological Association, Washington. The entry cited in the text is taken from the 1974 edition and varies slightly in this latest edition.
- Thomas, Alan R. (ed.) (1995), *Classification: Options and Opportunities*, Haworth, New York. Also published as *Cataloging and Classification Quarterly*, 19 (3/4).
- Thomas, Alan R. and Shearer, James R. (2000), *Internet Searching and Indexing: the Subject Approach*, Haworth, Binghamton, NY. Published simultaneously in *Journal of Internet Cataloging*, Vol. 2 (3/4).
- Unesco Thesaurus: a Structured List of Descriptors for Indexing and Retrieving Literature in the Fields of Education, Science, Culture, Communication and Information* (1994), New ed., Unesco, Paris. Previous edition compiled by Jean Aitchison, 1977.
- Universal Decimal Classification* (various dates), English text, (BS 1000). Available at: www.udconline.net.
- Vernon, K.D.C. and Lang, Valerie (1979), *The London Classification of Business Studies: a Classification and Thesaurus for Business Libraries*, 2nd ed. rev., K.G.B. Bakewell and David A. Cotton, Aslib, London.
- Vickery, B.C. (1960), *Faceted Classification: a Guide to Construction and Use of Special Schemes*, Aslib, London.
- Viet, Jean and Slype, Georges van (1984), *EUDISED Multilingual Thesaurus for Information Processing in the Field of Education*, New ed., English version, Mouton, Berlin. At head of title: Council of Europe. Commission of the European Communities.
- Wellisch, Hans H. (ed.) (1977), *The PRECIS Index System: Principles, Applications and Prospects*, H.W. Wilson, New York.

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